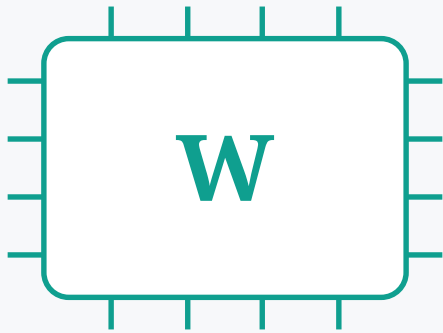


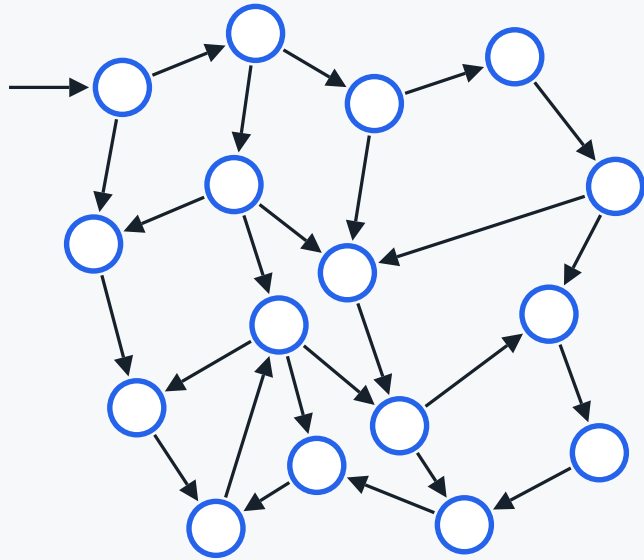


Certifying Symbolic Model Checking

Dr. Nils Froleys

KU LEUVEN

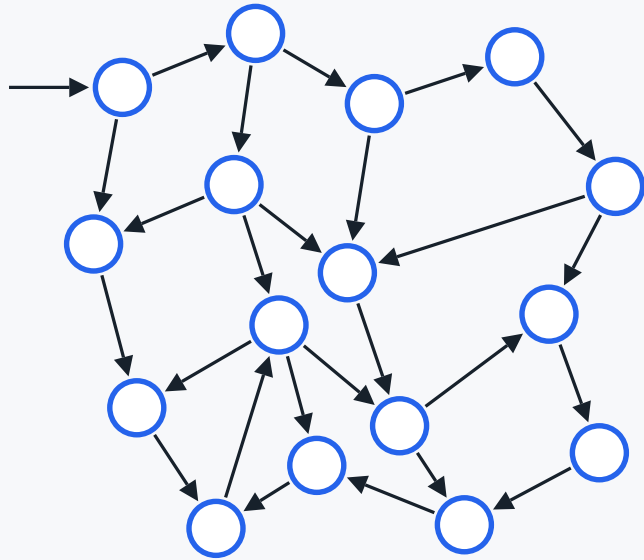
Symbolic State Space



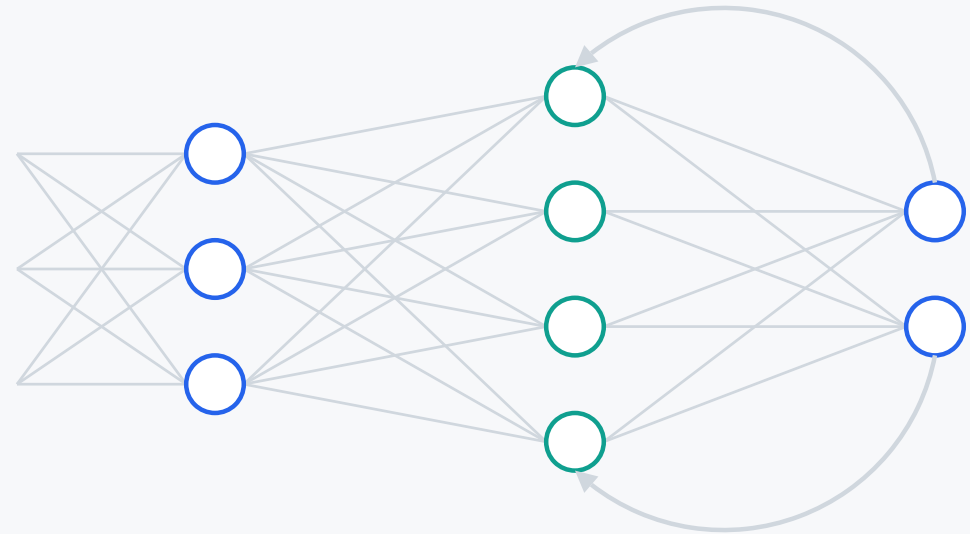
Program

```
long x = 0x5f3759df;  
long y = 0x9e3779b9;  
float z = 1.0f, h = 0.5f;  
  
while (1) {  
    x ^= x << 13;  
    y = * (long *) &z;  
    y = 0x5f3759df - (y >> 1);  
    h = (y & 0xff) * 0.00390625f;  
    z = z * (1.5f - h * z * z);  
    x = (x >> 1) ^ (y & 0x7fffffff);  
    assert(z >= 0.0f);  
}
```

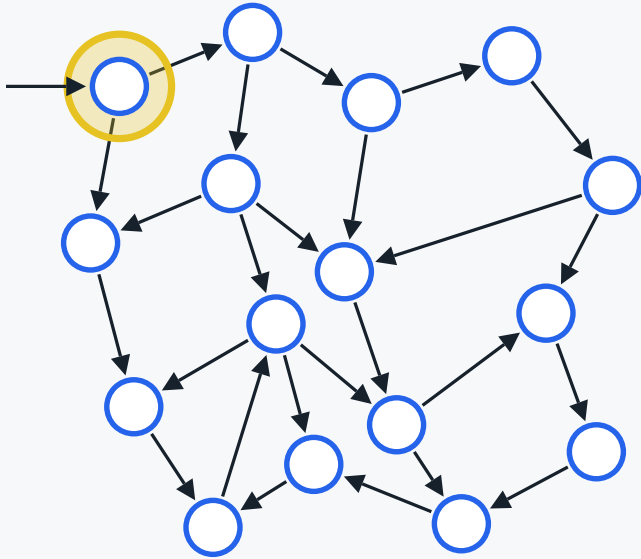
Symbolic State Space



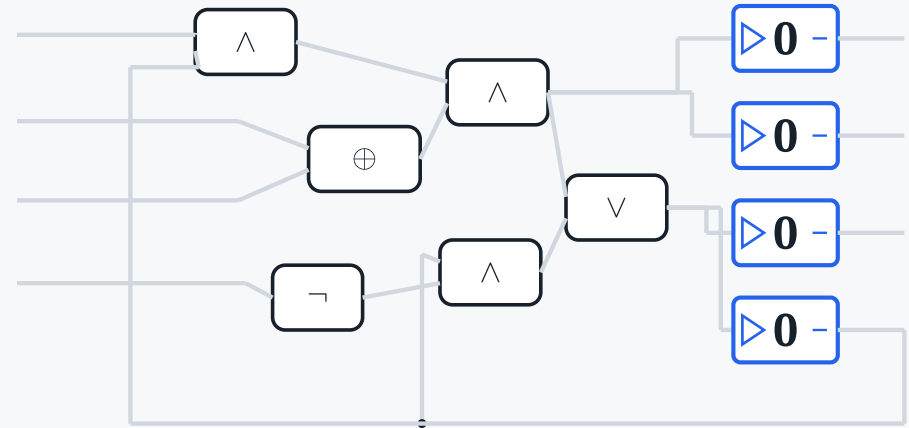
Recurrent Neural Network



Symbolic State Space

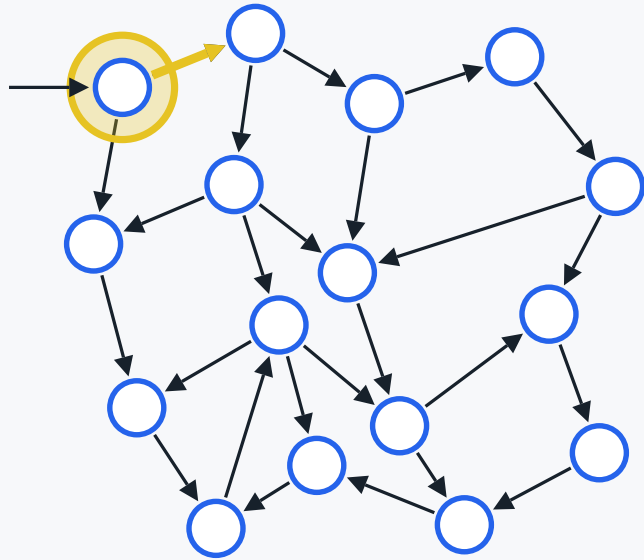


Logic Circuit

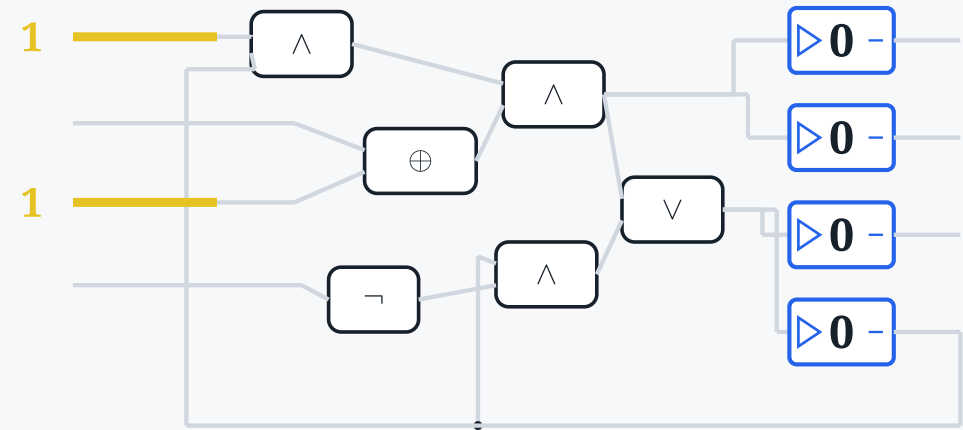


$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$

Symbolic State Space

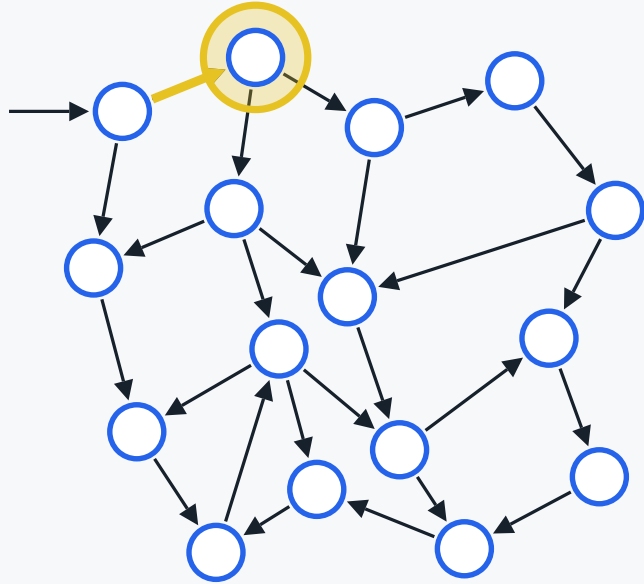


Logic Circuit

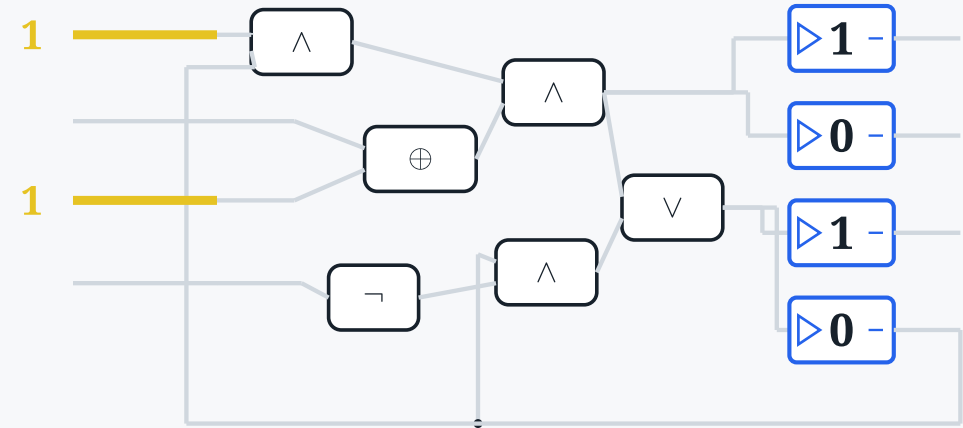


$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$

Symbolic State Space

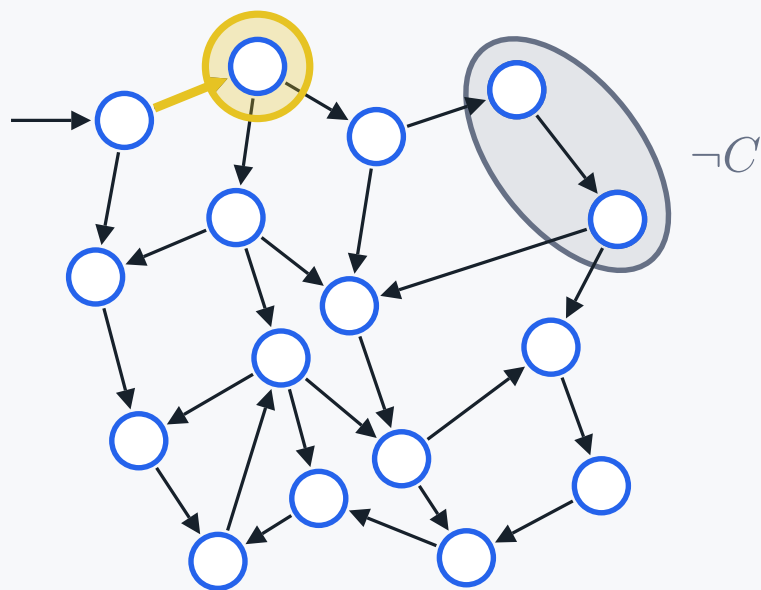


Logic Circuit

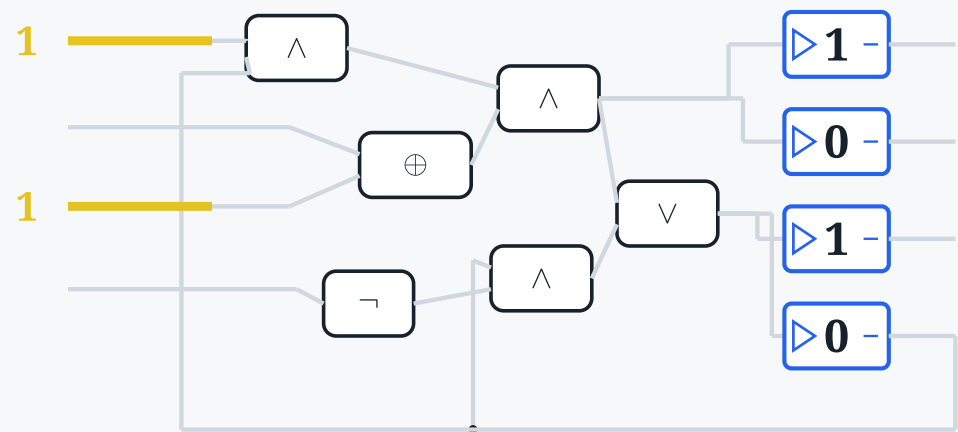


$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$

Symbolic State Space

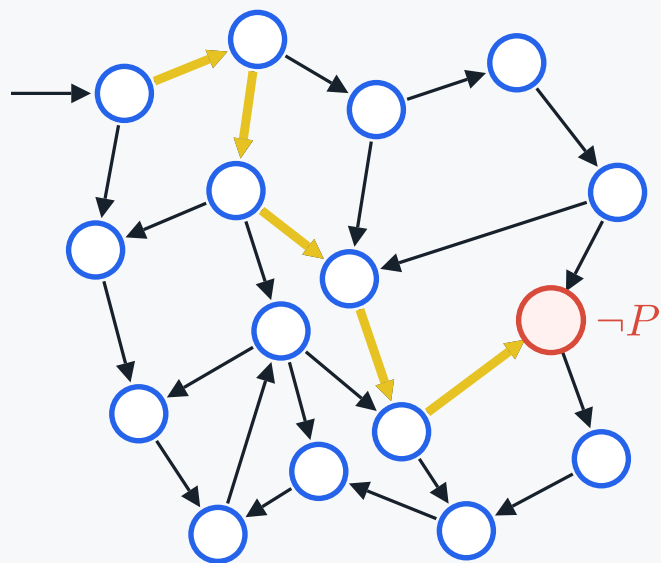


Logic Circuit

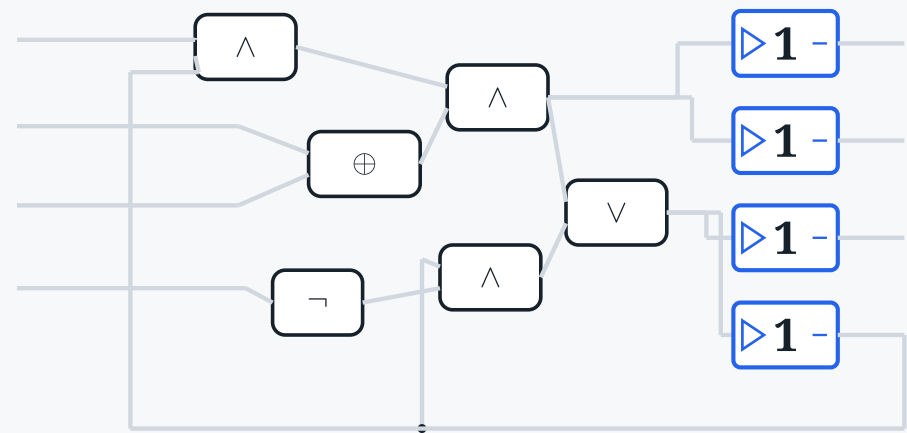


$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$

Symbolic State Space

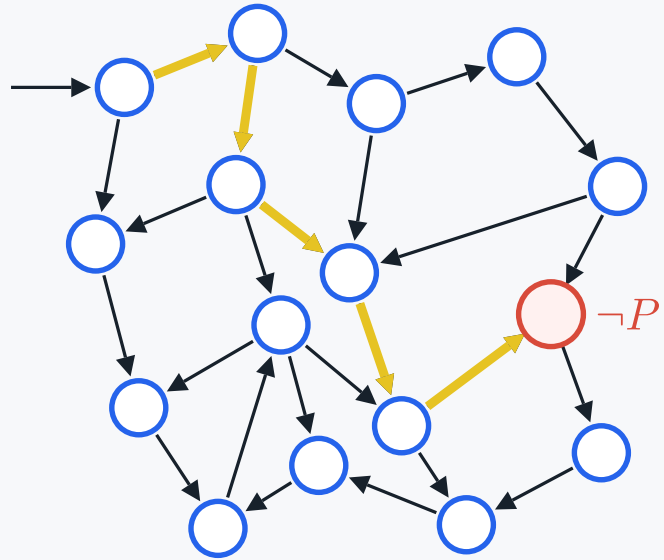


Logic Circuit

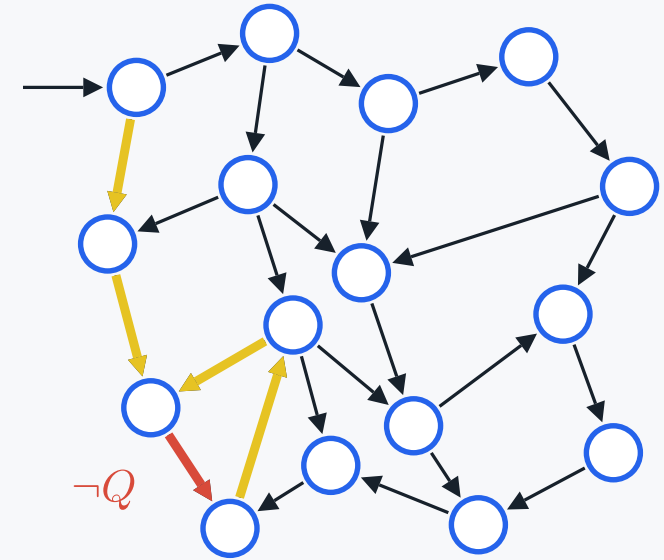


$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$

Symbolic State Space

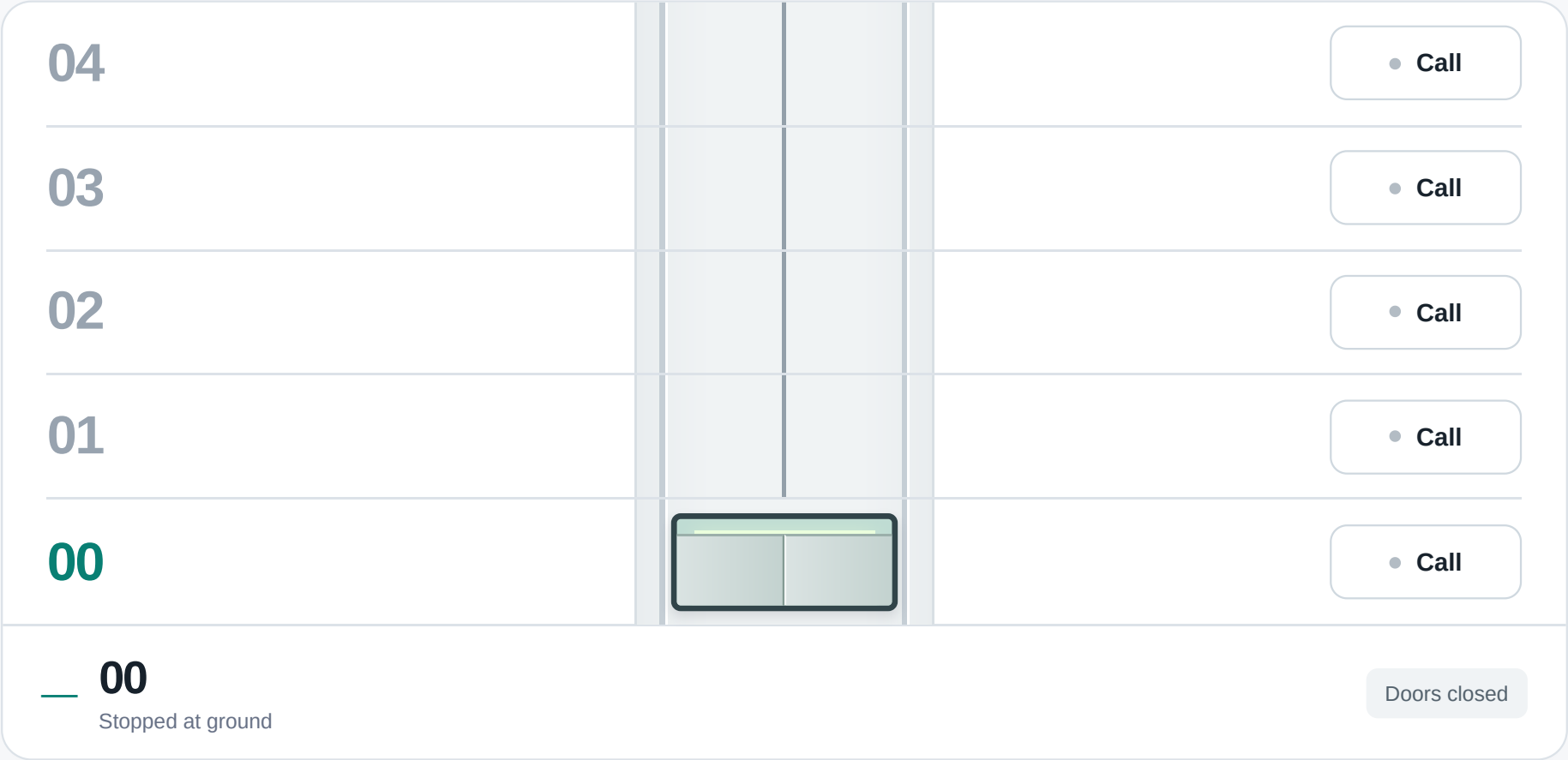


Safety



Liveness

Elevator Example



CONTROLLER

Simple · lowest floor first

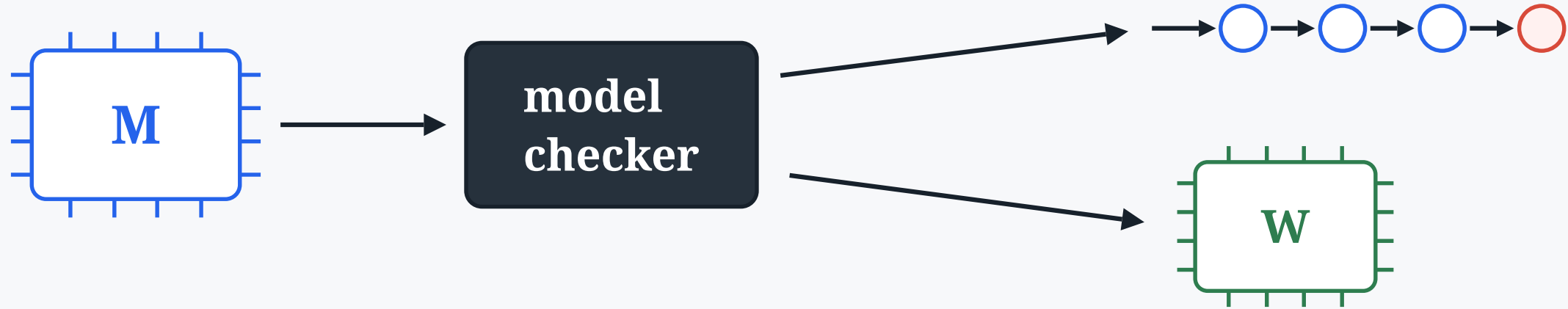
SPEED

$\frac{1}{2}\times$ 1× 2× 4×

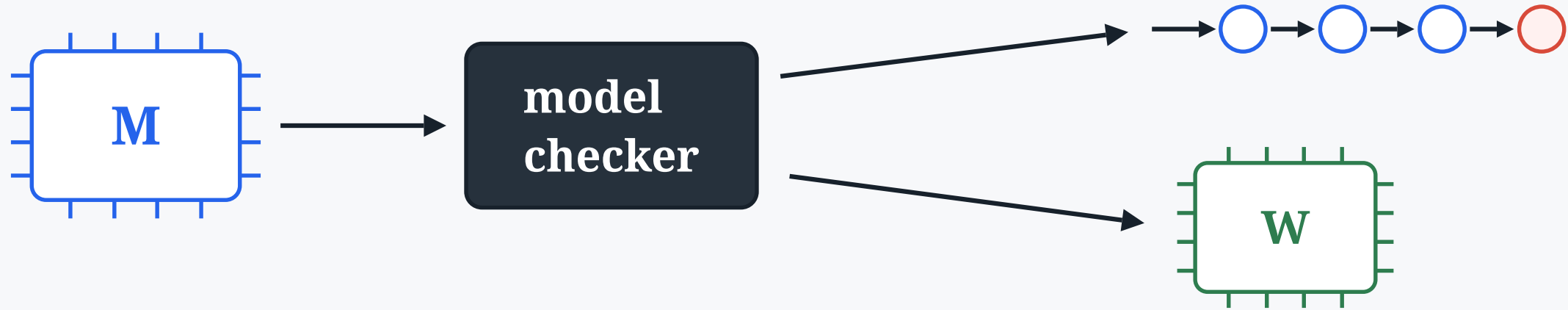


► Explore safety & liveness

Certificates



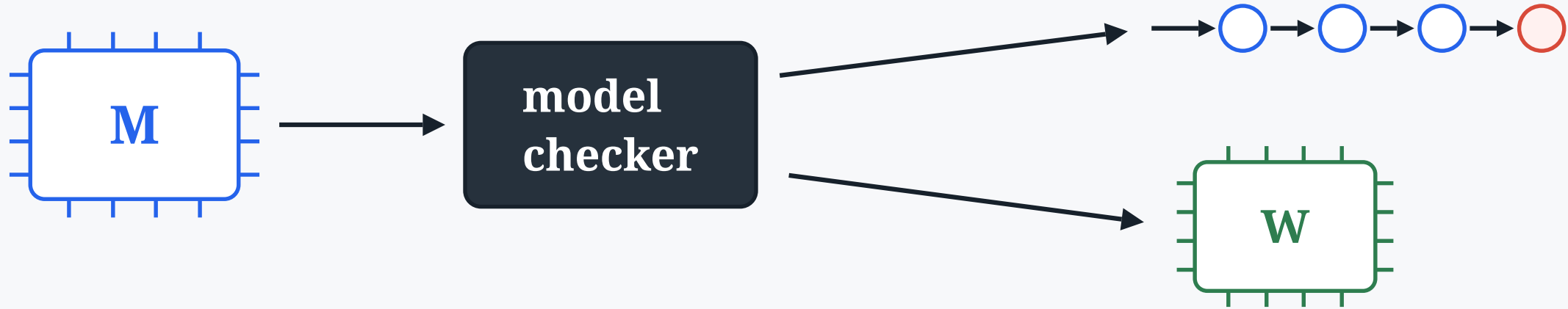
Certificates



PROMPT

Hey, ChatGPT
write a model checker!
Make no mistakes!

Certificates



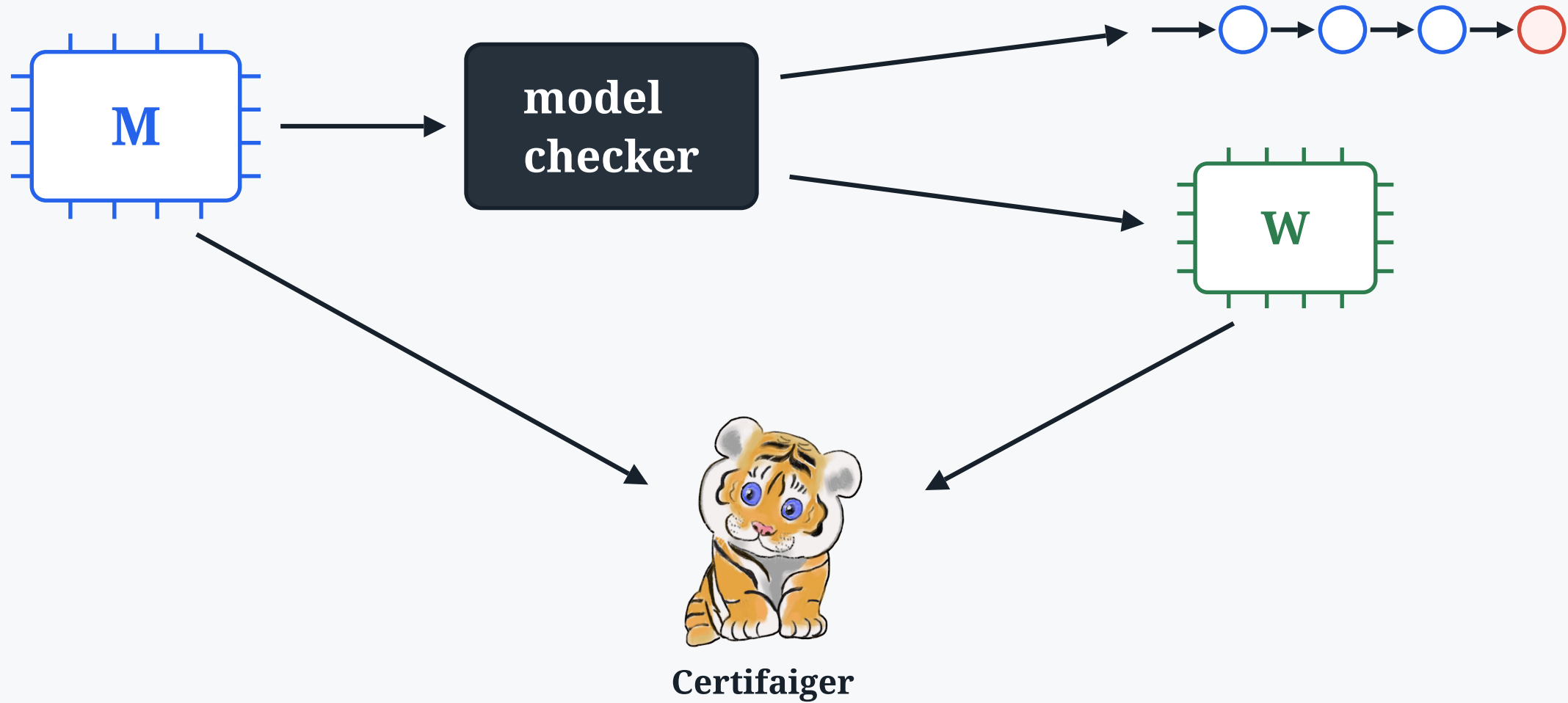
PROMPT

Hey, ChatGPT
write a model checker!
Make no mistakes!

model_checker.py

```
while time < 59:58
    if BMC finds counterexample
        return Unsafe
return Safe
```

Certificates



Certification Gap

Safety

- [\[CAV21\]](#) IC3, BDD, Interpolation
- [\[FMCAD22\]](#) k-Induction
- [\[FMCAD23\]](#) Temporal Decomposition
- [\[IJCAR24\]](#) Cone of Influence, Phase Abstraction
- [\[CAV25\]](#) Case Study: Certifying Model Checking Competition
- [\[FM26\]](#) Uniqueness Constraints, Constraint Mining
- [\[IJCAR26\]](#) Higher Level Reasoning

Liveness

- [\[CAV26\]](#) k-Liveness
- [\[CAV26\]](#) Stabilizer Extraction
- [\[CAV26\]](#) Liveness to Safety
- [\[CAV26\]](#) rLive

Certification Gap

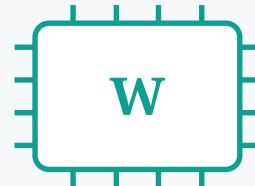
Safety

- [CAV21] IC3, BDD, Interpolation
- [FMCAD22] k-Induction
- [FMCAD23] Temporal Decomposition
- [IJCAR24] Cone of Influence, Phase Abstraction
- [CAV25] Case Study: Certifying Model Checking Competition
- [FM26] Uniqueness Constraints, Constraint Mining
- [IJCAR26] Higher Level Reasoning

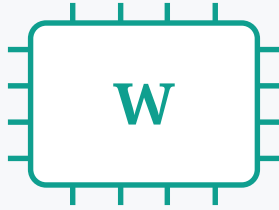
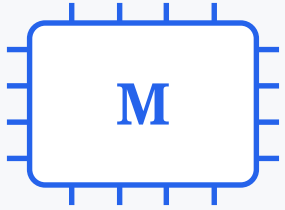
Liveness

- [CAV26] k-Liveness
- [CAV26] Stabilizer Extraction
- [CAV26] Liveness to Safety
- [CAV26] rLive

one generic certificate



Certificate Checking



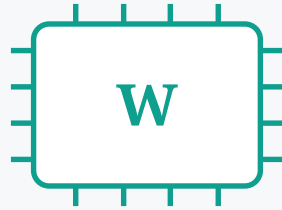
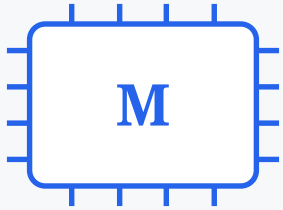
Simulates



Inductive

Ranked

Certificate Checking



Simulates

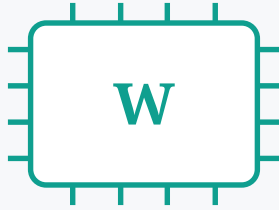
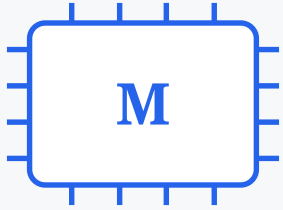
W is safe and live $\Rightarrow M$ is safe and live



Inductive

Ranked

Certificate Checking



Simulates

W is safe and live $\Rightarrow M$ is safe and live



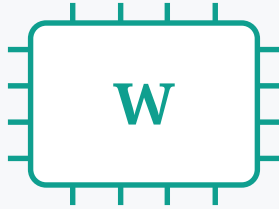
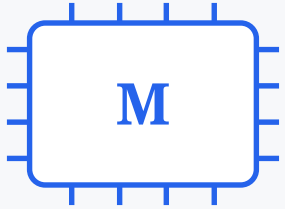
Inductive

W is safe

Ranked

W is live

Certificate Checking



$\langle I, L, R, F, C, P, Q \rangle$

$\langle I', L', R', F', C', P', Q' \rangle$

Simulates



Inductive

Ranked

Stratified $\mathcal{R}'\{L' \setminus L\}$ is acyclic

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

8 SAT Checks

Proof Format

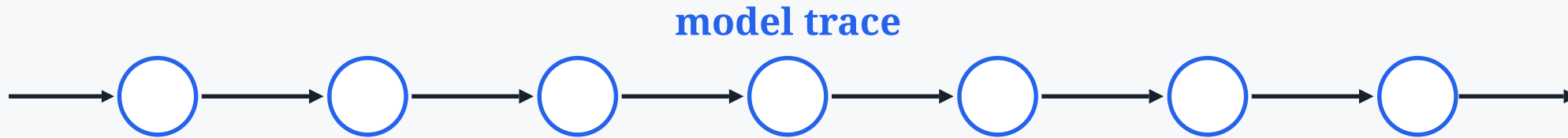
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

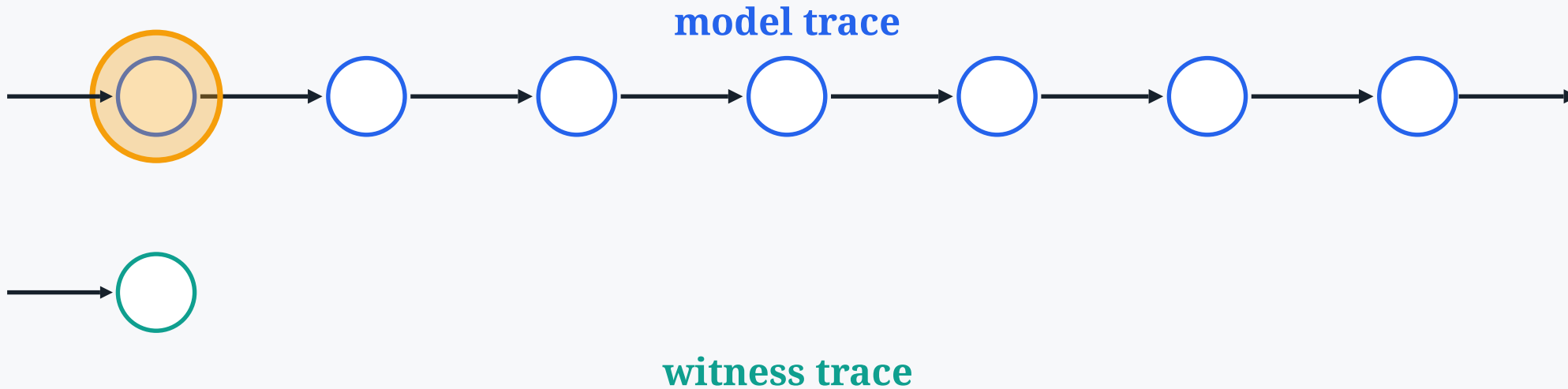
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

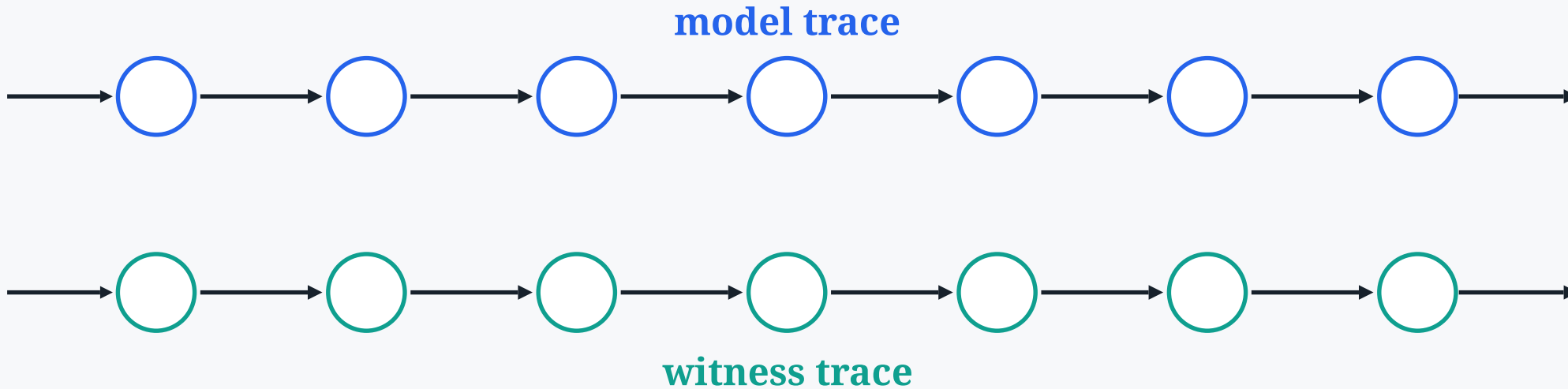
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

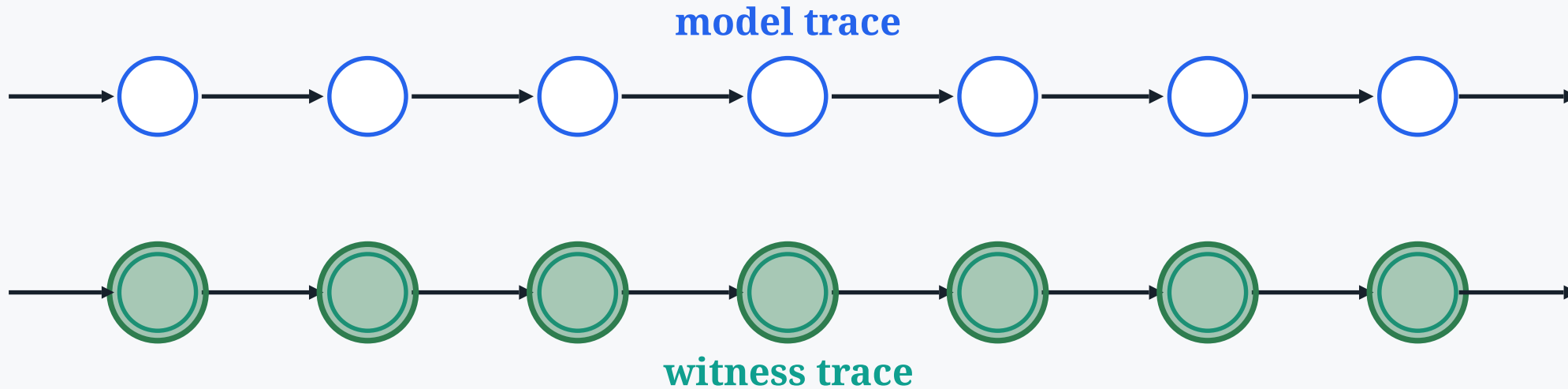
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

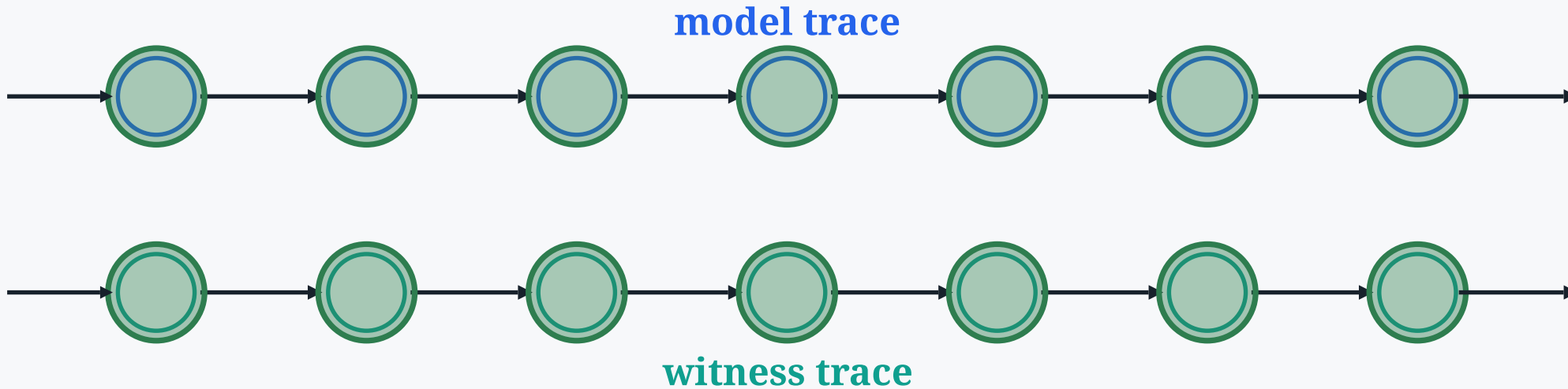
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

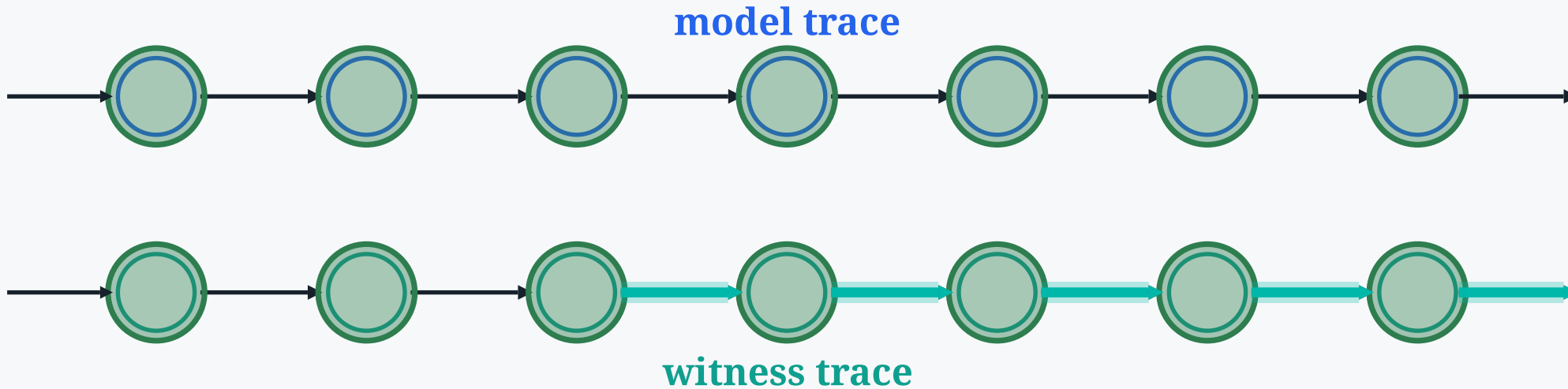
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



Proof Format

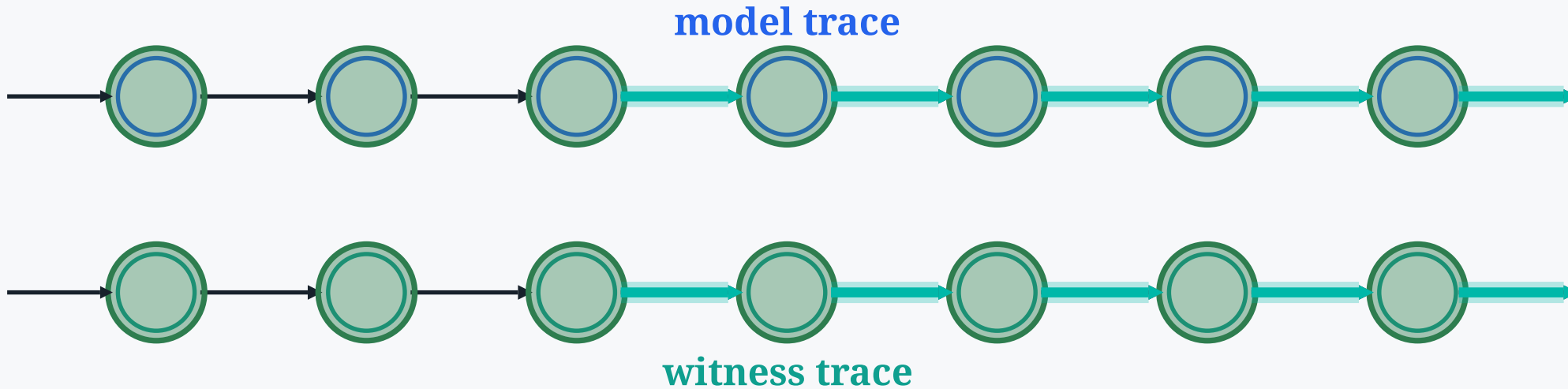
Simulation

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

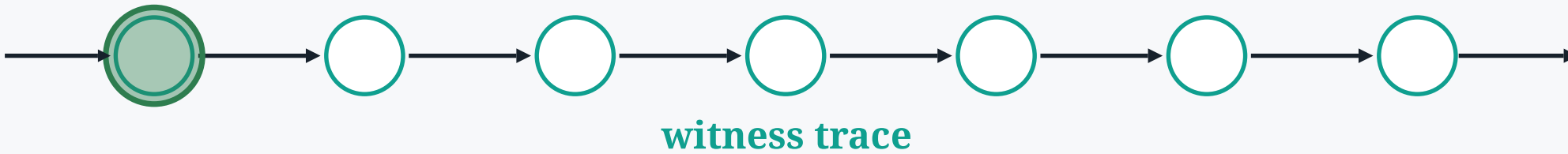


Proof Format

Inductive

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$



Proof Format

Inductive

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

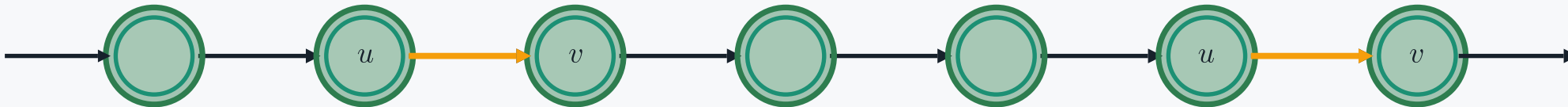


Proof Format

Ranked

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

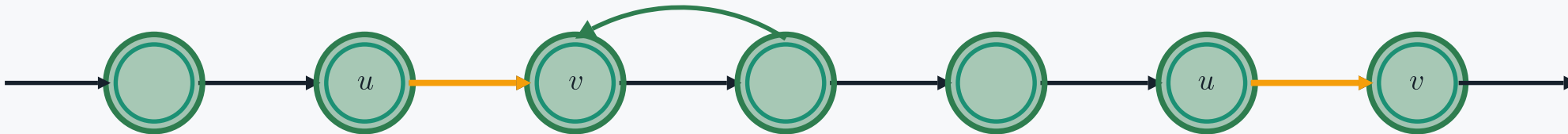


Proof Format

Ranked

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

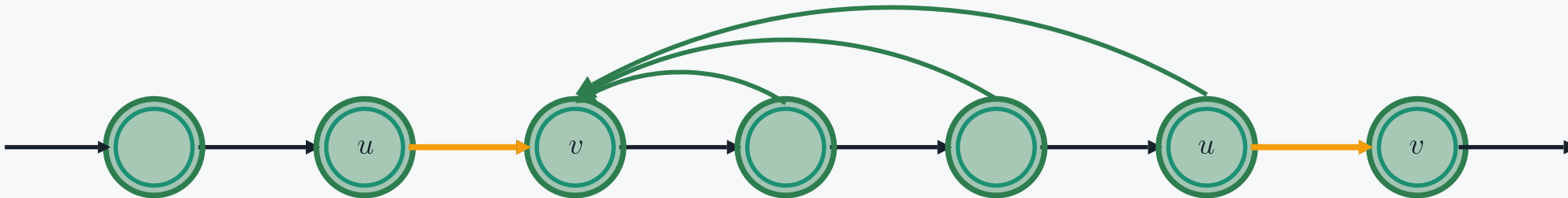


Proof Format

Ranked

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

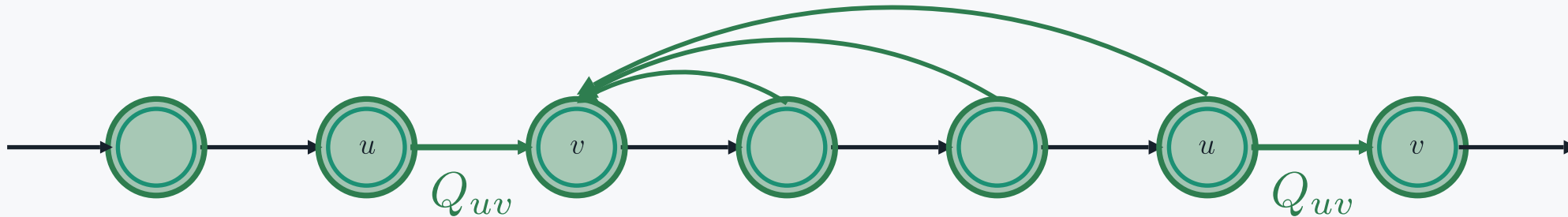


Proof Format

Ranked

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

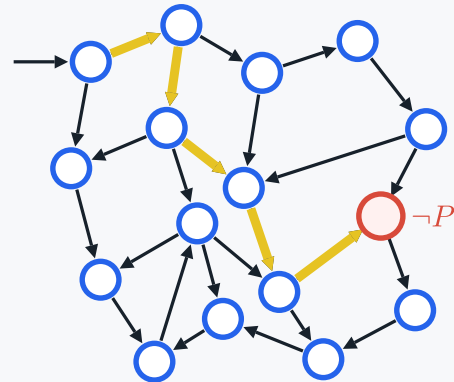
Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



Liveness Certification

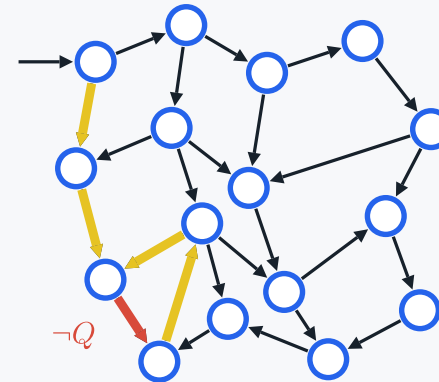
Safety

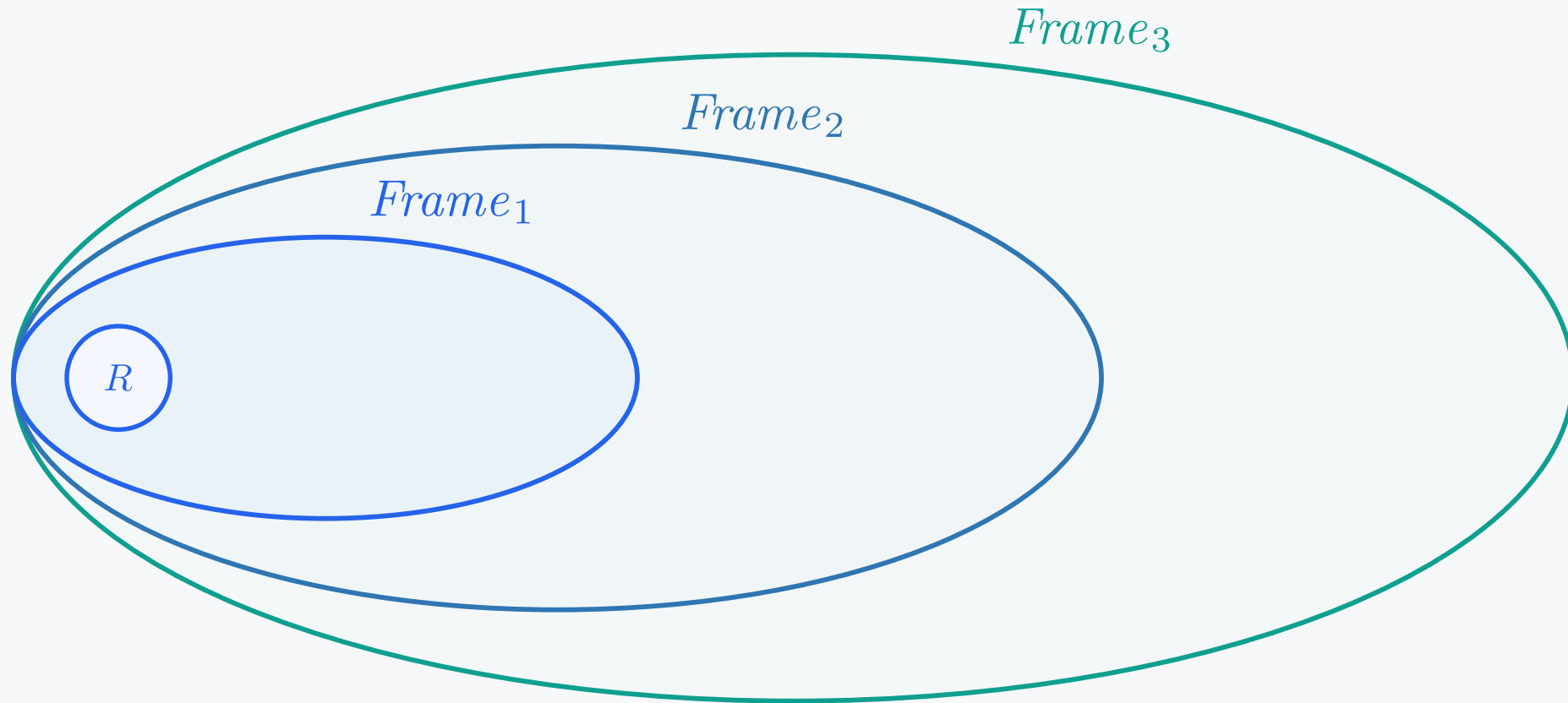
- [CAV21] IC3, BDD, Interpolation
- [FMCAD22] k-Induction
- [FMCAD23] Temporal Decomposition
- [IJCAR24] Cone of Influence, Phase Abstraction
- [CAV25] Case Study: Certifying Model Checking Competition
- [FM26] Uniqueness Constraints, Constraint Mining
- [IJCAR26] Higher Level Reasoning



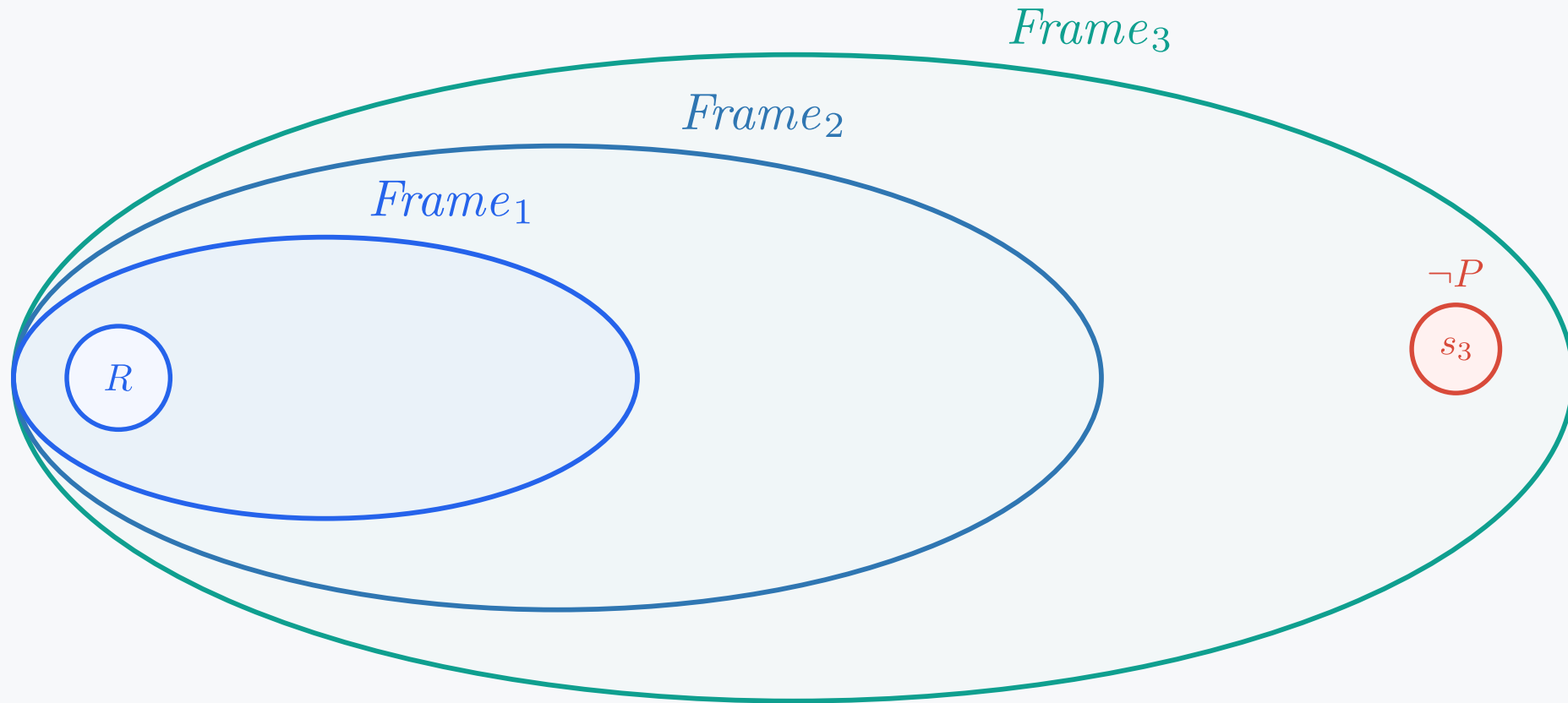
Liveness

- [CAV26] k-Liveness
- [CAV26] Stabilizer Extraction
- [CAV26] Liveness to Safety
- [CAV26] rLive



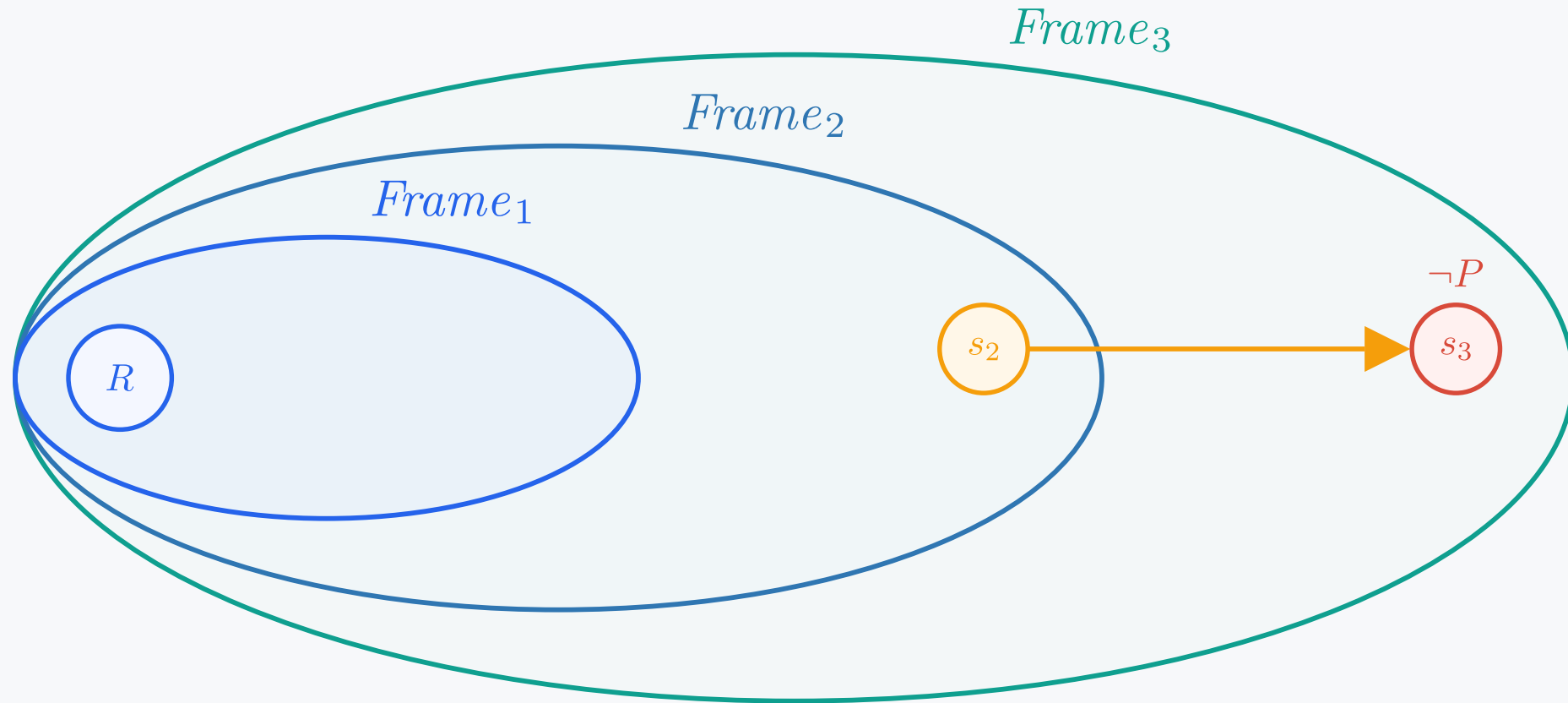


$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

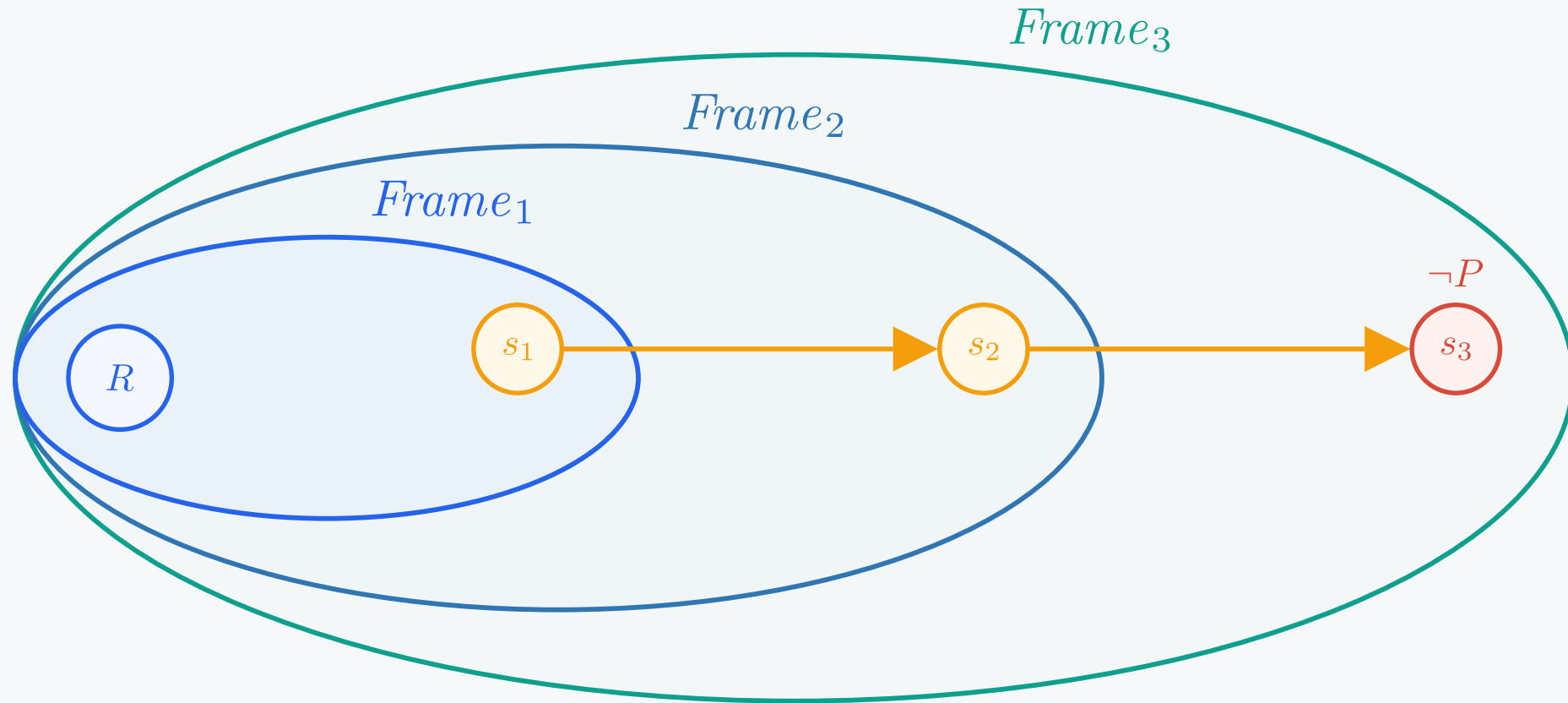


$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

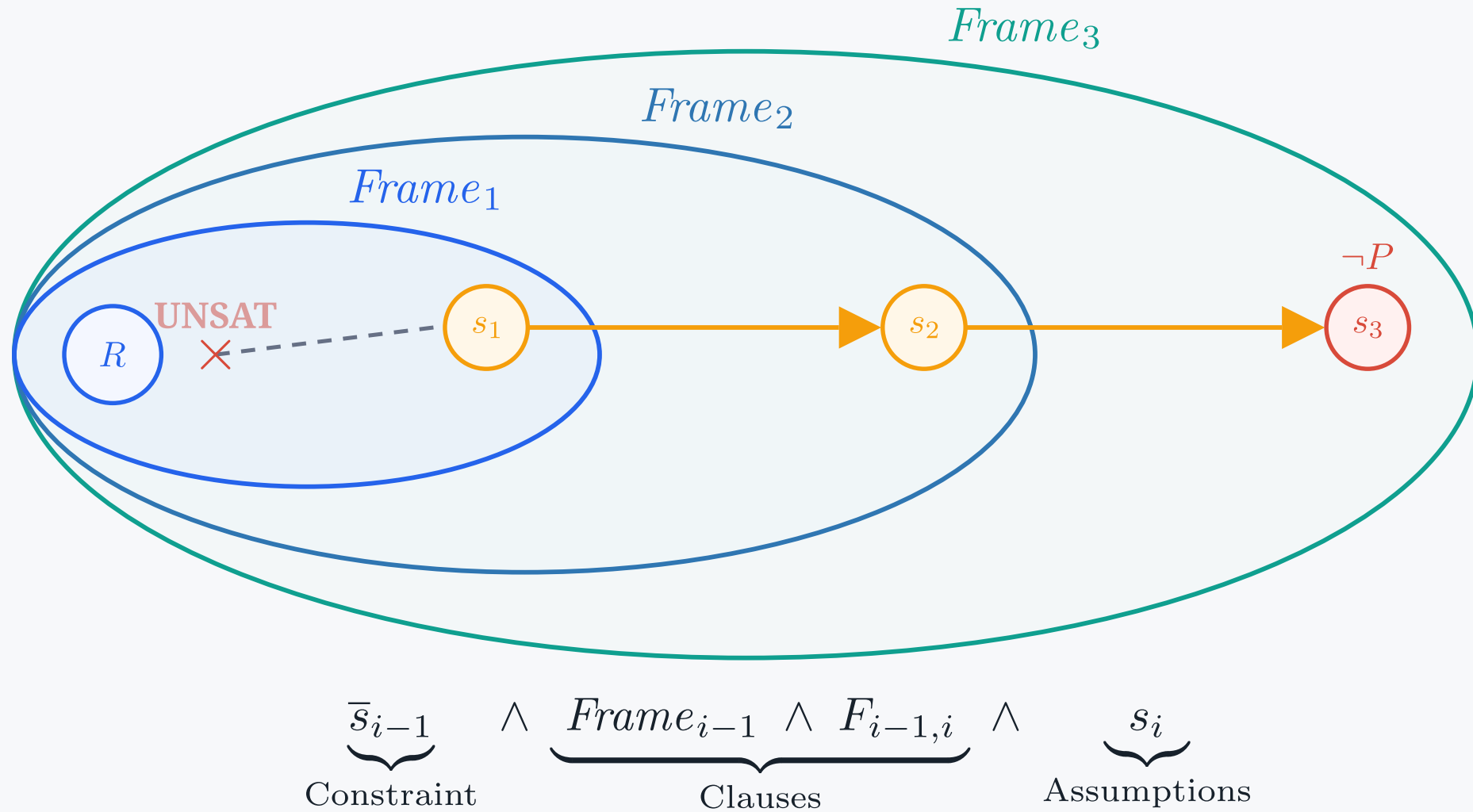
Froleyks & Biere [FMCAD21]

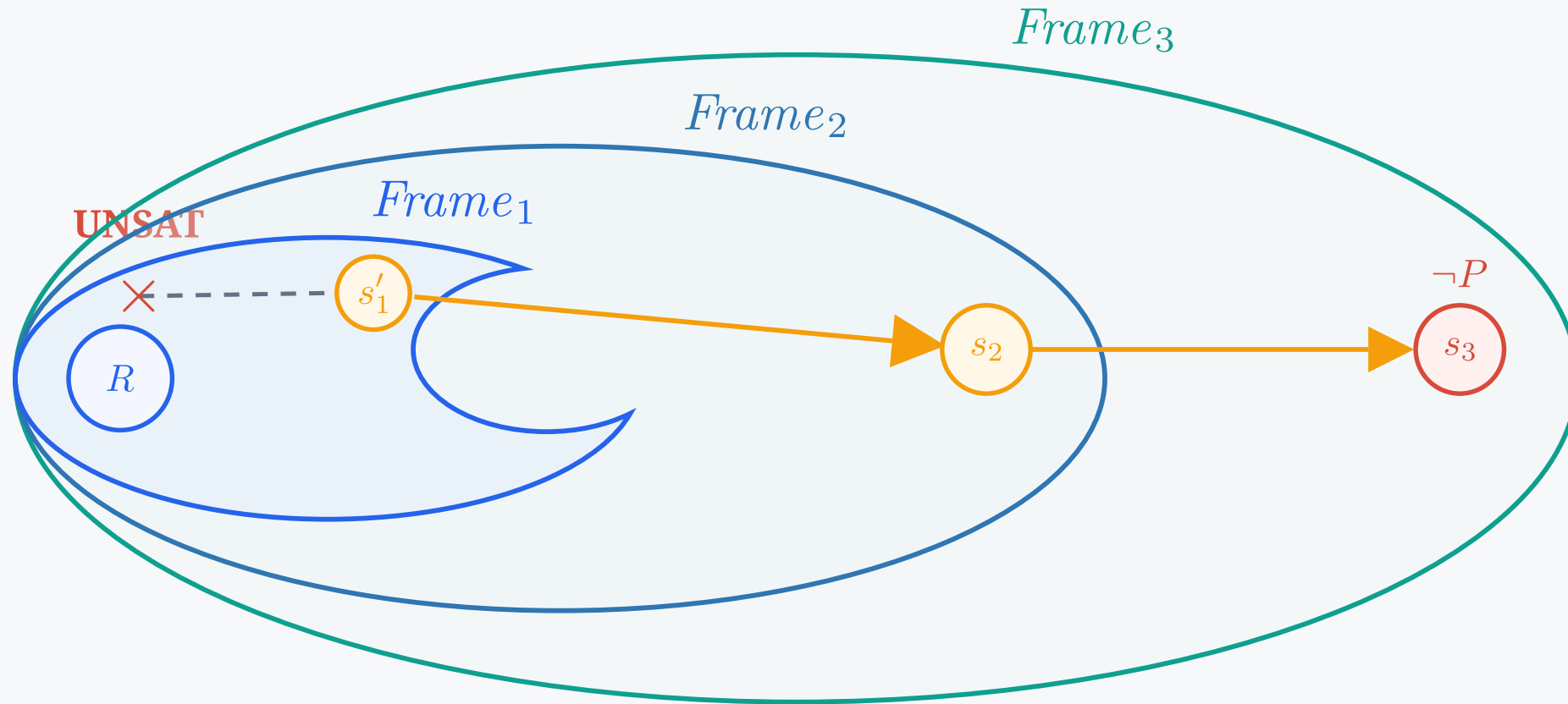


$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$



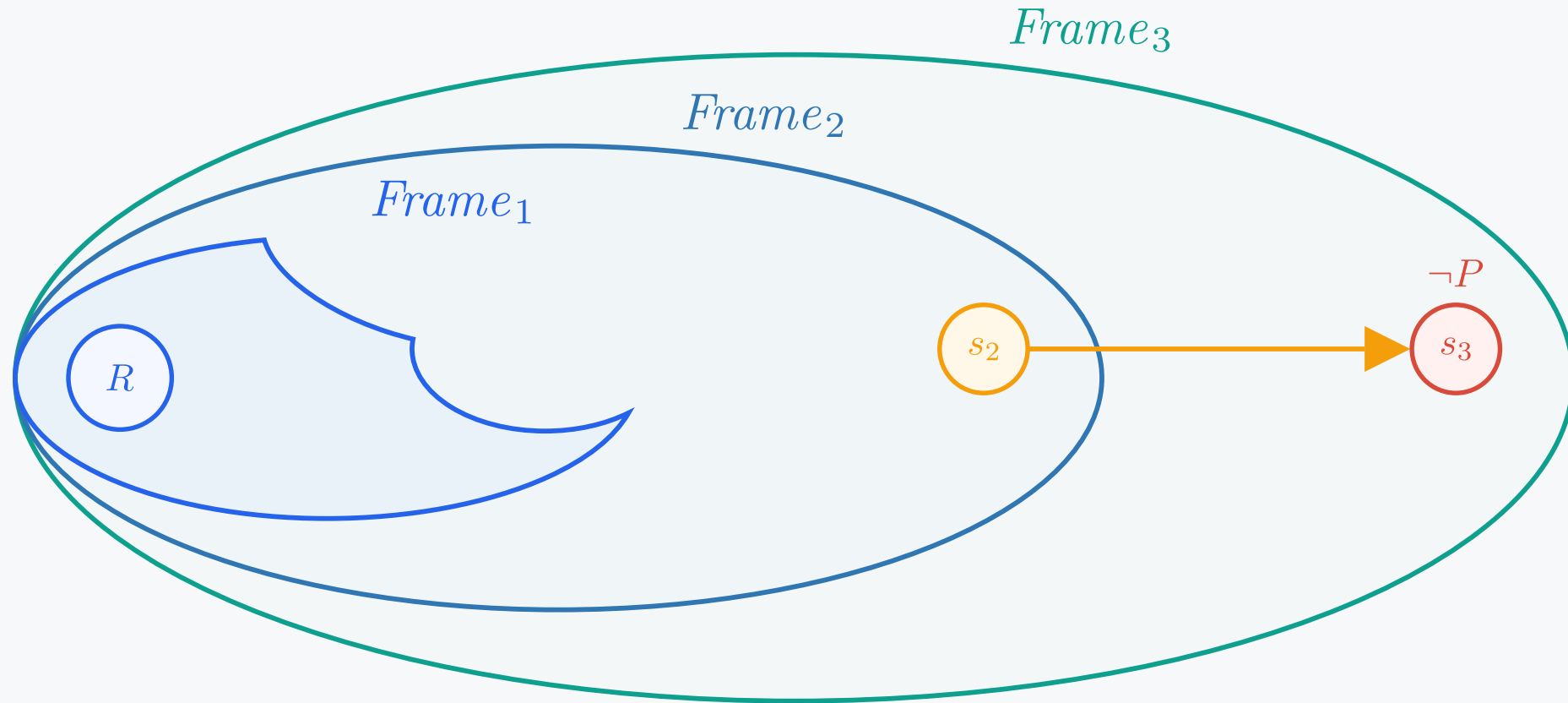
$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$



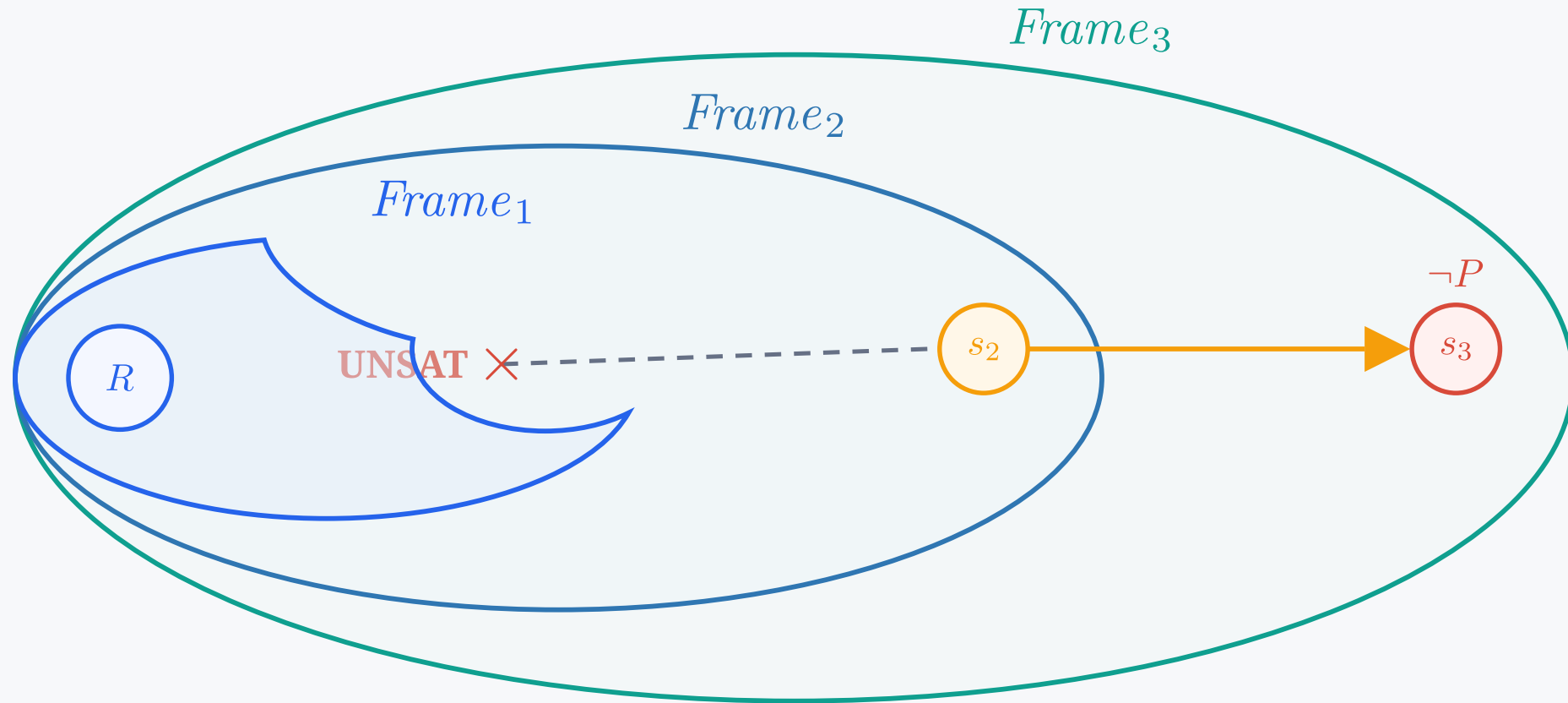


$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

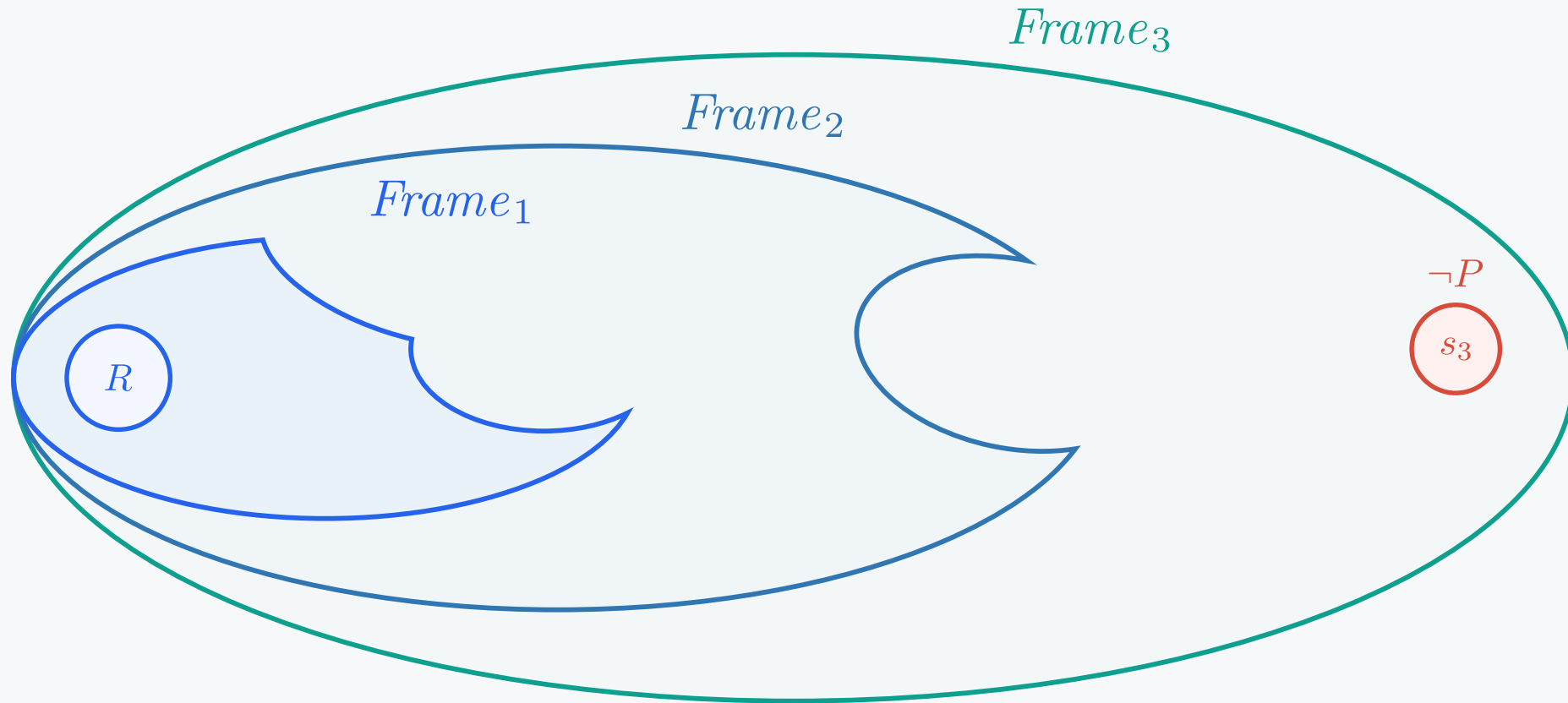
Froleyks & Biere [FMCAD21]



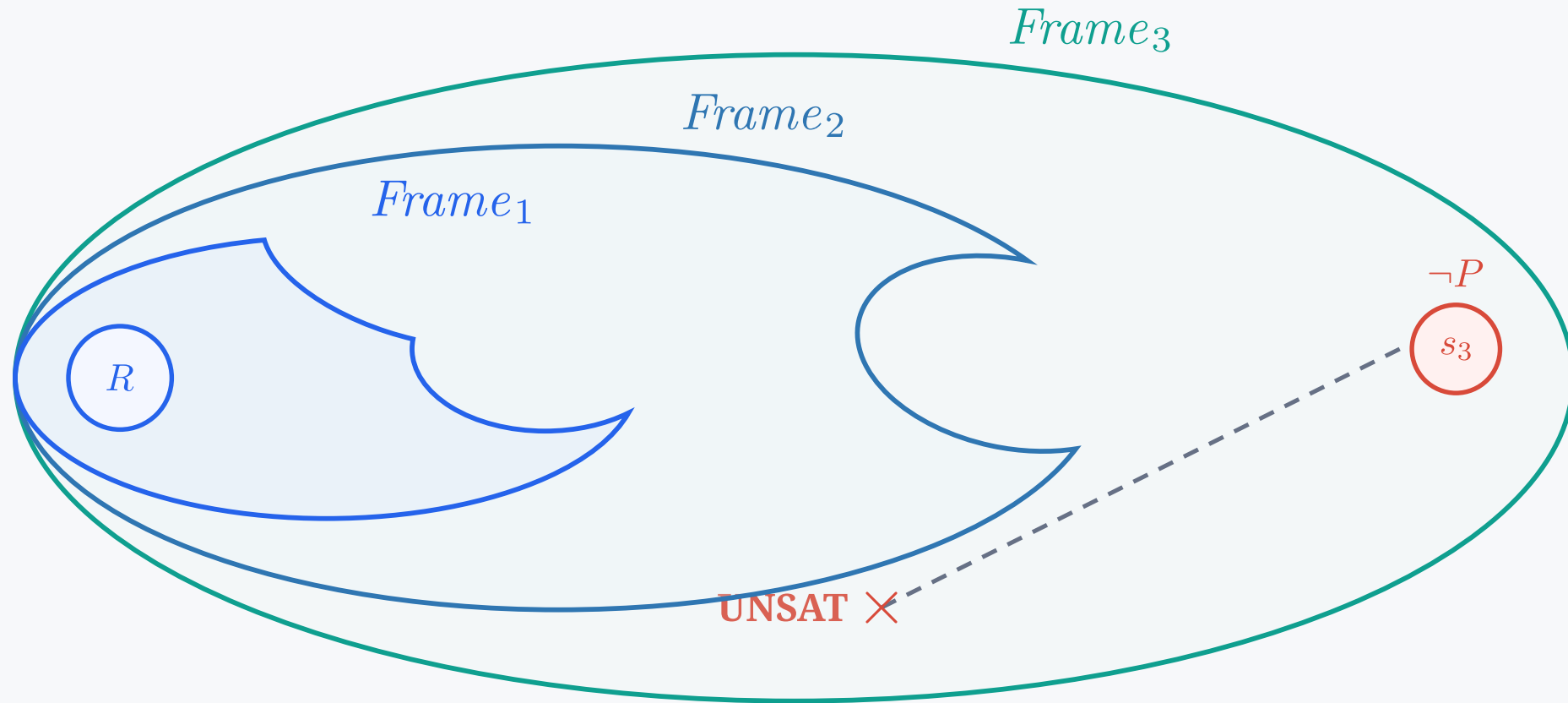
$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$



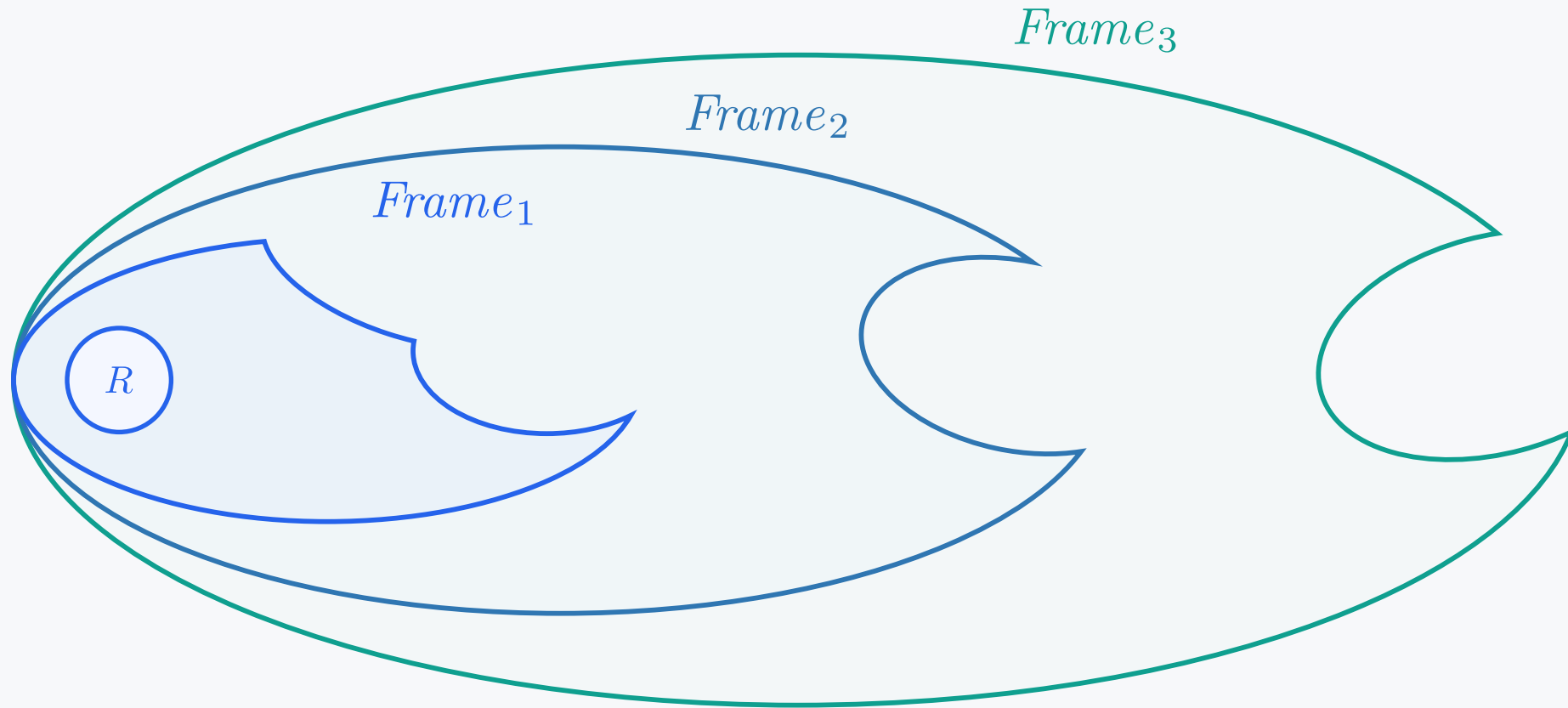
$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$



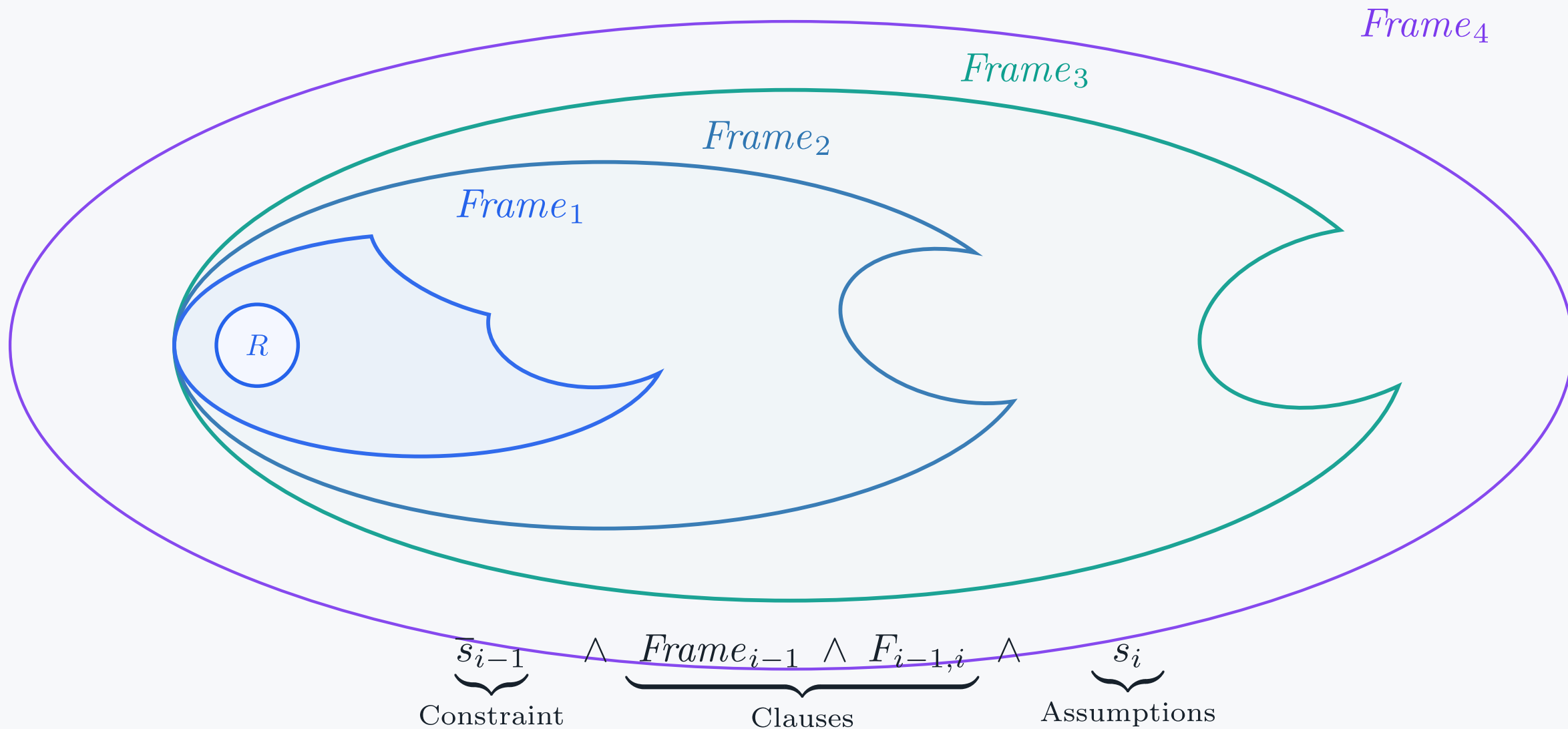
$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

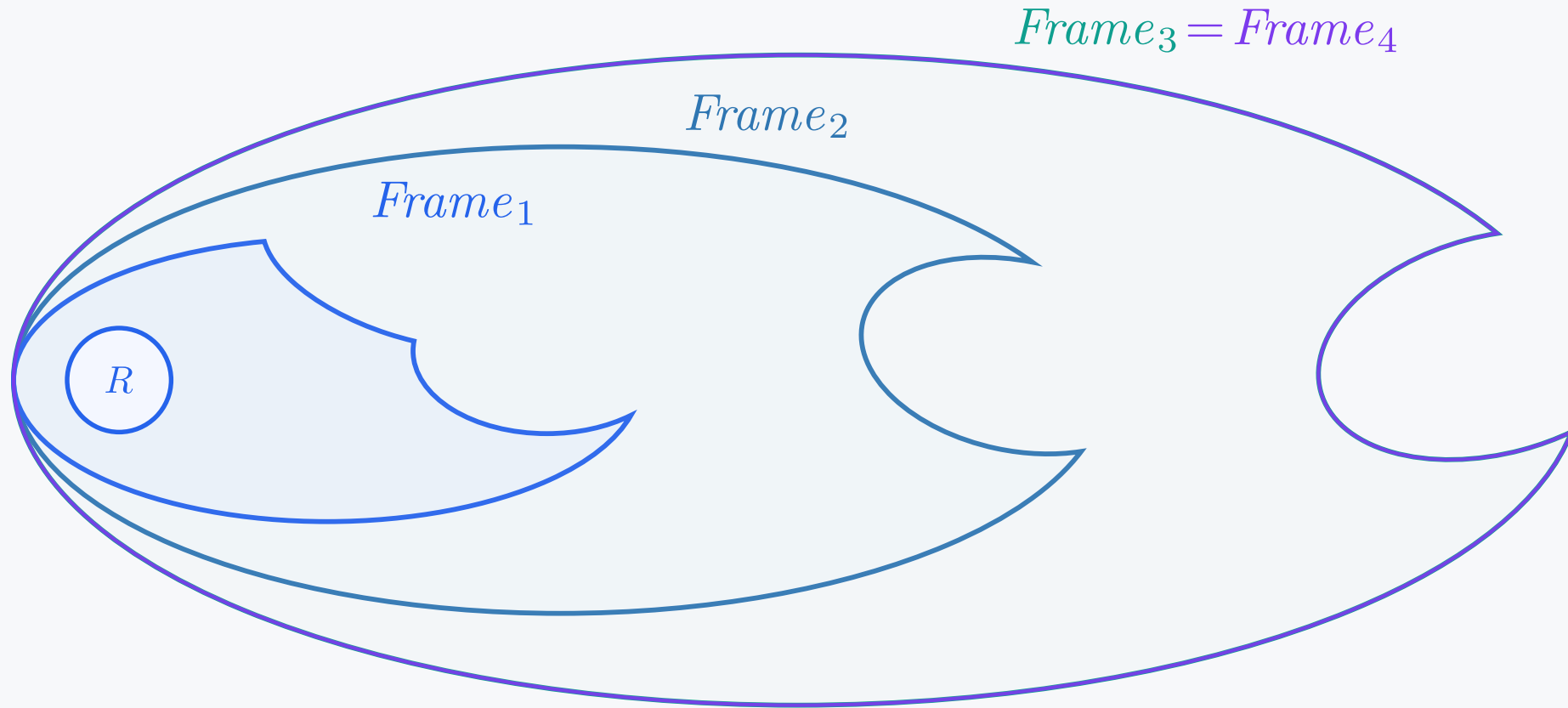


$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

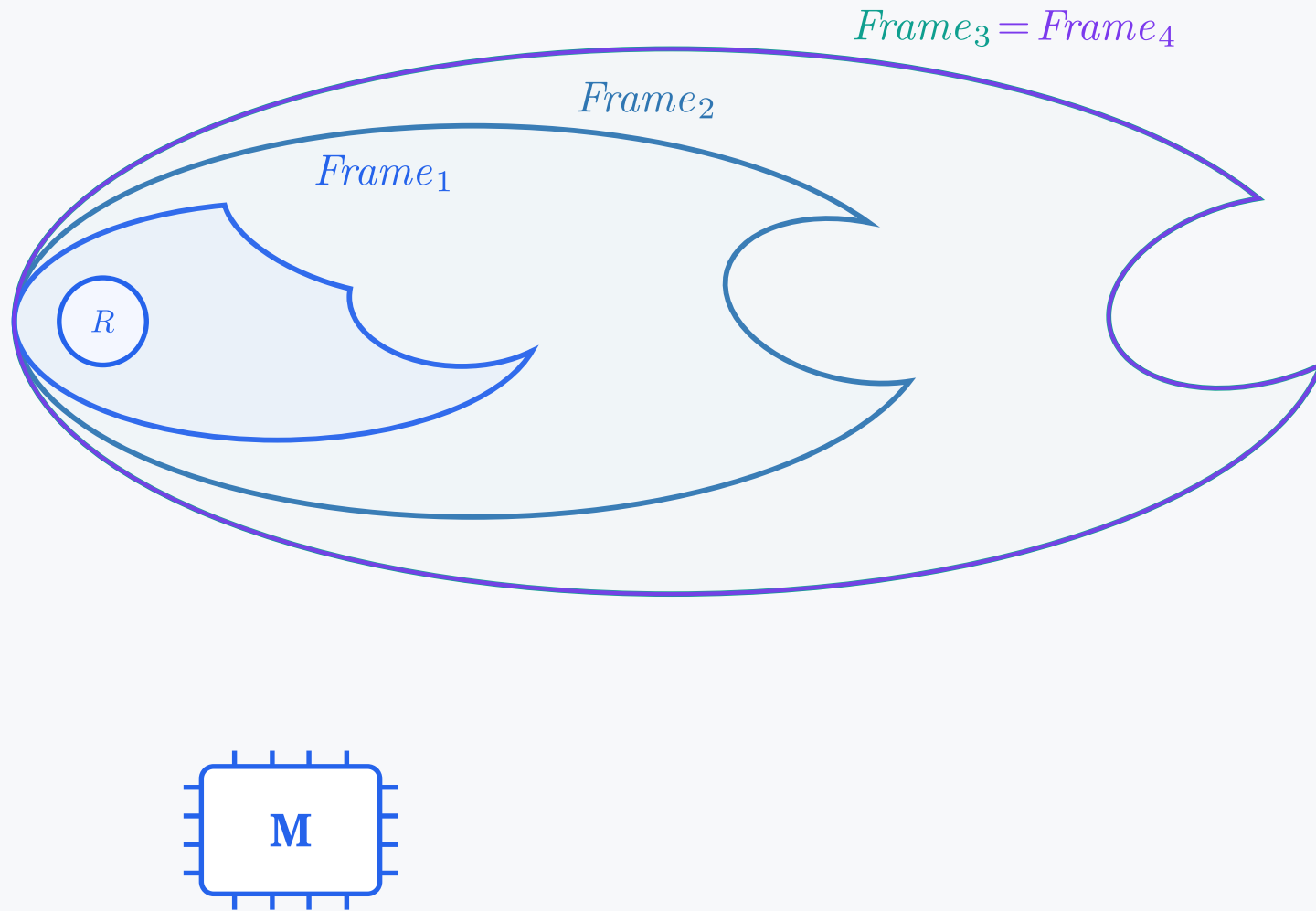


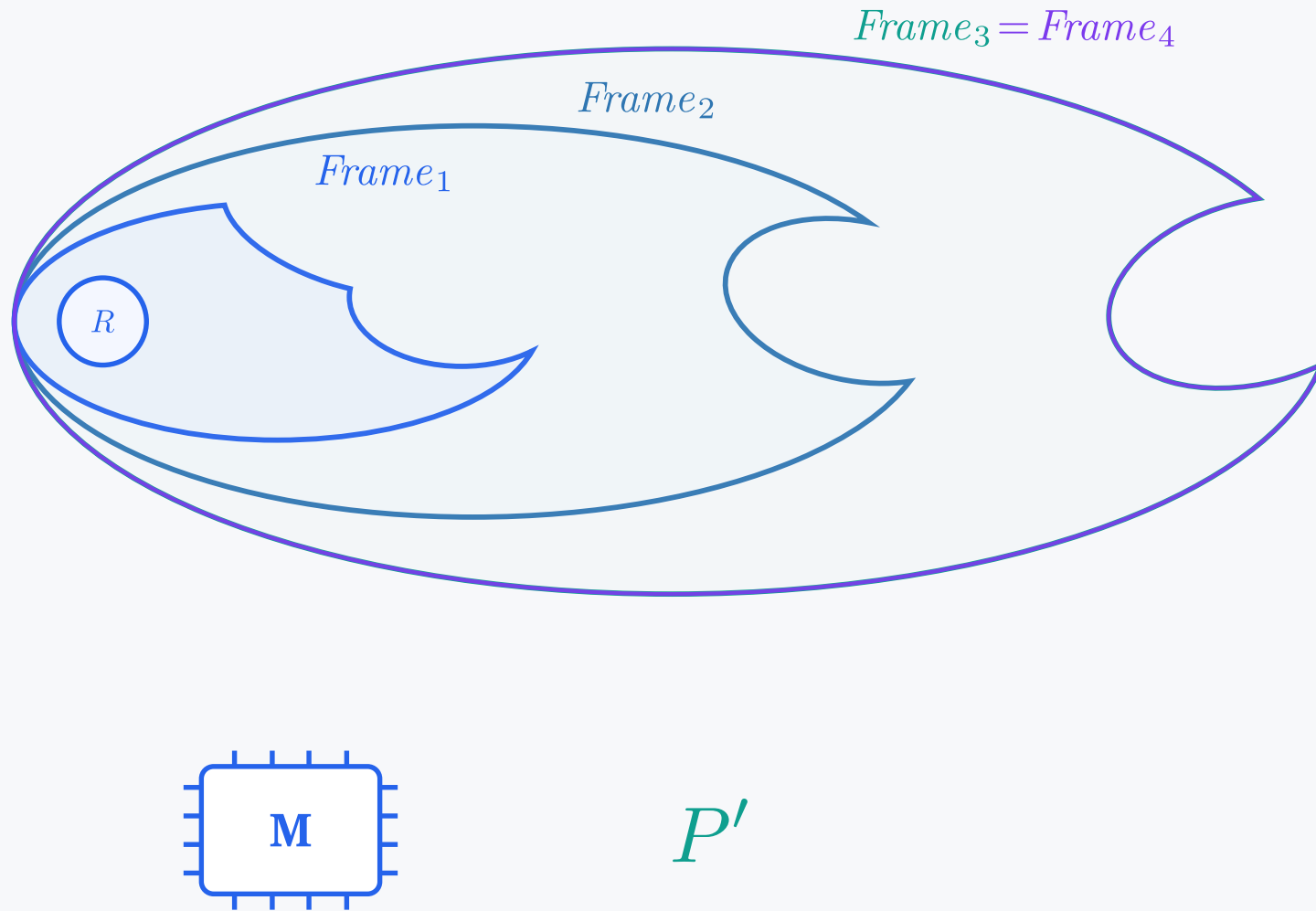
$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$

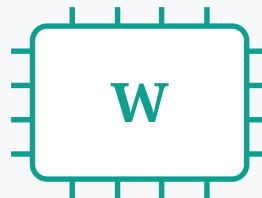
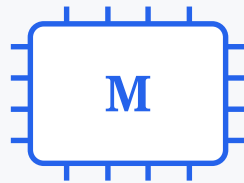
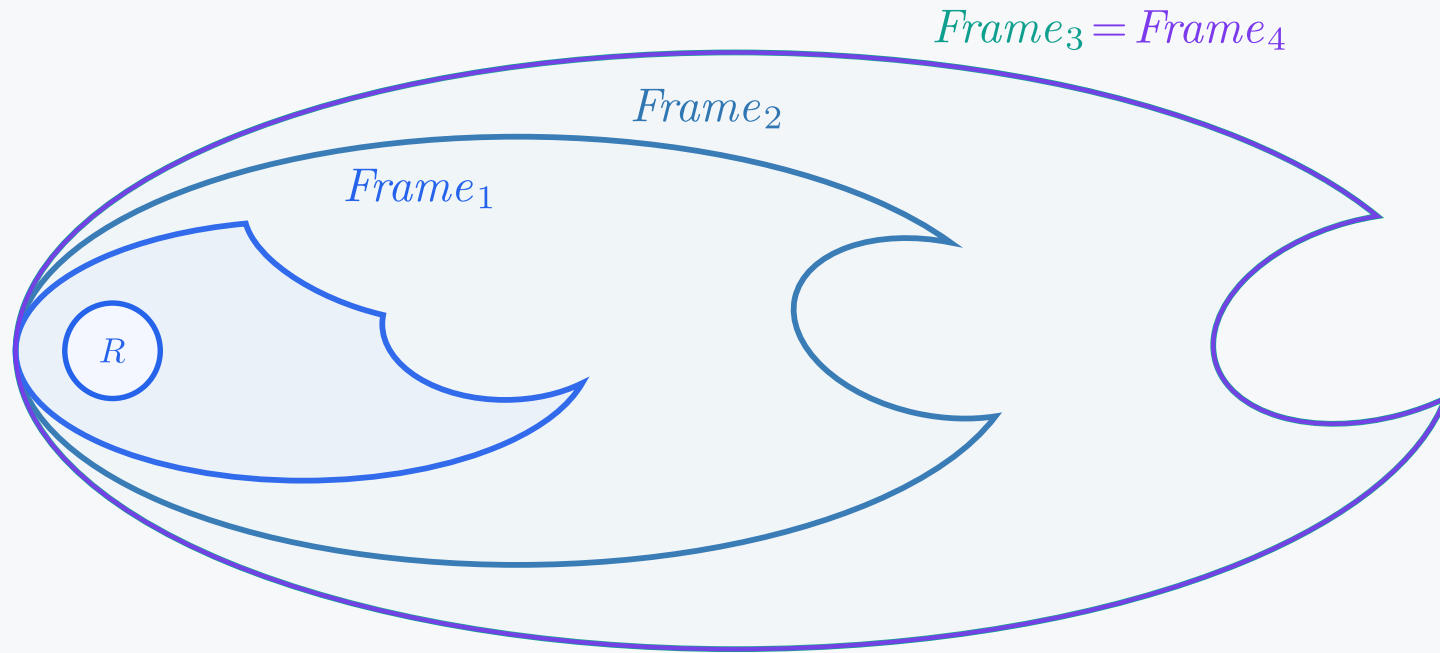




$$\underbrace{\bar{s}_{i-1}}_{\text{Constraint}} \wedge \underbrace{Frame_{i-1} \wedge F_{i-1,i}}_{\text{Clauses}} \wedge \underbrace{s_i}_{\text{Assumptions}}$$







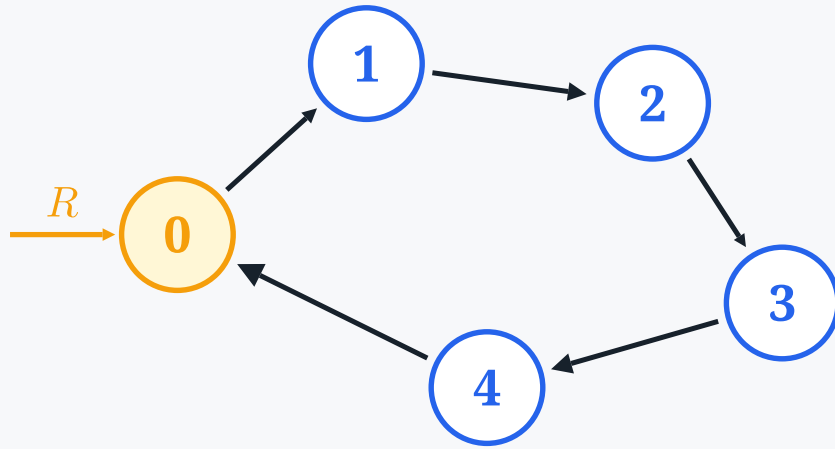
Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

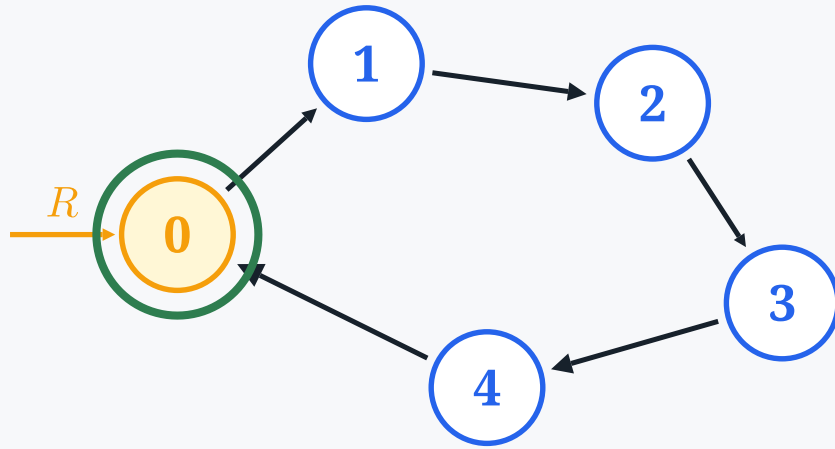
k-Induction

Eén & Sörensson [BMC03]



k-Induction

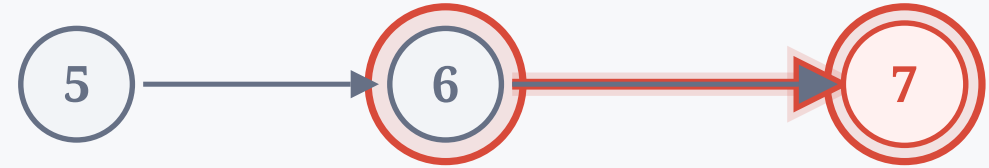
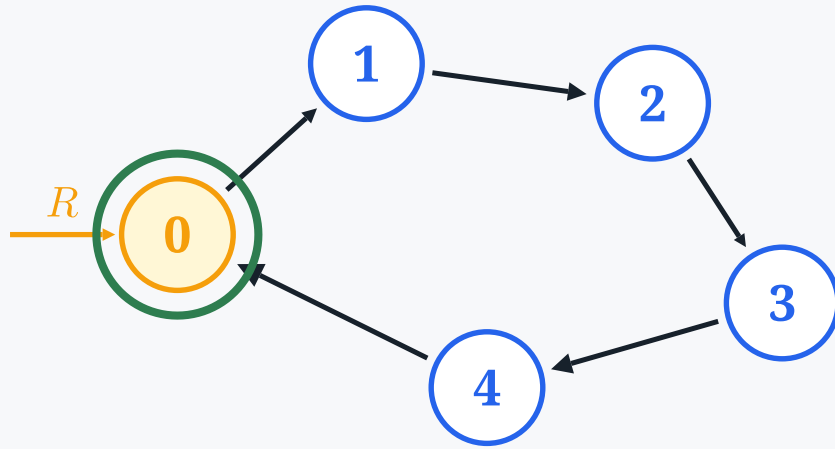
Eén & Sörensson [BMC03]



$R_0 \wedge \neg P_0$
Assumption

k-Induction

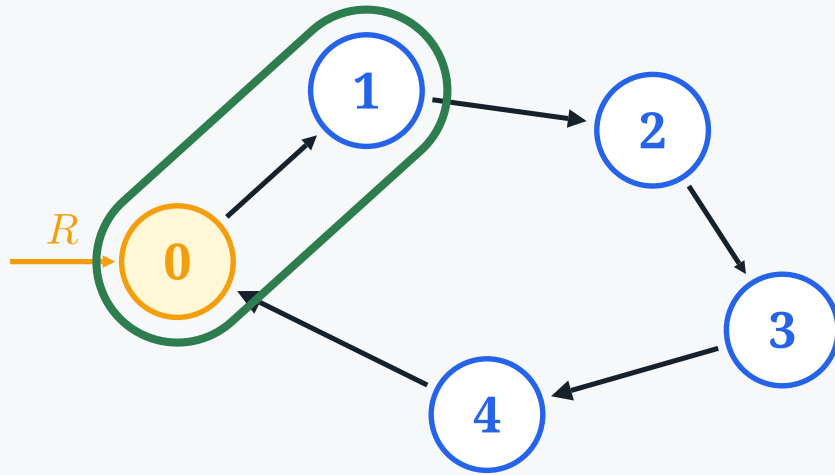
Eén & Sörensson [BMC03]



$$P_0 \wedge F_{0,1} \wedge \neg P_1$$

k-Induction

Eén & Sörensson [BMC03]

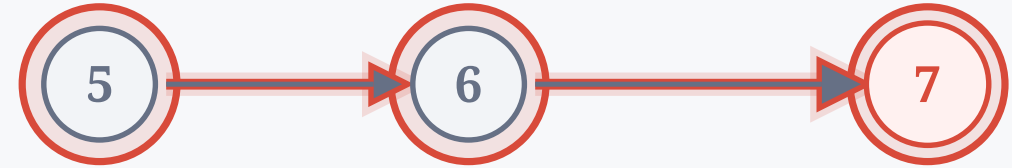
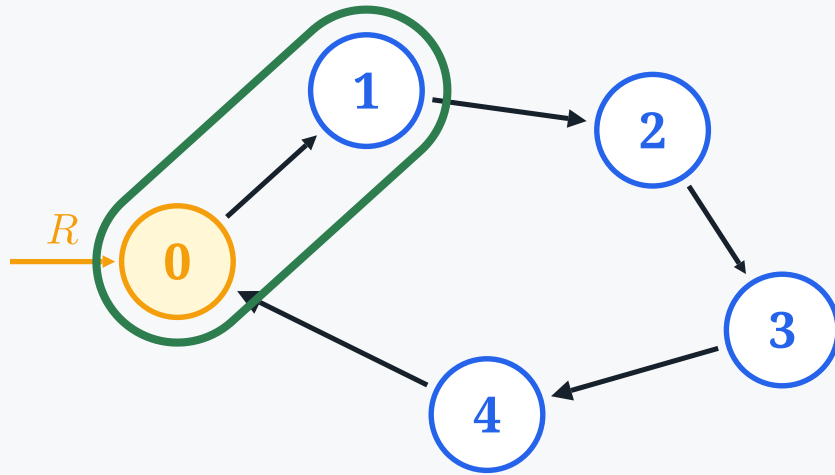


k = 2 **BMC** **UNSAT**

$\underbrace{R_0}_{\text{Assumption}} \wedge P_0 \wedge F_{0,1} \wedge \neg P_1$

k-Induction

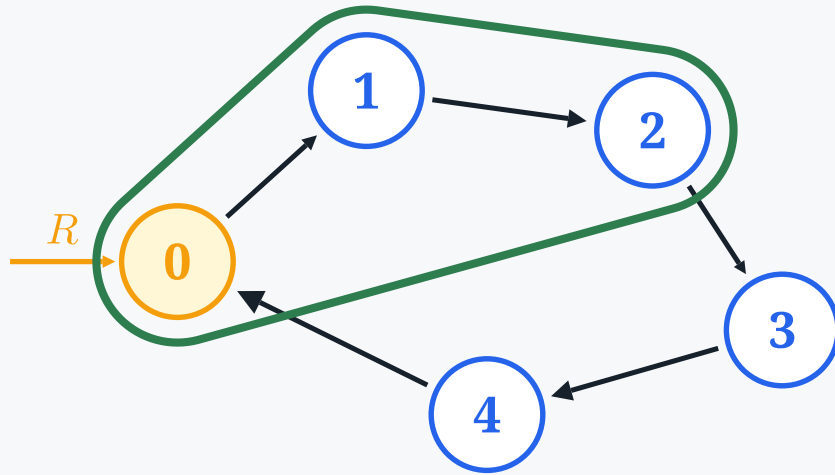
Eén & Sörensson [BMC03]



$$P_0 \wedge F_{0,1} \wedge P_1 \wedge F_{1,2} \wedge \neg P_2$$

k-Induction

Eén & Sörensson [BMC03]



$k = 3$

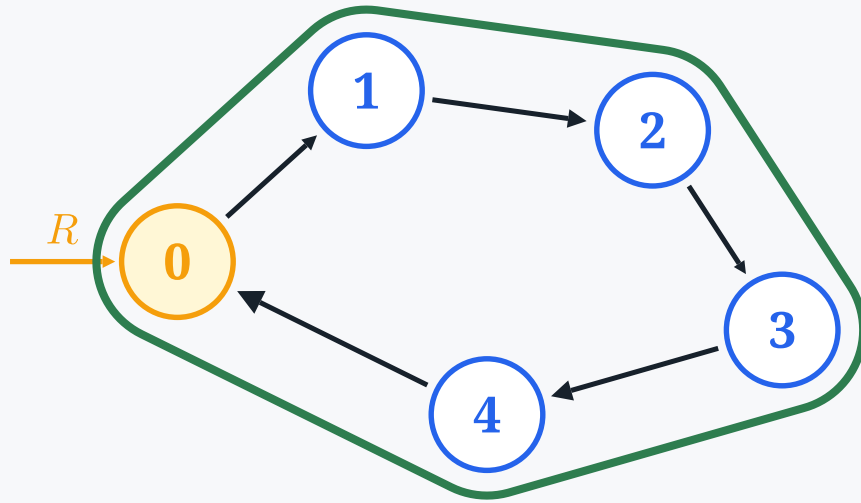
BMC

UNSAT

$$\underbrace{R_0}_{\text{Assumption}} \wedge P_0 \wedge F_{0,1} \wedge P_1 \wedge F_{1,2} \wedge \neg P_2$$

k-Induction

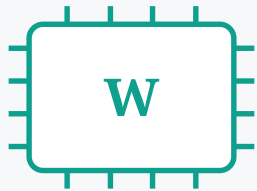
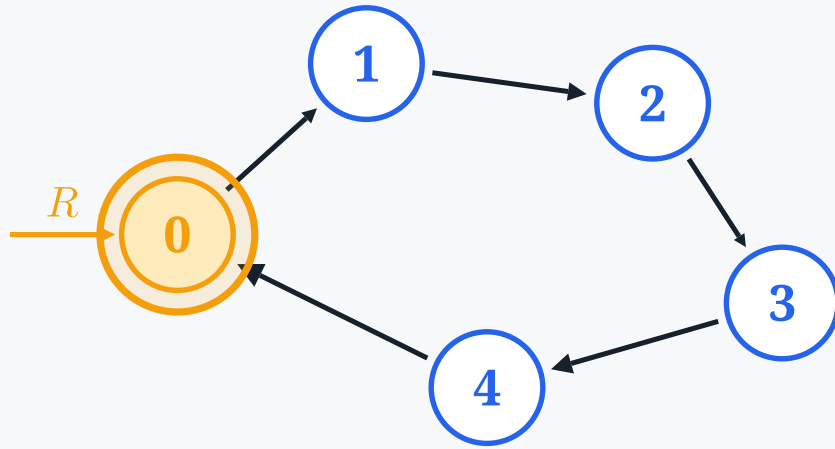
Eén & Sörensson [BMC03]



$$P_0 \wedge F_{0,1} \wedge P_1 \wedge F_{1,2} \wedge P_2 \wedge F_{2,3} \wedge \neg P_3$$

k-Induction

Eén & Sörensson [BMC03]



oldest

simulating state

Reset

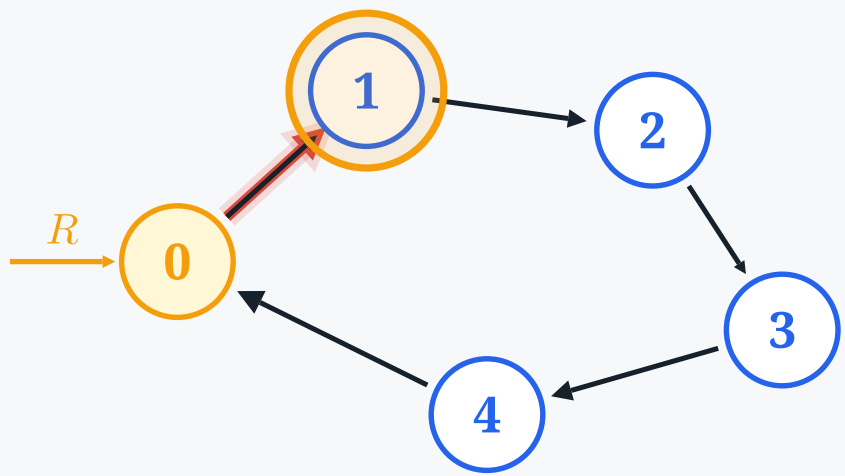
$$\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$$

Transition

$$F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$$

k-Induction

Eén & Sörensson [BMC03]

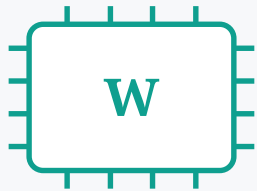
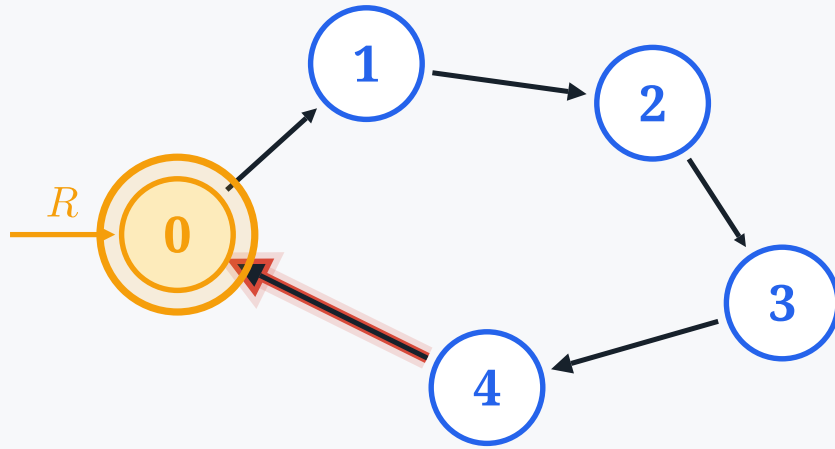


Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

k-Induction

Eén & Sörensson [BMC03]



oldest

simulating state

Reset

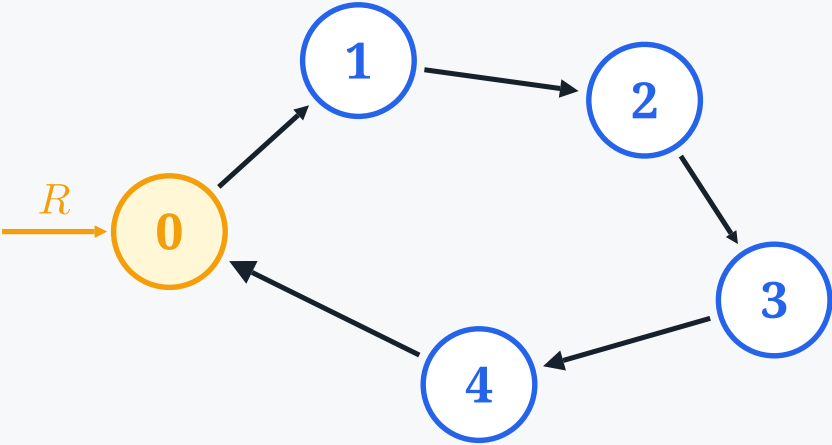
$$\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$$

Transition

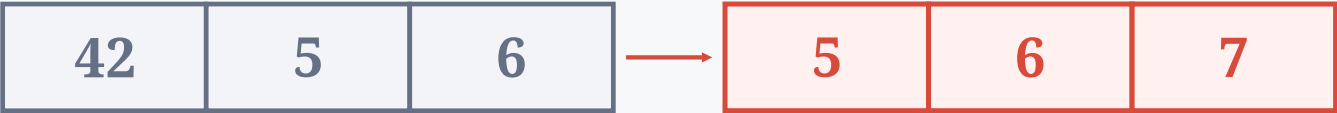
$$F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$$

k-Induction

Eén & Sörensson [BMC03]

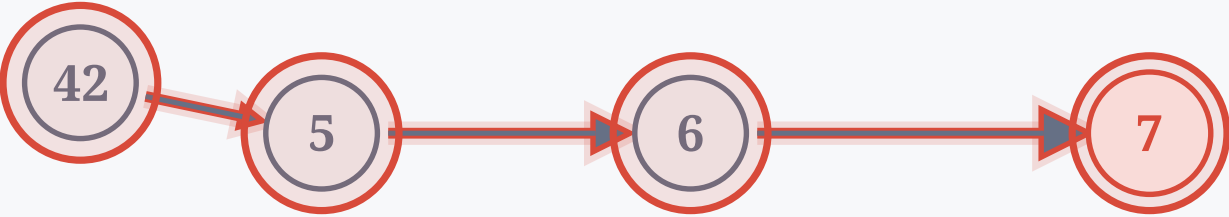
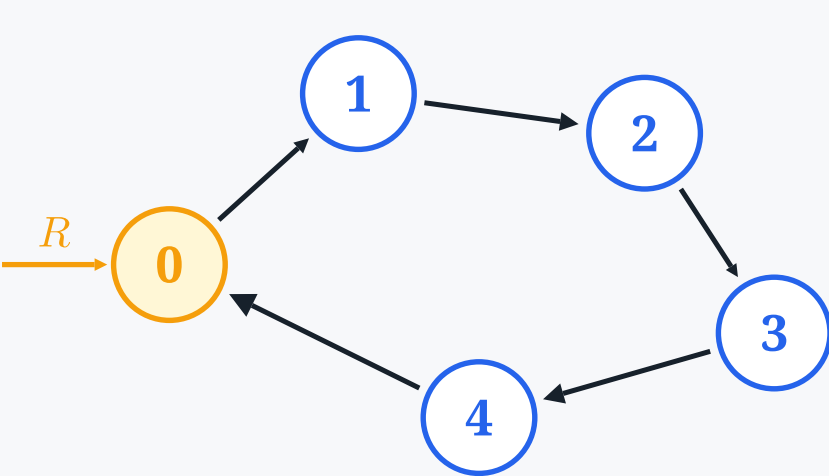


$$\begin{array}{ll} P' = P_0 \wedge P_1 \wedge P_2 & \text{if initialized} \\ \wedge F_{0,1} \wedge F_{1,2} & \\ \wedge R & \text{if not} \end{array}$$



Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$
Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

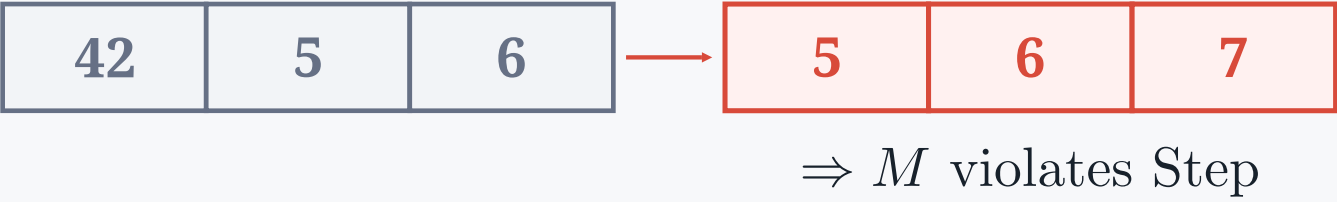
k-Induction



$$P' = P_0 \wedge P_1 \wedge P_2$$
$$\wedge F_{0,1} \wedge F_{1,2}$$
$$\wedge R$$

if initialized

if not

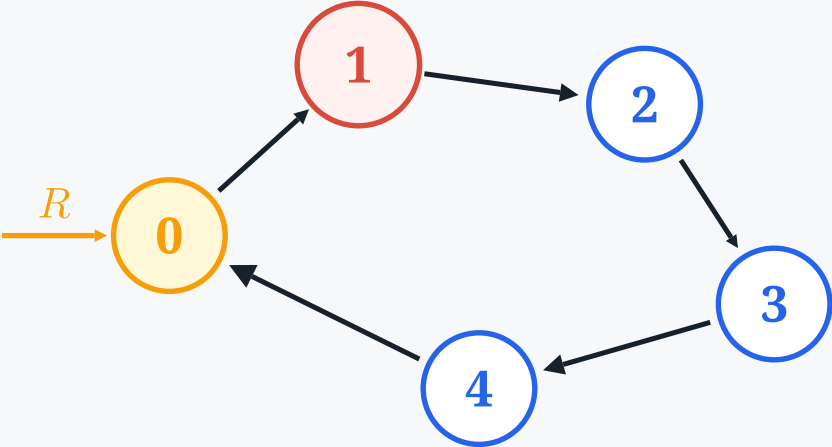


Base

Induction

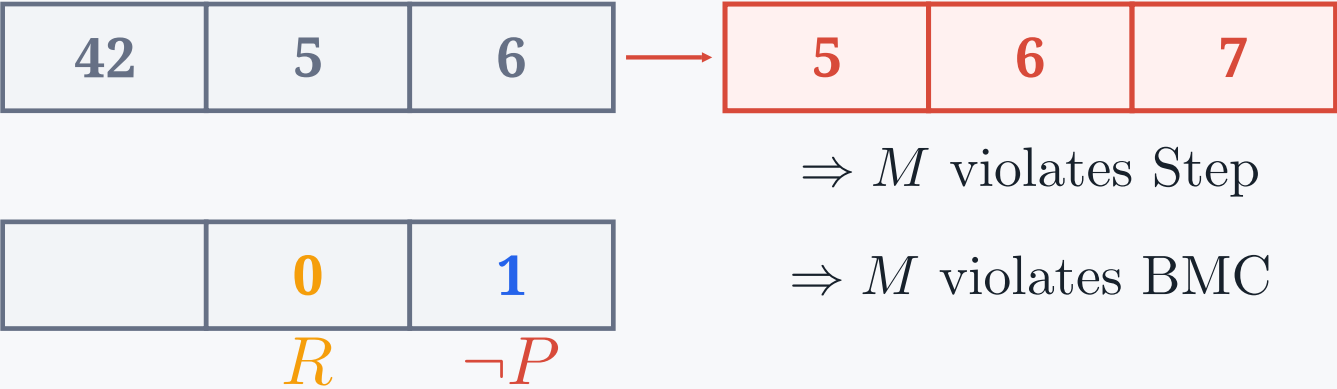
$$\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$$
$$F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$$

k-Induction



$$\begin{array}{ll} P' = P_0 \wedge P_1 \wedge P_2 & \text{if initialized} \\ \wedge F_{0,1} \wedge F_{1,2} & \\ \wedge R & \text{if not} \end{array}$$

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$
Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$



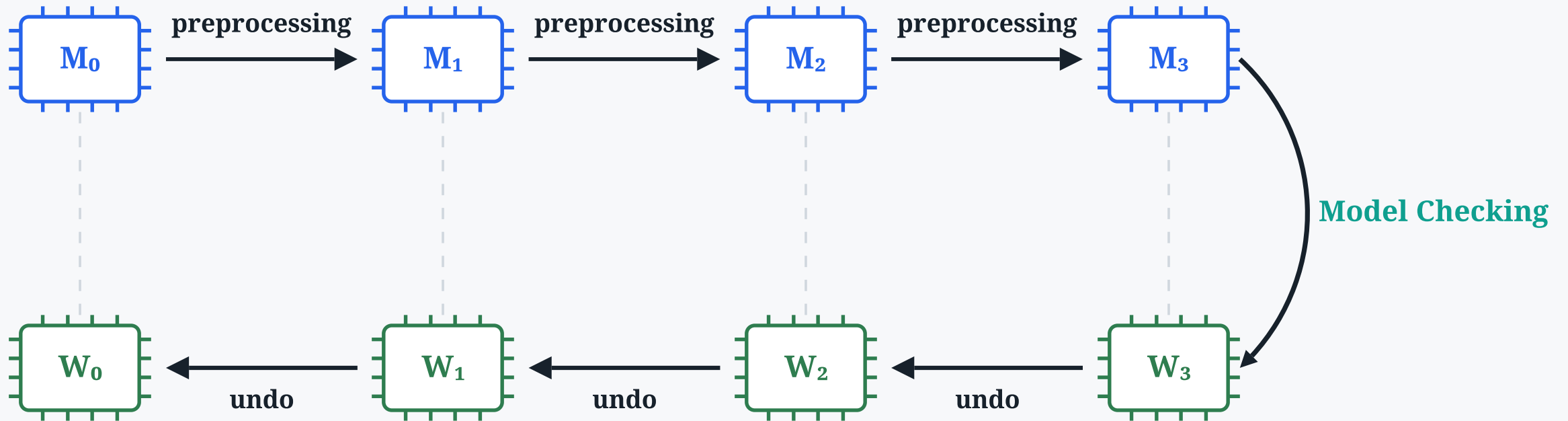
Certifying Preprocessing

Yu et al. [FMCAD23]



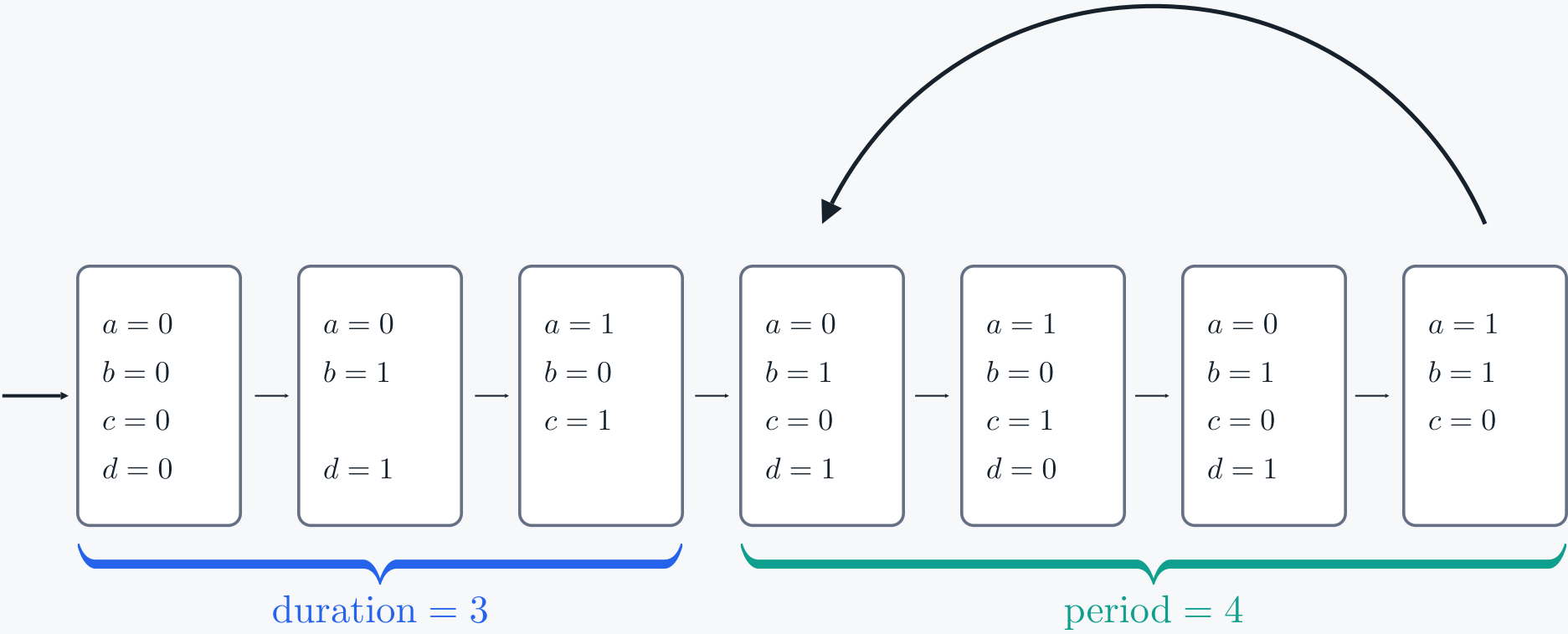
Certifying Preprocessing

Yu et al. [FMCAD23]



Phase Abstraction — Cube Simulation

Froleyks et al. [IJCAR24]



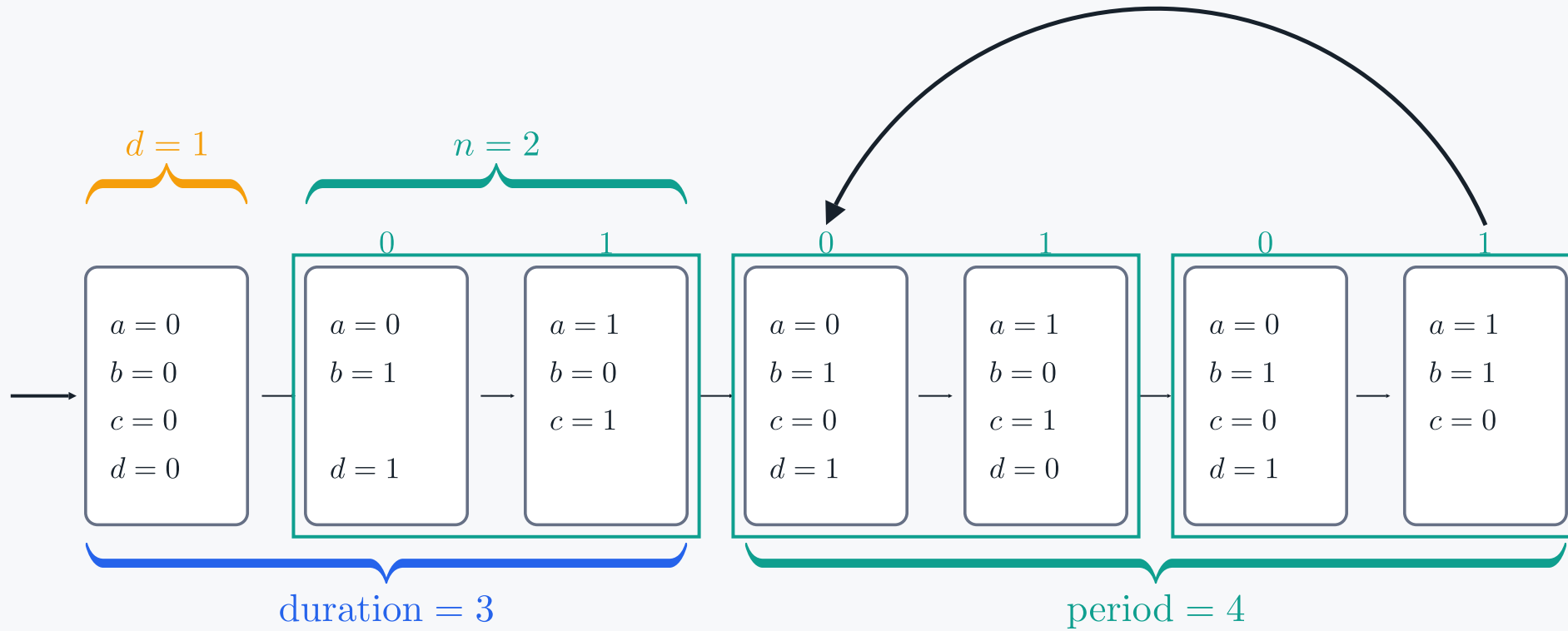
$$R \Rightarrow C_0$$
$$C_{i-1} \wedge F_{i-1,i} \Rightarrow C_i$$

Ternary Simulation
Froleyks et al. [MBMV24]

Backbones
Biere et al. [SAT23]
Froleyks et al. [FMCAD23]

Phase Abstraction — Cube Simulation

Froleyks et al. [IJCAR24]



$$\begin{aligned}\lambda_a &= (1, [0, 1]) \\ \lambda_b &= (1, [1, b]) \\ \lambda_c &= (1, [c, \neg b]) \\ \lambda_d &= (1, [1, d])\end{aligned}$$

$$R \Rightarrow C_0$$

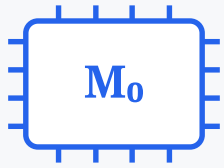
$$C_{i-1} \wedge F_{i-1,i} \Rightarrow C_i$$

Ternary Simulation
Froleyks et al. [MBMV24]

Backbones
Biere et al. [SAT23]
Froleyks et al. [FMCAD23]

Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



$$\lambda_a = (1, [0, 1])$$

$$\lambda_b = (1, [1, b])$$

$$\lambda_c = (1, [c, \neg b])$$

$$\lambda_d = (1, [1, d])$$



Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



$$\lambda_a = (1, [0, 1])$$

$$\lambda_b = (1, [1, b])$$

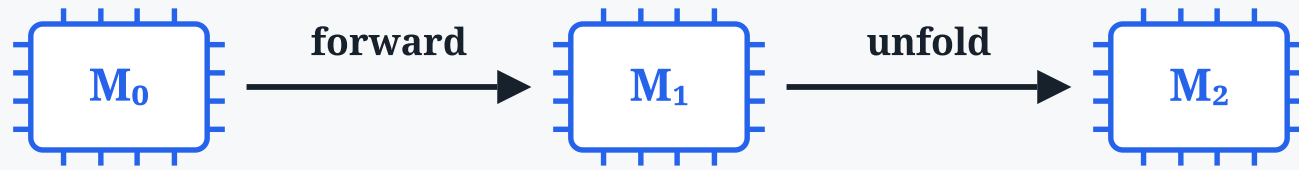
$$\lambda_c = (1, [c, \neg b])$$

$$\lambda_d = (1, [1, d])$$



Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



$$\lambda_a = (1, [0, 1])$$

$$\lambda_b = (1, [1, b])$$

$$\lambda_c = (1, [c, \neg b])$$

$$\lambda_d = (1, [1, d])$$



Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



$$\lambda_a = (1, [0, 1])$$

$$\lambda_b = (1, [1, b])$$

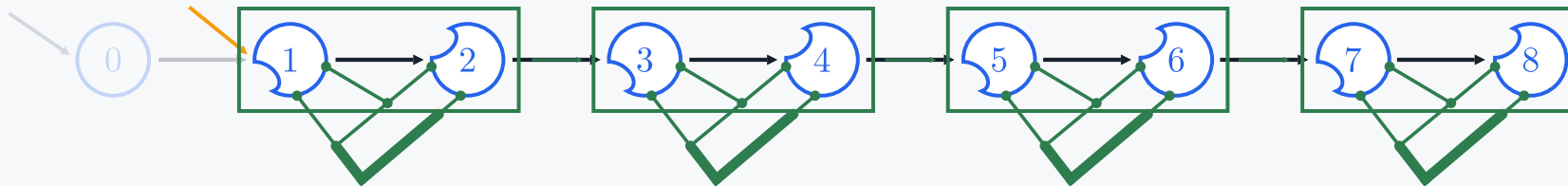
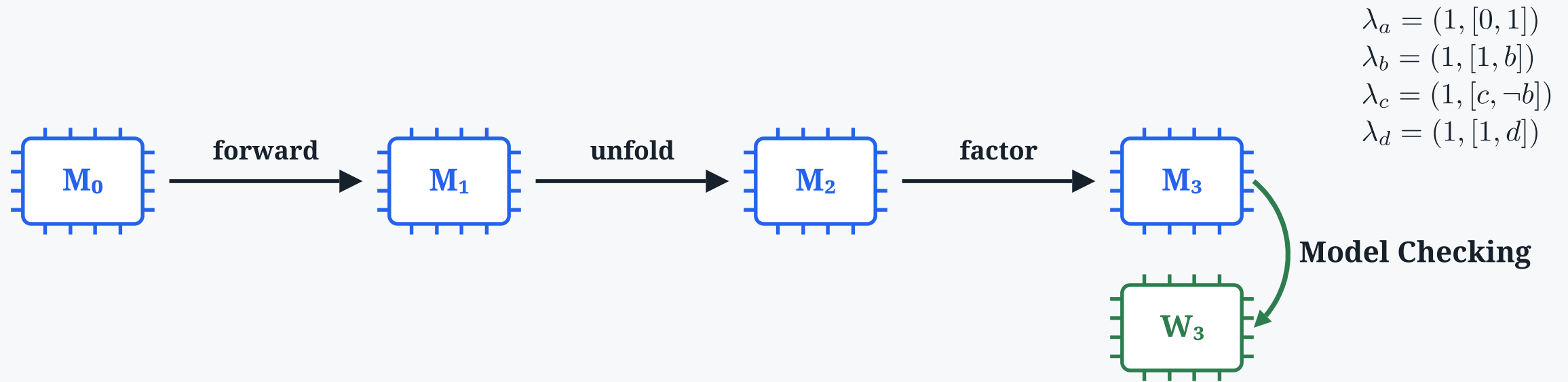
$$\lambda_c = (1, [c, \neg b])$$

$$\lambda_d = (1, [1, d])$$



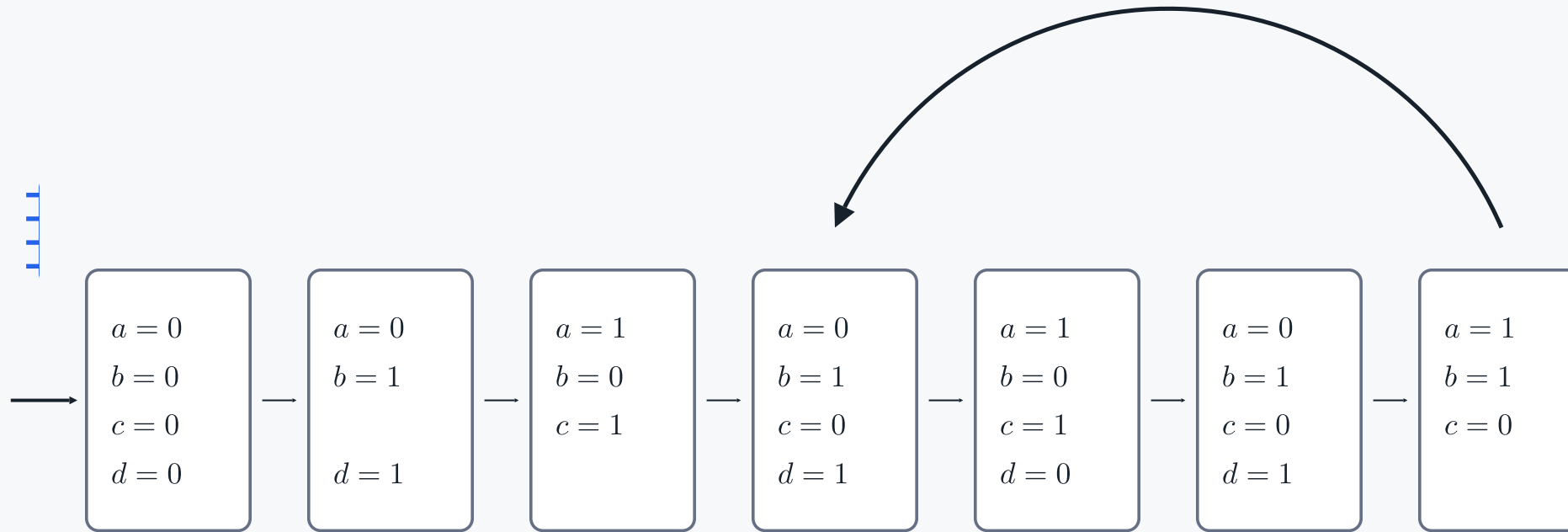
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



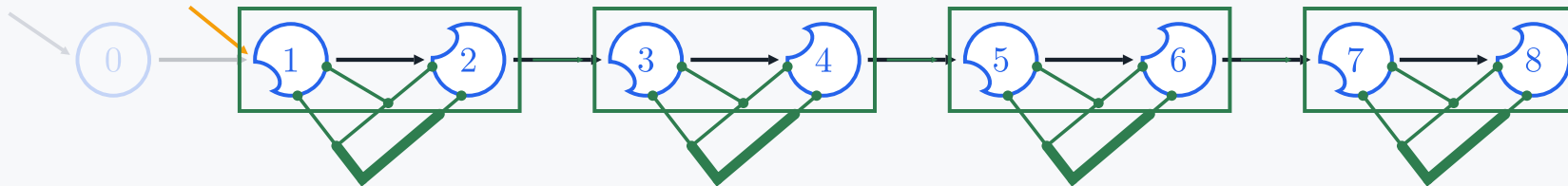
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



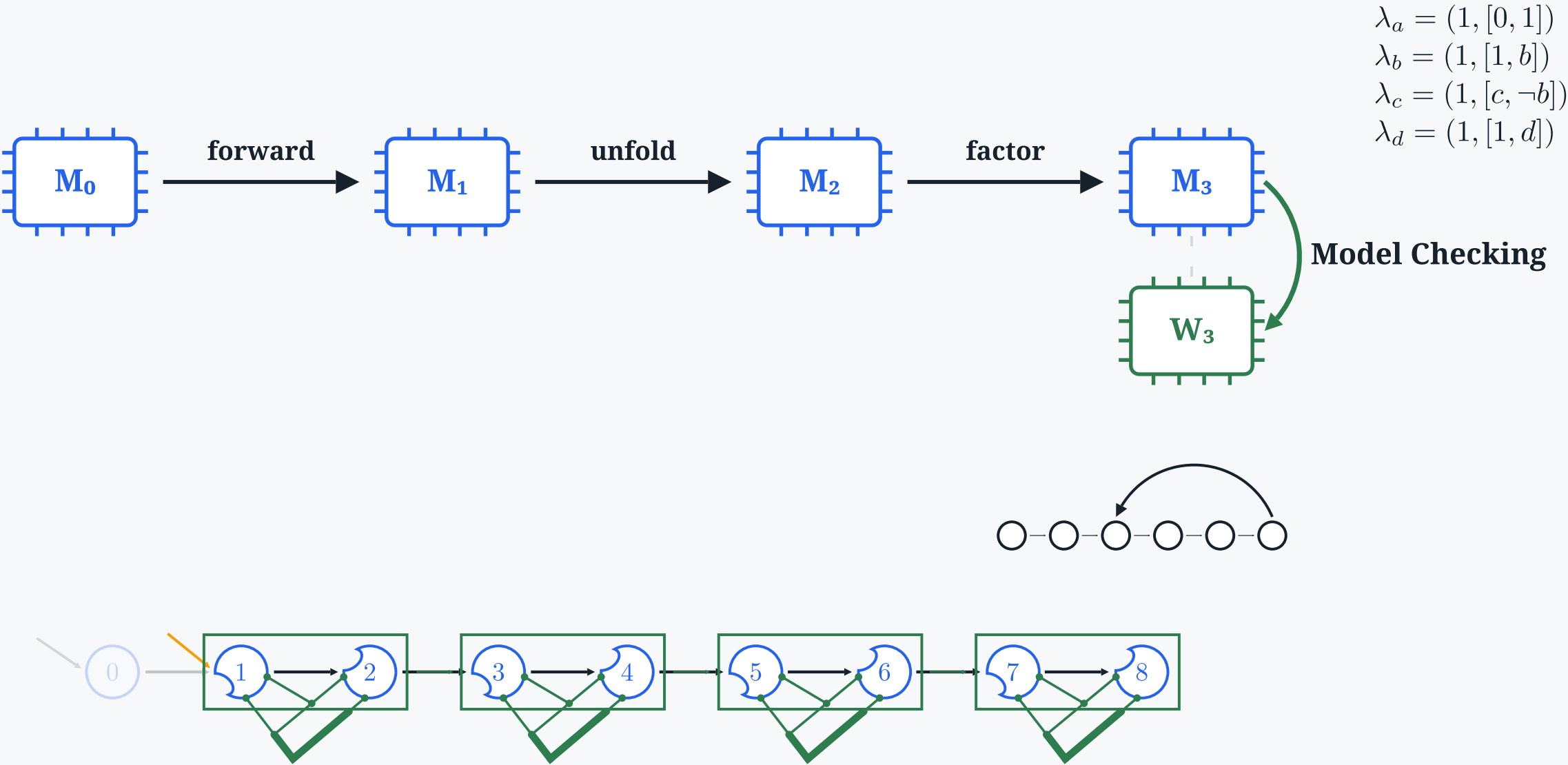
$$\begin{aligned}\lambda_a &= (1, [0, 1]) \\ \lambda_b &= (1, [1, b]) \\ \lambda_c &= (1, [c, \neg b]) \\ \lambda_d &= (1, [1, d])\end{aligned}$$

Model Checking



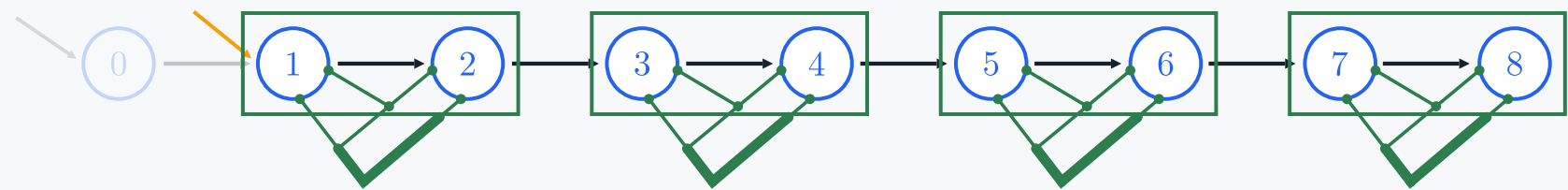
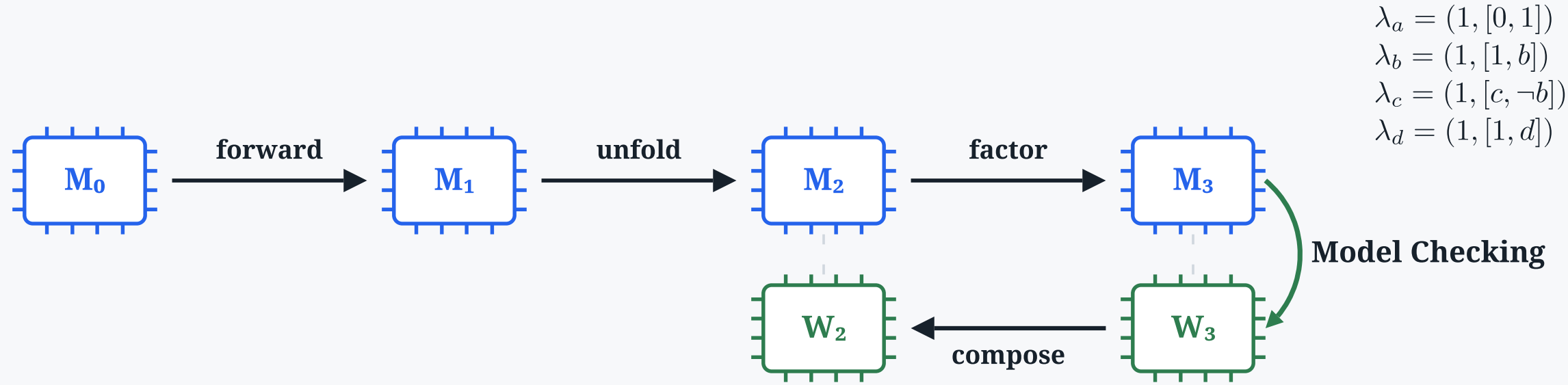
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



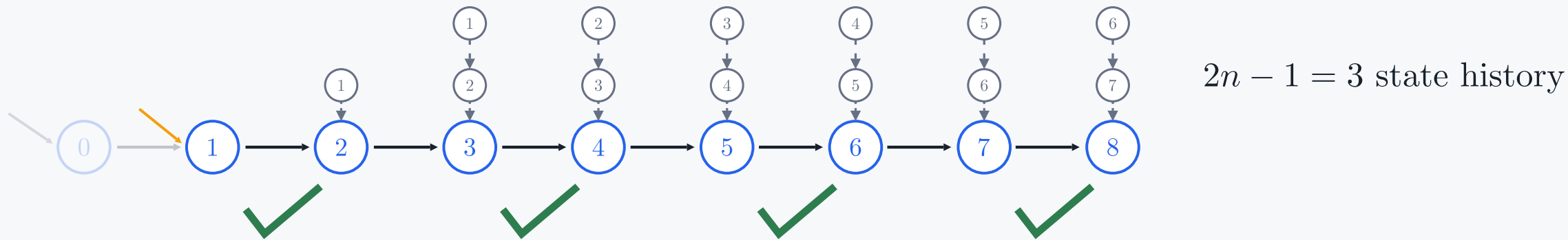
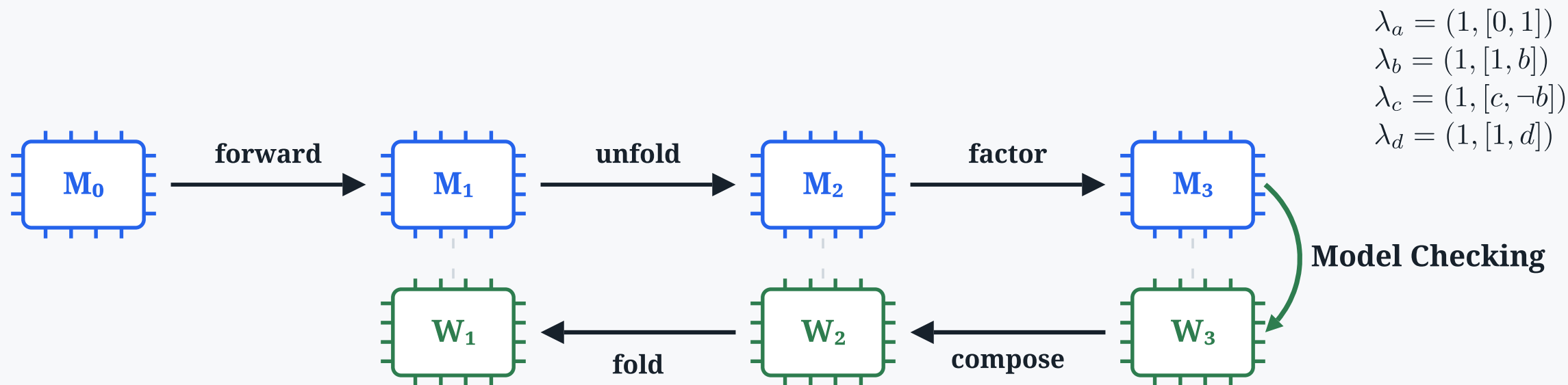
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



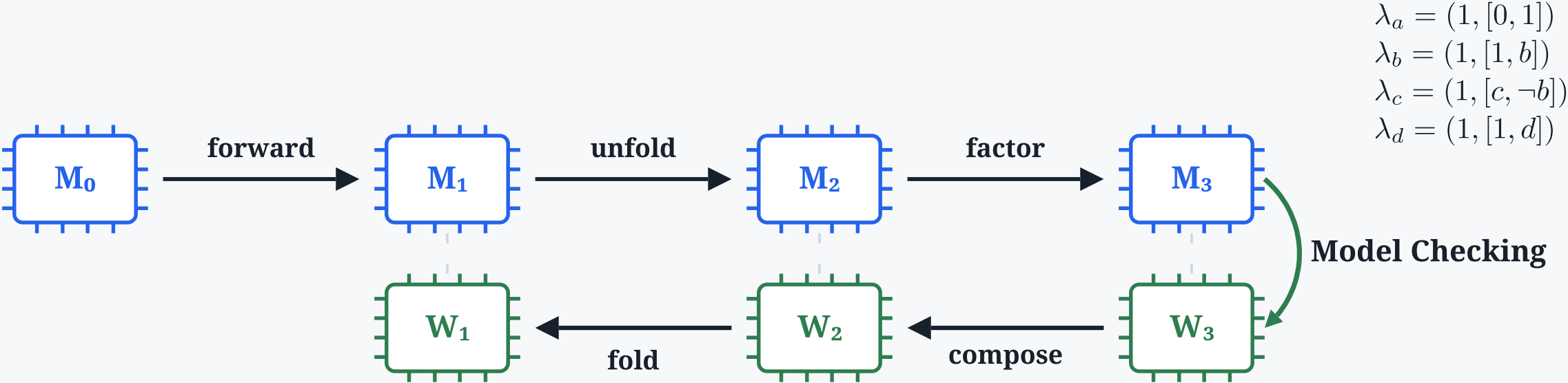
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]

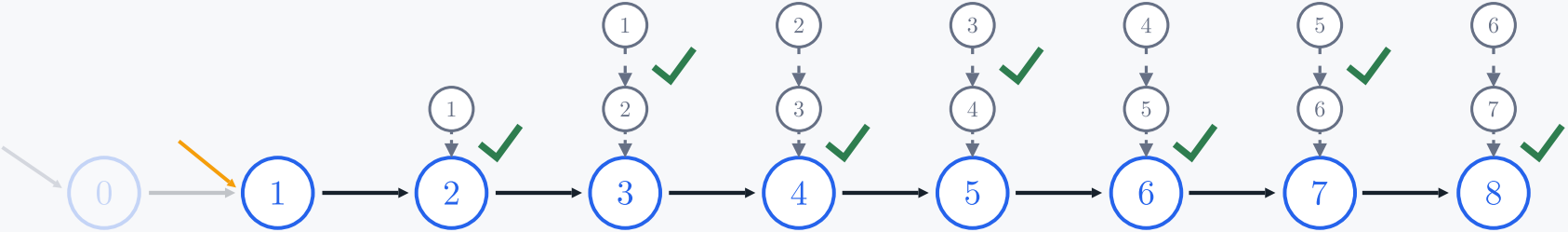


Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



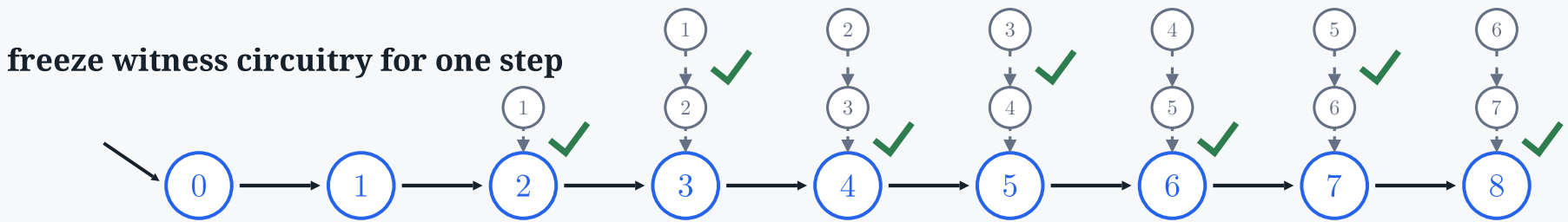
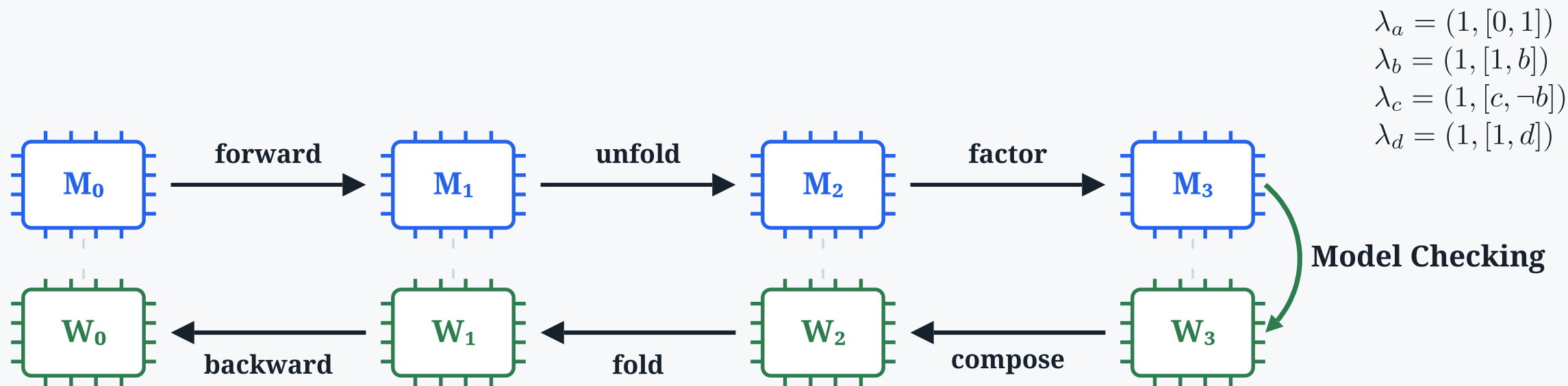
$$\begin{aligned}\lambda_a &= (1, [0, 1]) \\ \lambda_b &= (1, [1, b]) \\ \lambda_c &= (1, [c, \neg b]) \\ \lambda_d &= (1, [1, d])\end{aligned}$$



$2n - 1 = 3$ state history

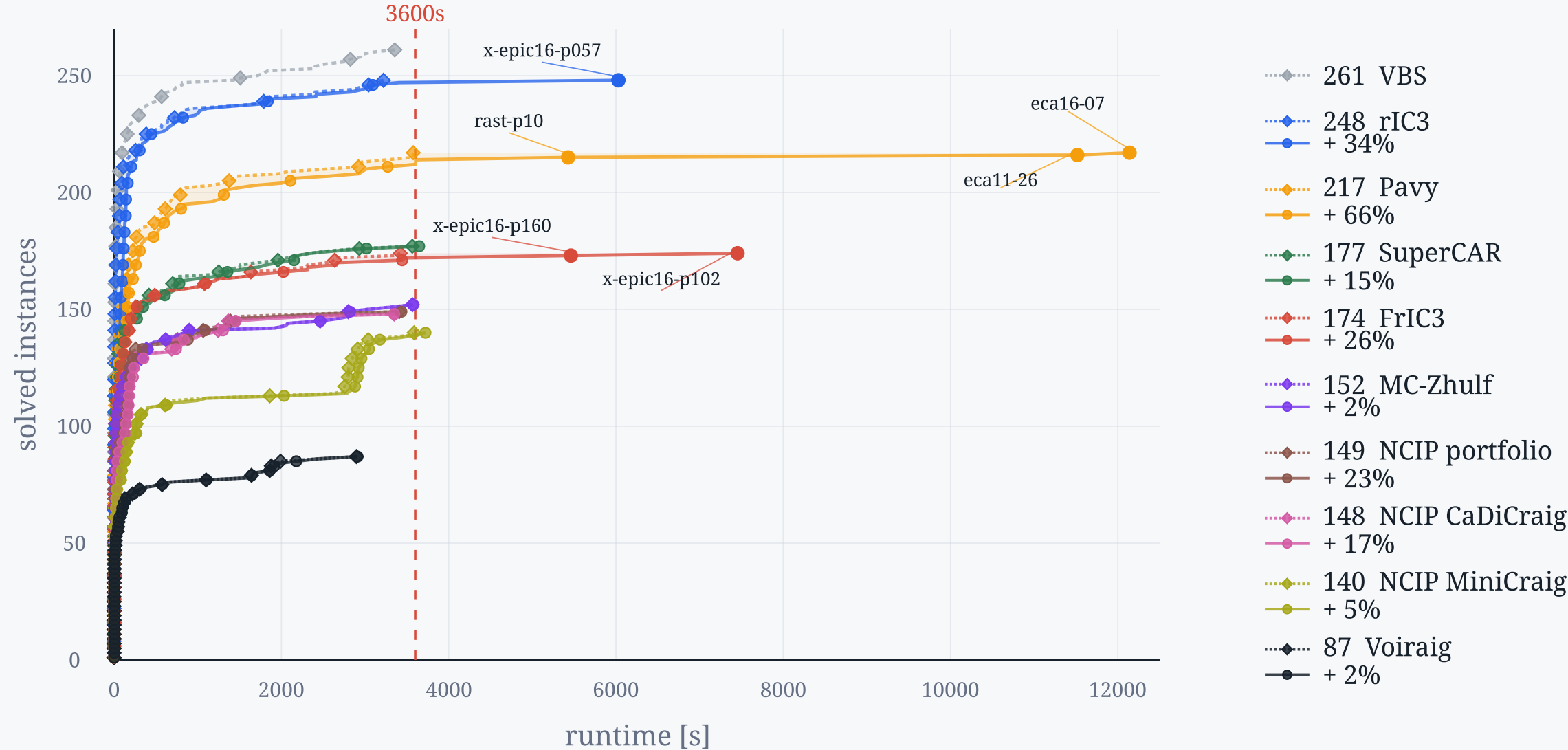
Phase Abstraction — Transform

Froleyks et al. [IJCAR24]



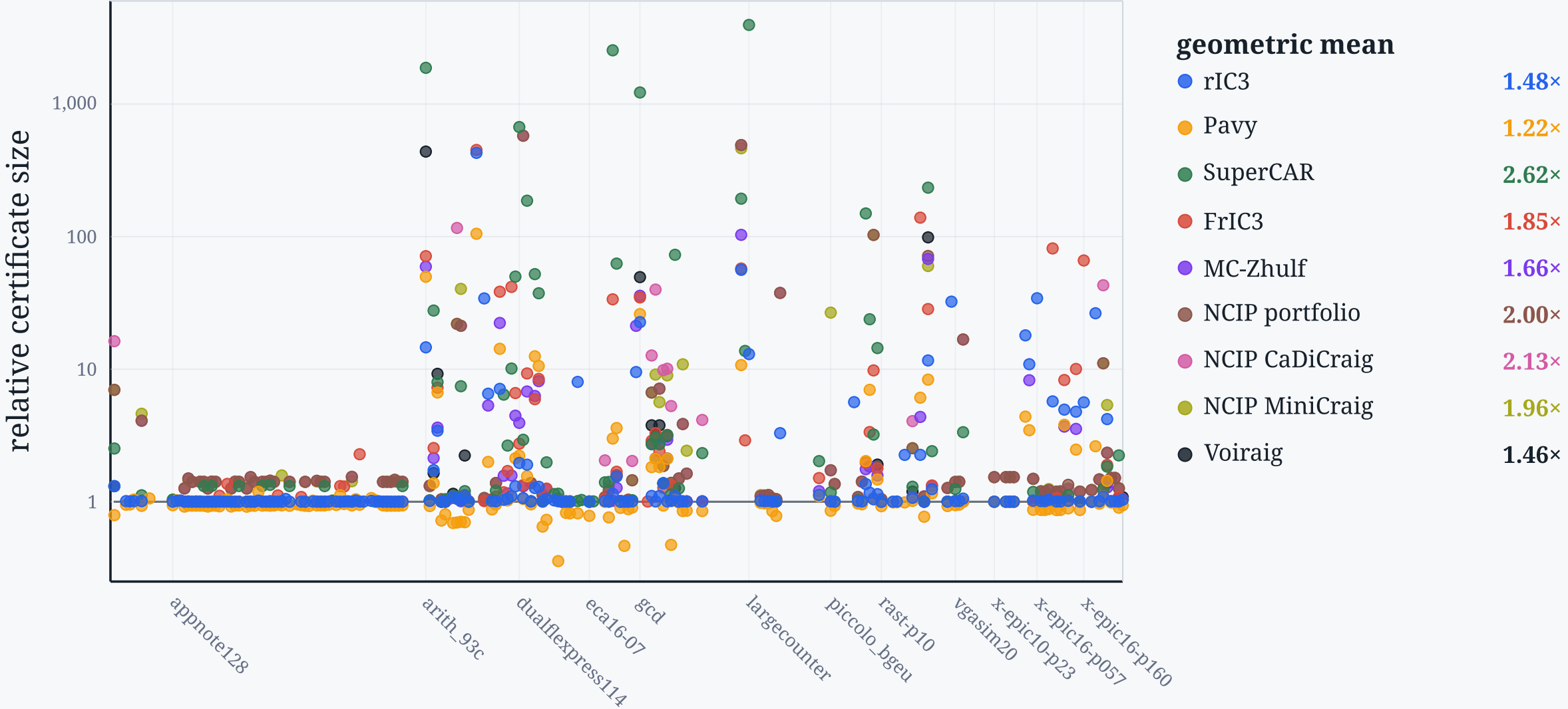
Certificates are Fast to Check

HWMCC24 · 261 instances



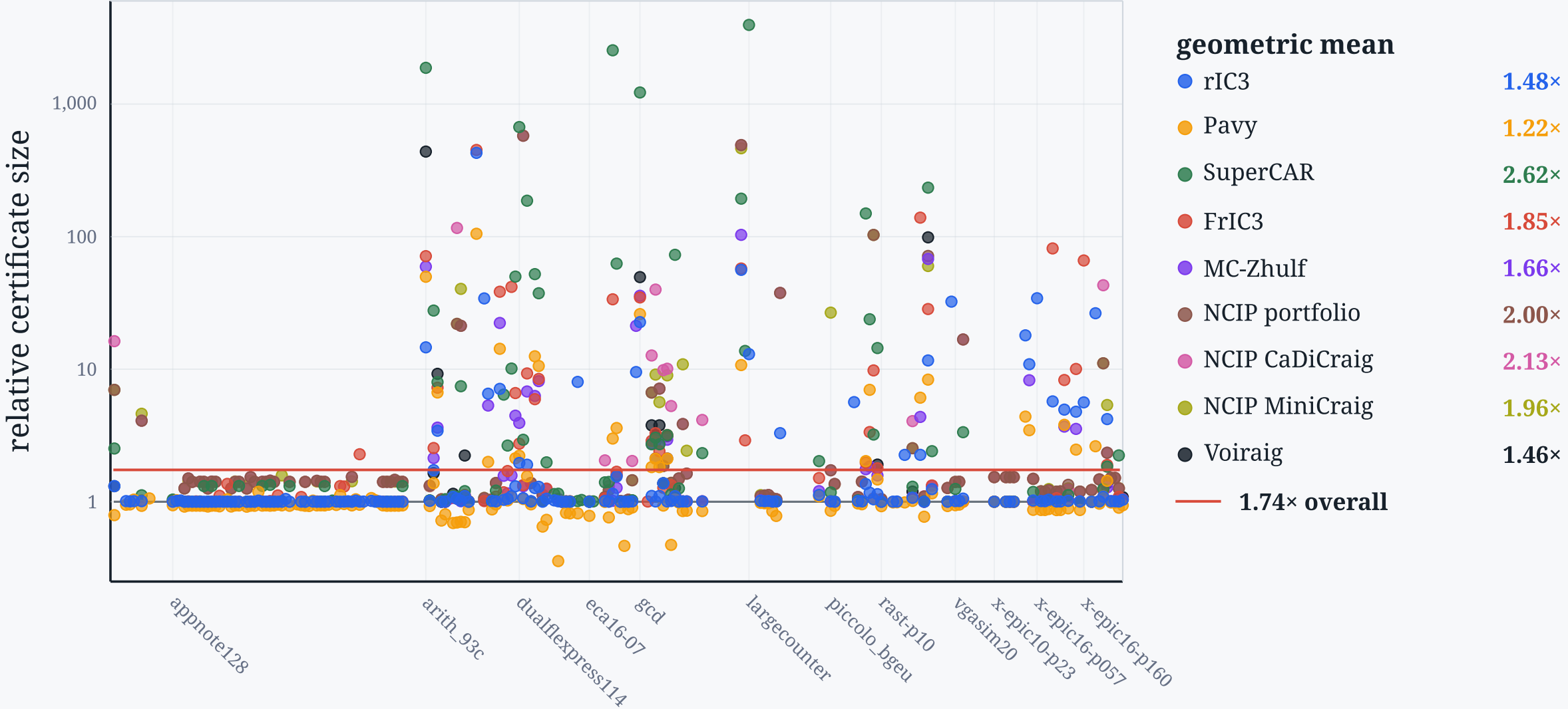
Certificates are Small

HWMCC24 · 1,062 certificates



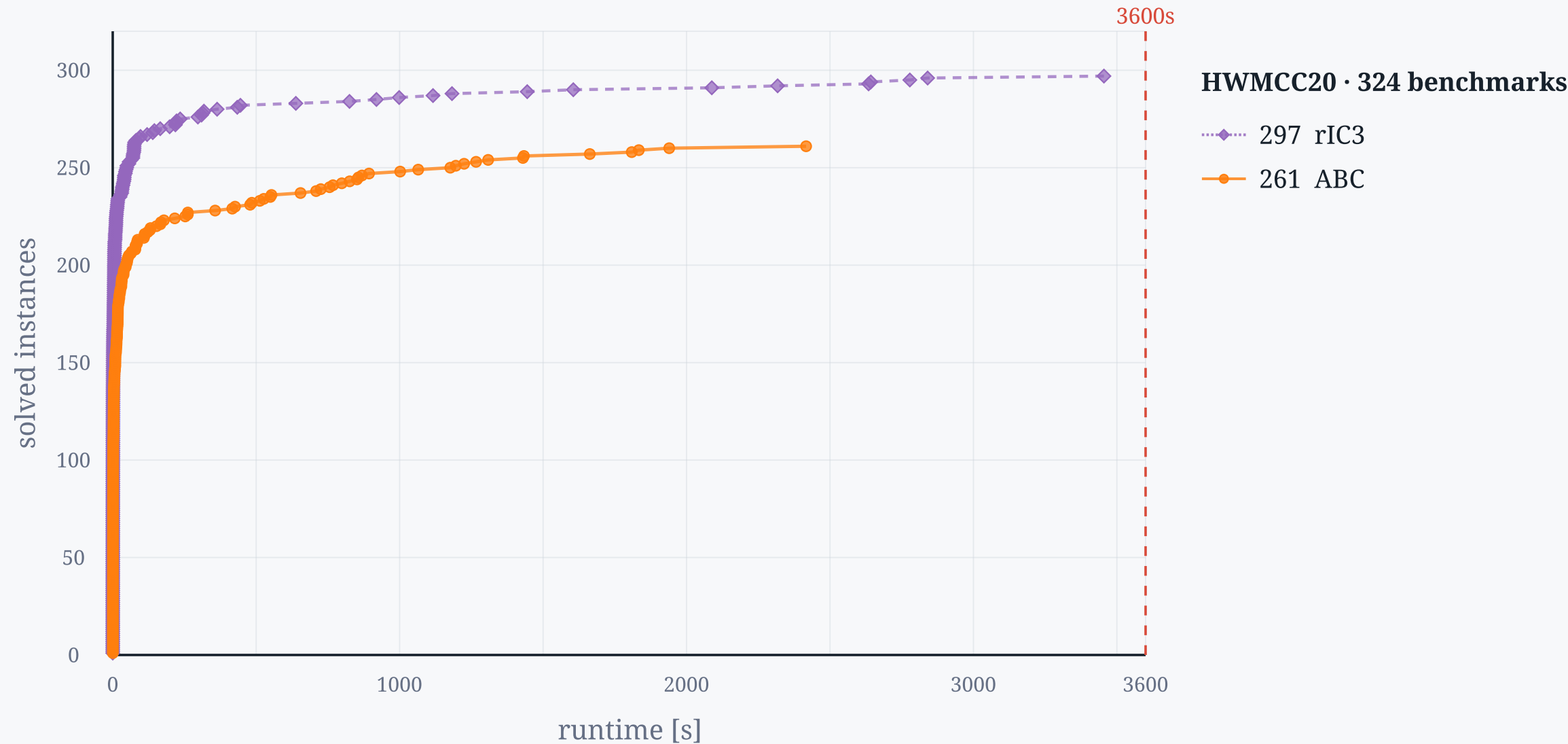
Certificates are Small

HWMCC24 · 1,062 certificates



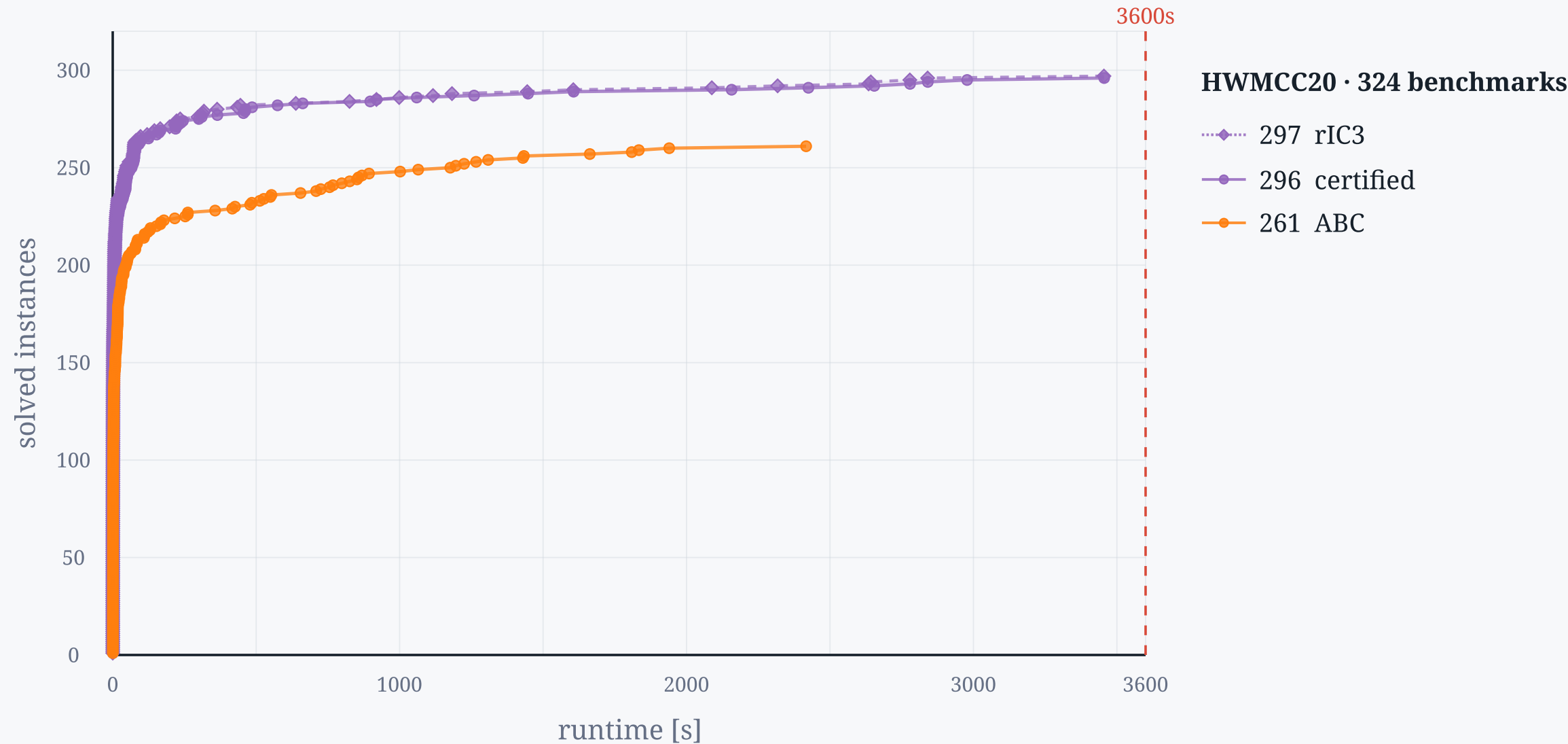
Certificates are Low Overhead

HWMCC20 · HWMCC24



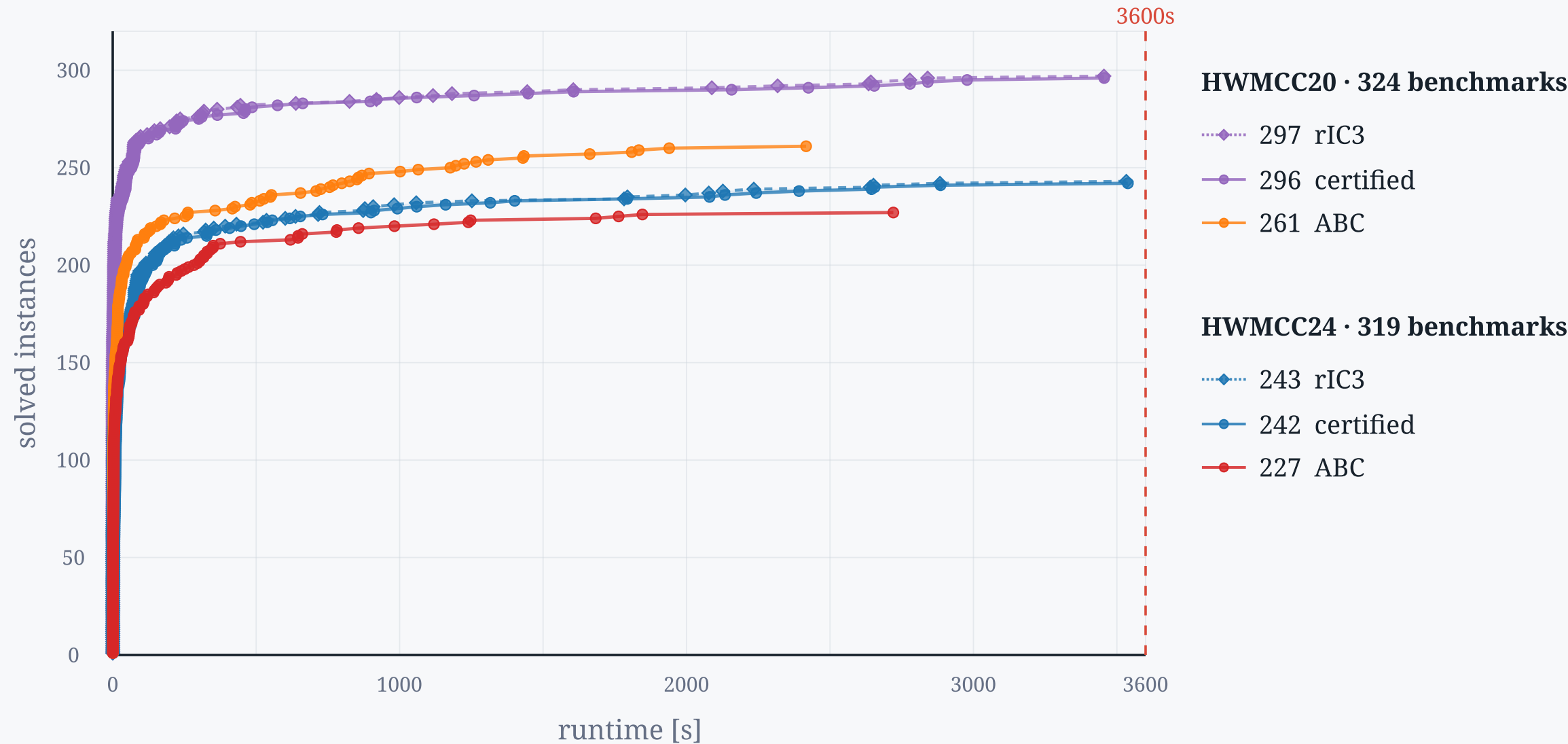
Certificates are Low Overhead

HWMCC20 · HWMCC24



Certificates are Low Overhead

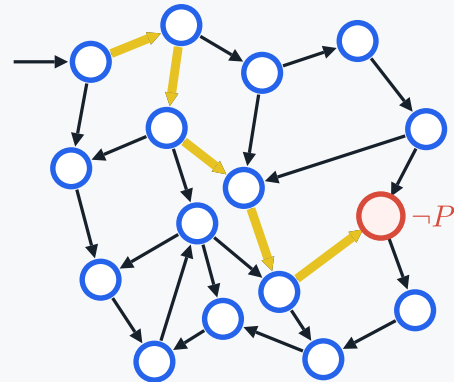
HWMCC20 · HWMCC24



Liveness Certification

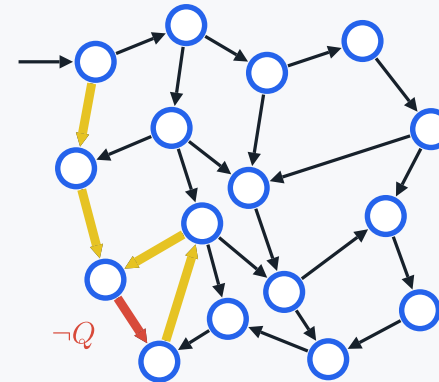
Safety

- [CAV21] IC3, BDD, Interpolation
- [FMCAD22] k-Induction
- [FMCAD23] Temporal Decomposition
- [IJCAR24] Cone of Influence, Phase Abstraction
- [CAV25] Case Study: Certifying Model Checking Competition
- [FM26] Uniqueness Constraints, Constraint Mining
- [IJCAR26] Higher Level Reasoning



Liveness

- [CAV26] k-Liveness
- [CAV26] Stabilizer Extraction
- [CAV26] Liveness to Safety
- [CAV26] rLive



k-Liveness

Claessen & Sörensson [FMCAD12]



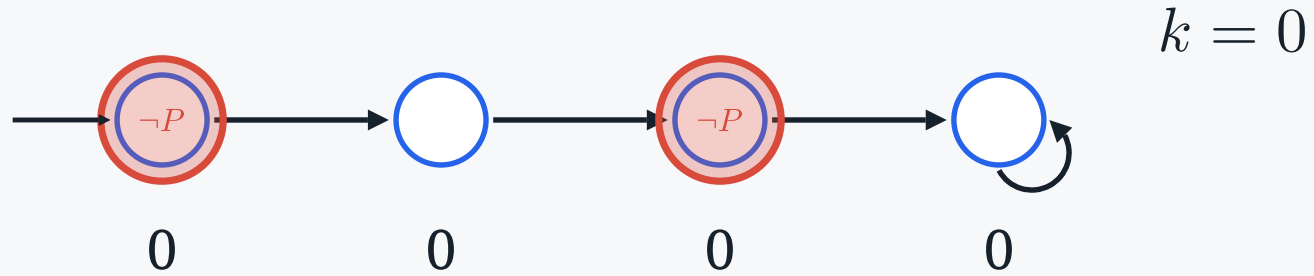
k-Liveness

Claessen & Sörensson [FMCAD12]



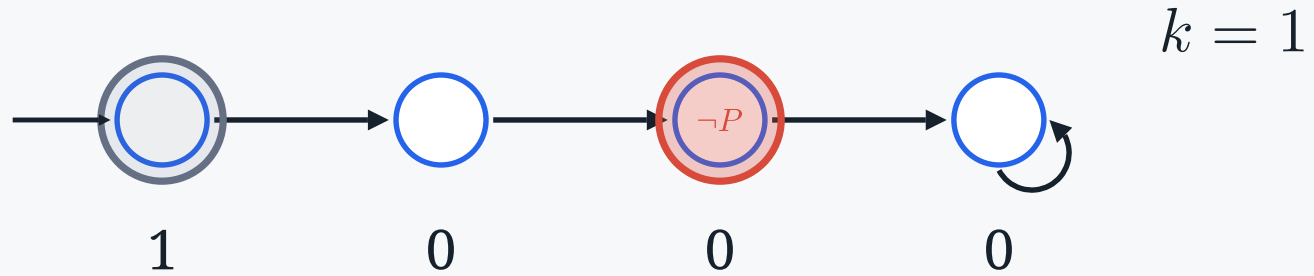
k-Liveness

Claessen & Sörensson [FMCAD12]



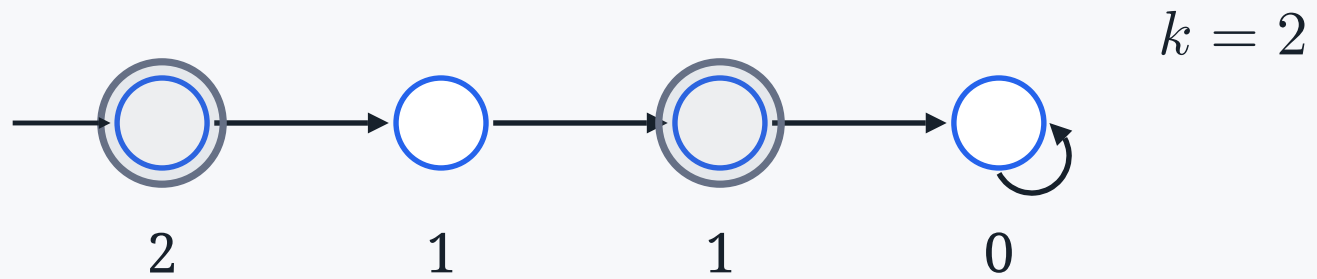
k-Liveness

Claessen & Sörensson [FMCAD12]

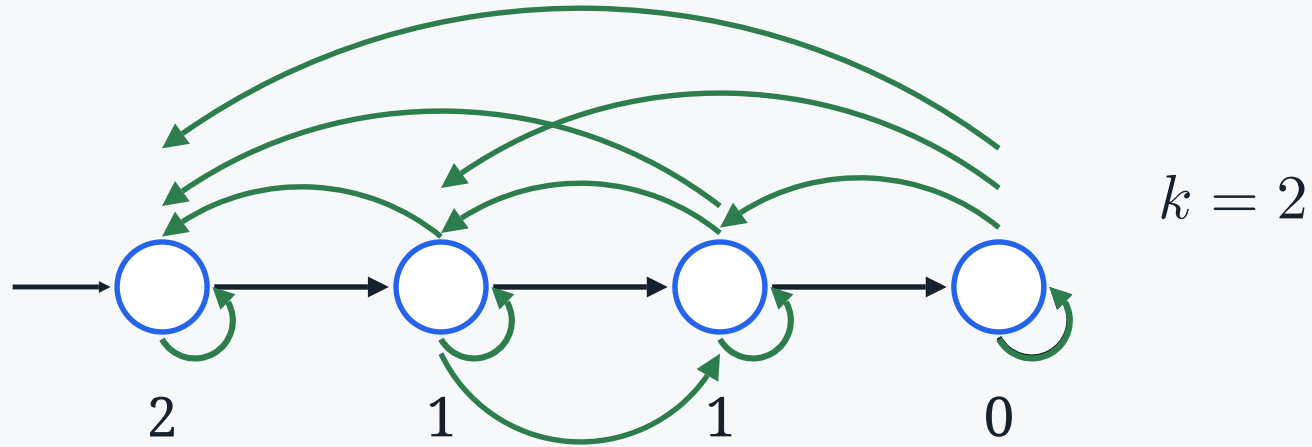


k-Liveness

Claessen & Sörensson [FMCAD12]



k-Liveness

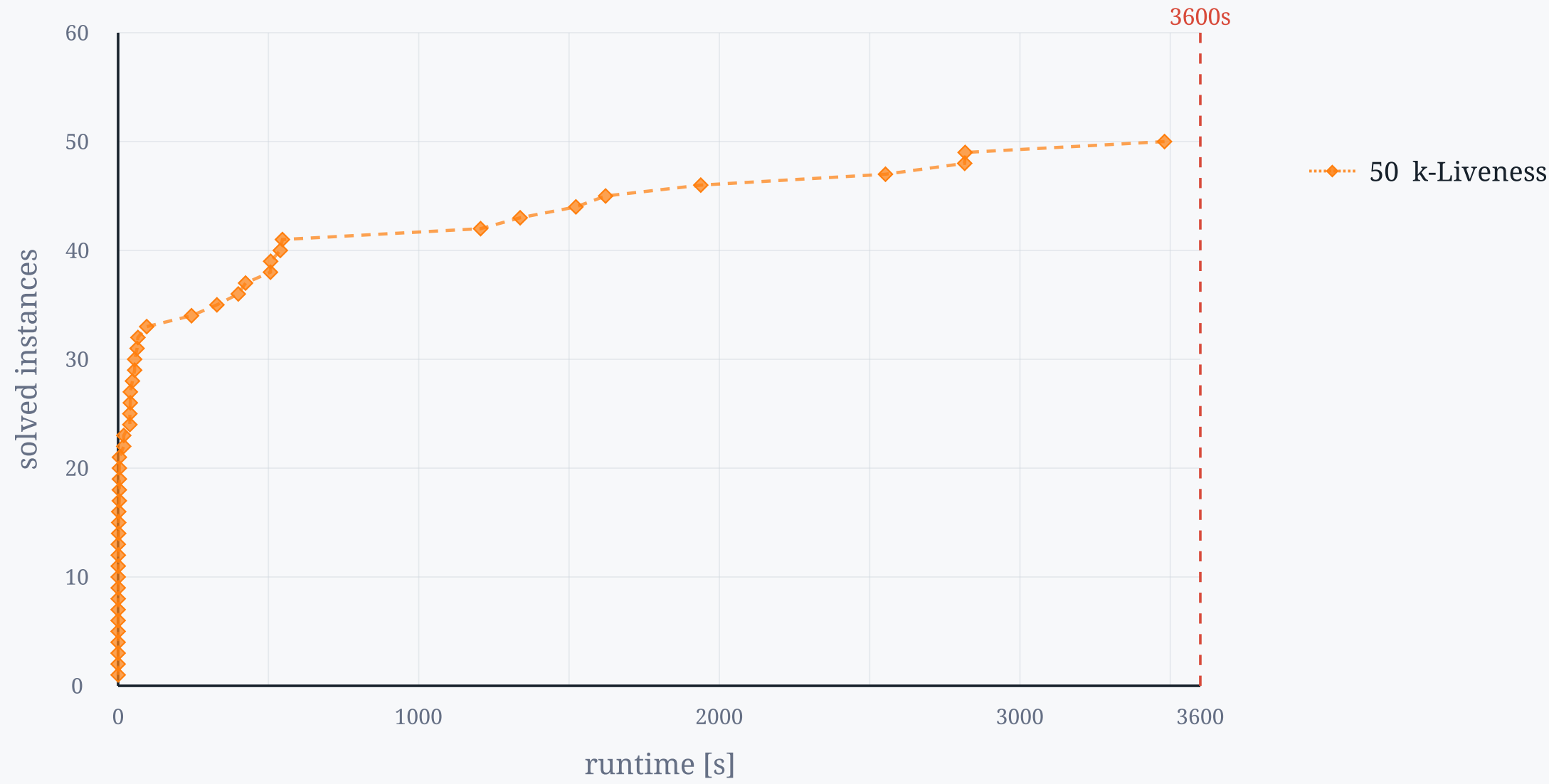


Liveness $\bigwedge_{x \in \{s,t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

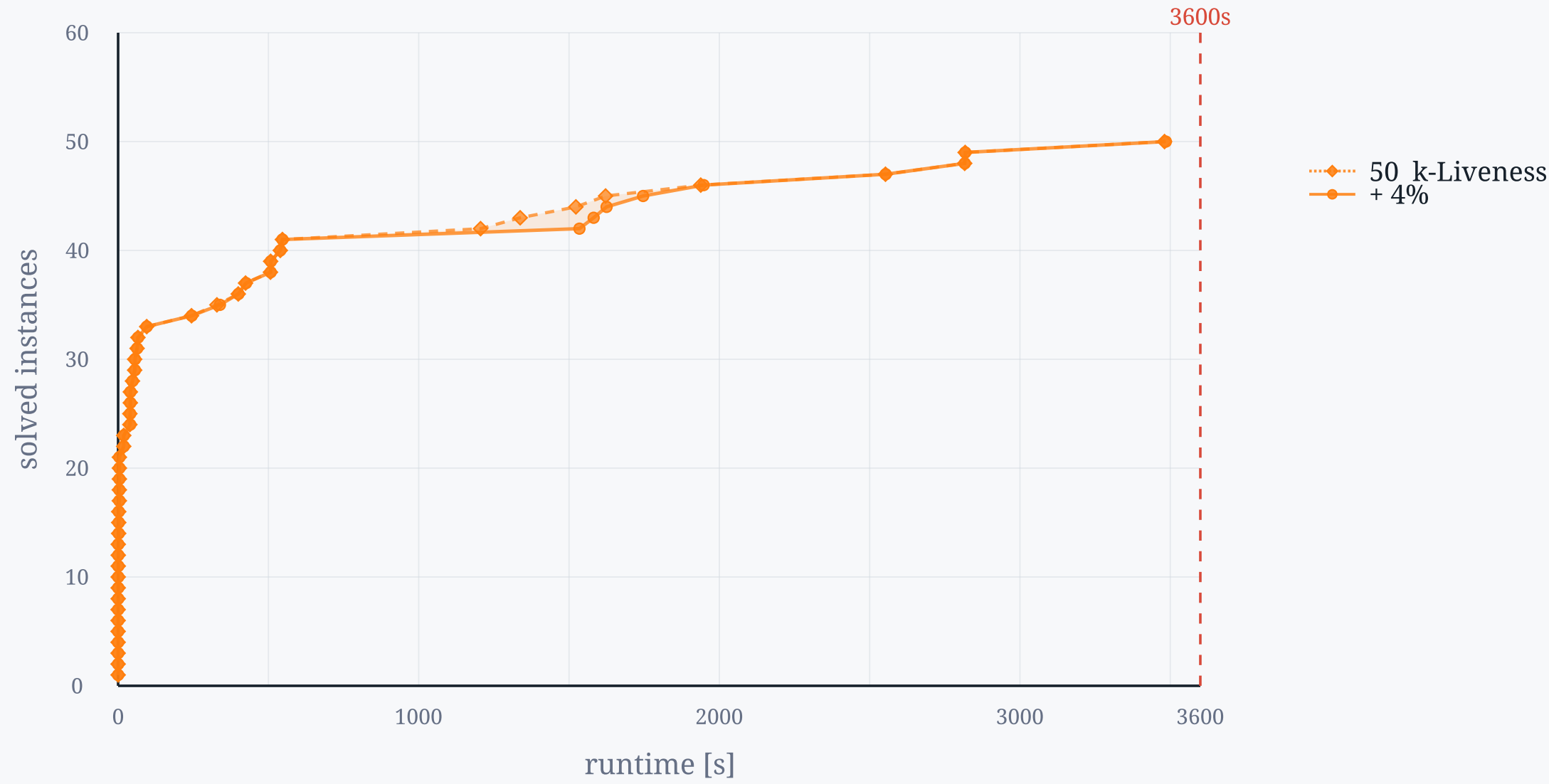
Decrease $\bigwedge_{x \in \{s,t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s,t,u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

HWMCC25

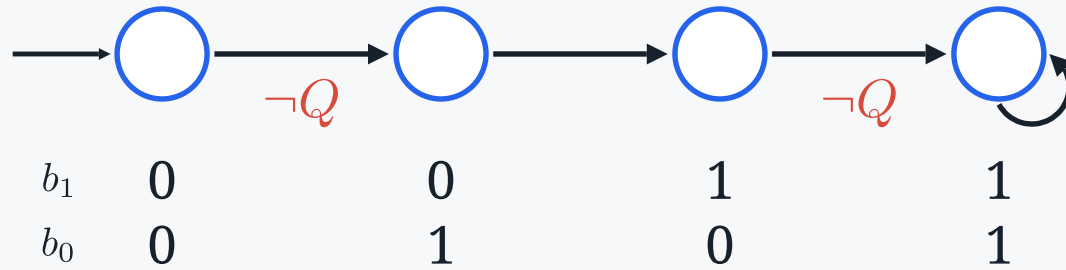


HWMCC25



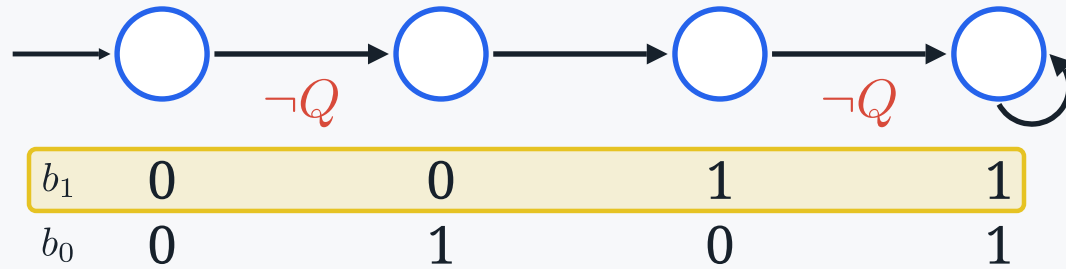
Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



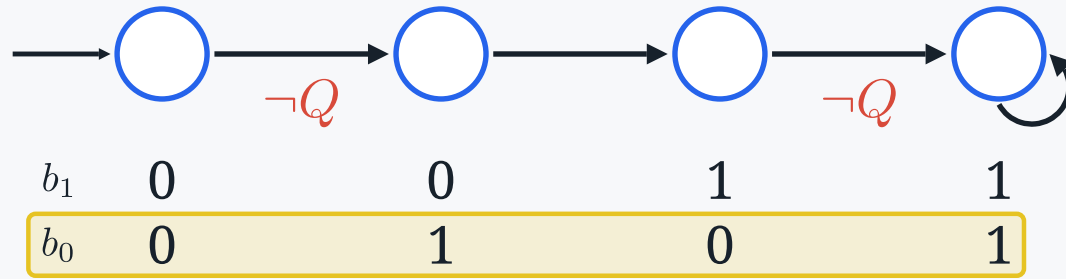
Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

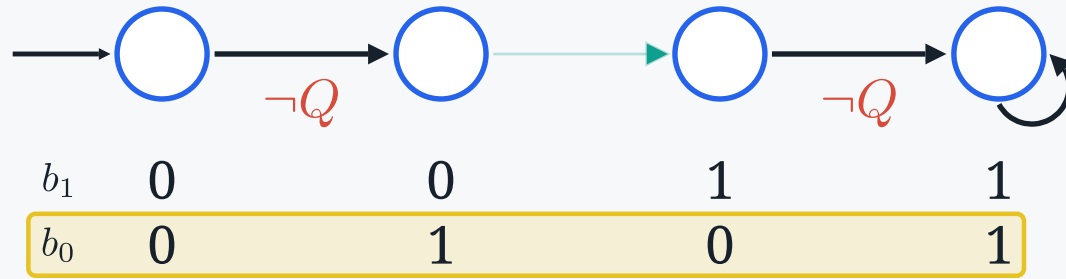
Stabilizer Extraction



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

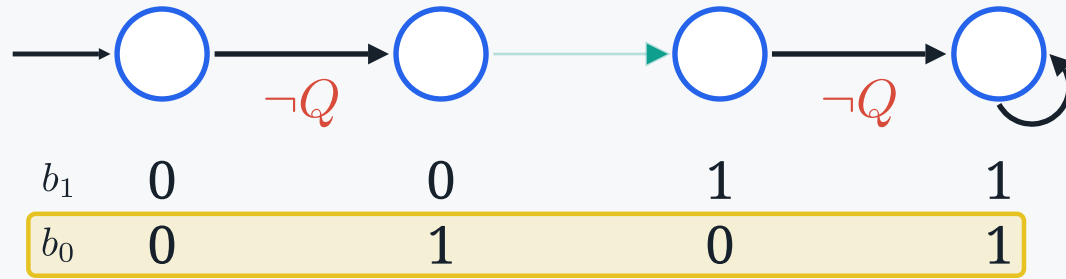
Stabilizer Extraction



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

Stabilizer Extraction



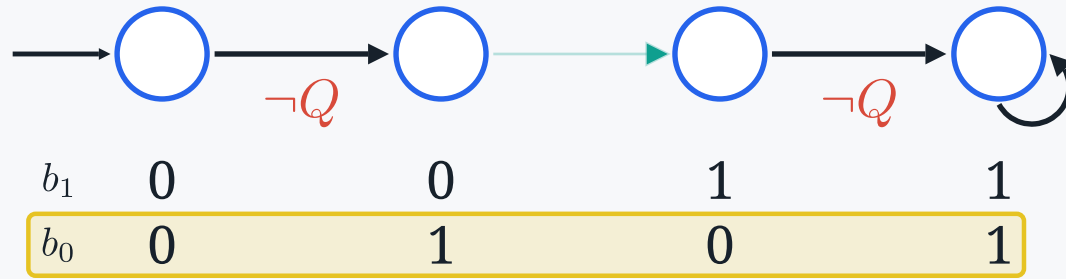
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Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



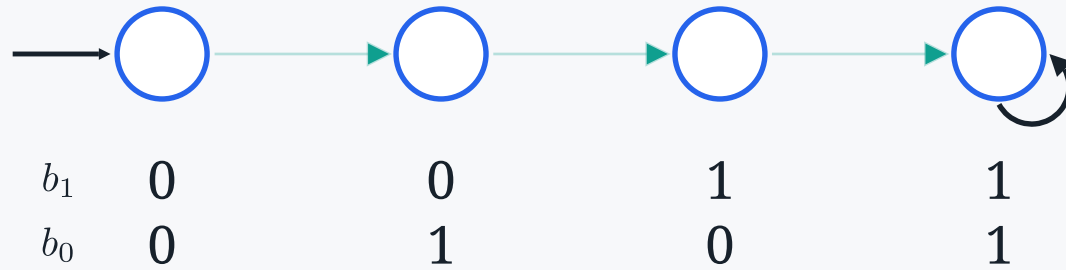
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

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Stabilizer Extraction

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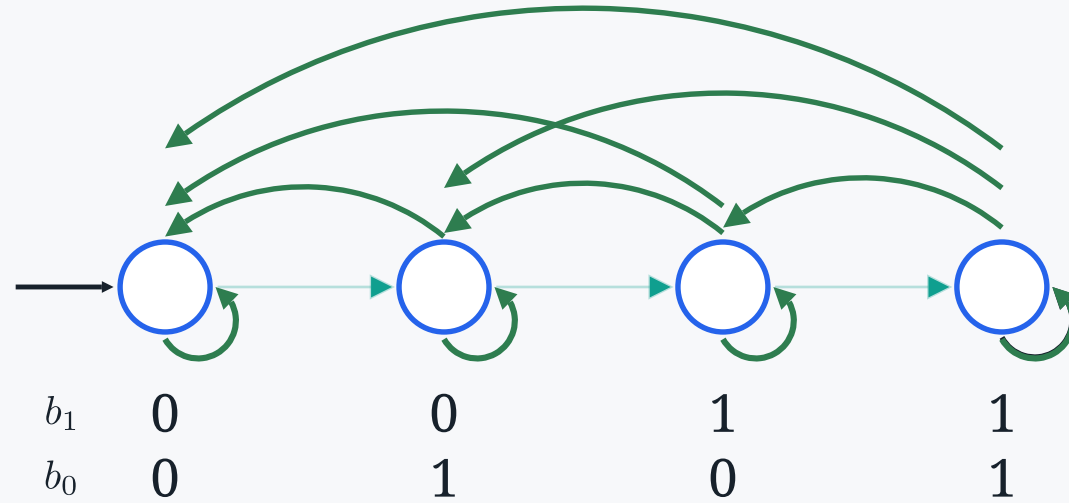
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$$(b_s^1 \Leftrightarrow b_t^1) \wedge (b_s^0 \Leftrightarrow b_t^0) \Rightarrow Q$$

Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



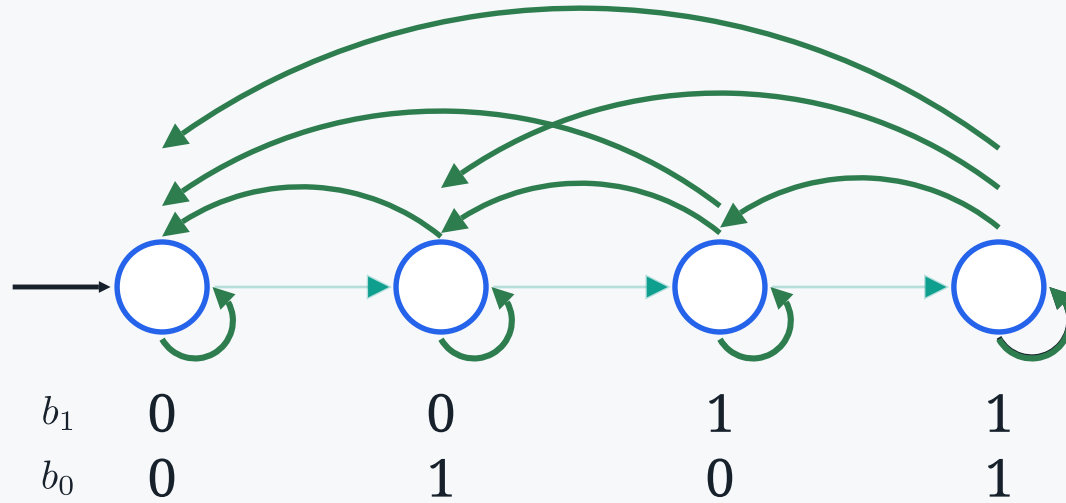
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

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Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



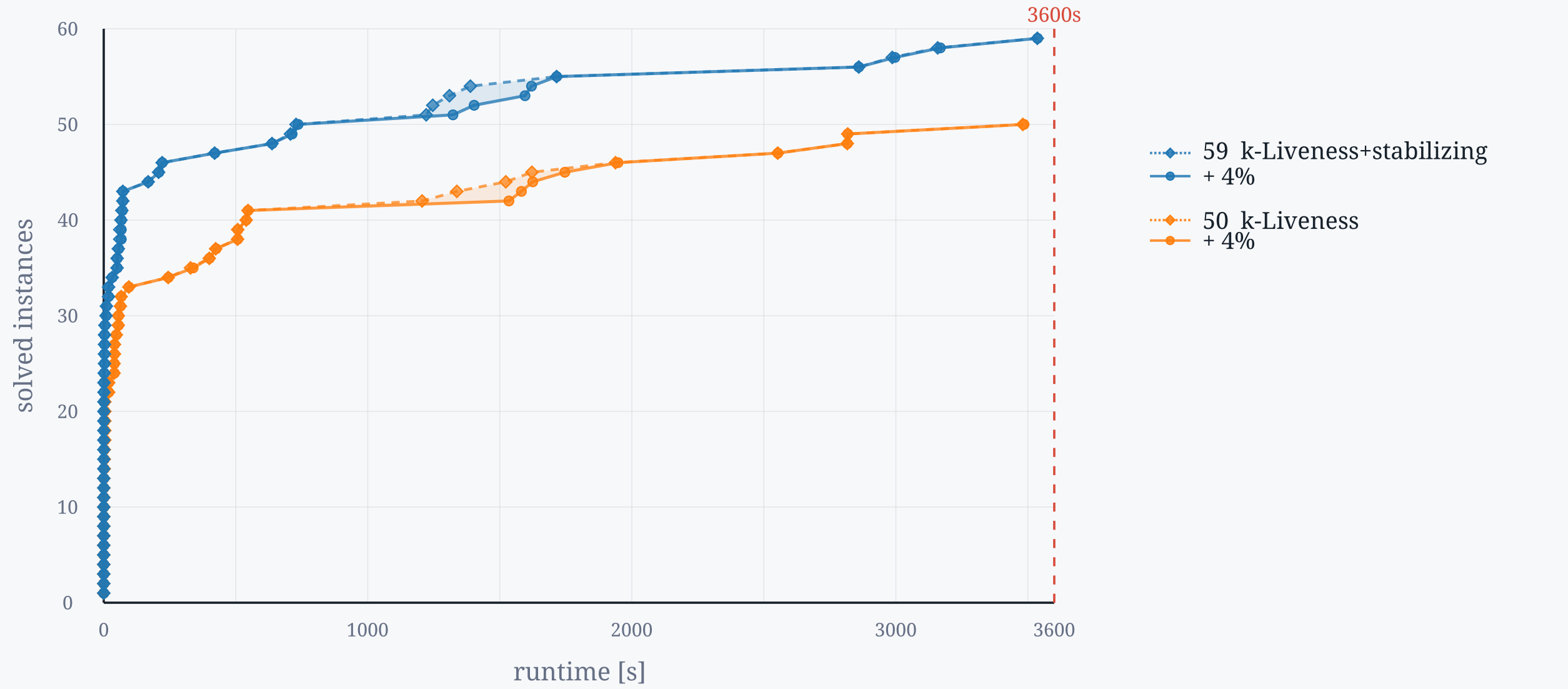
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$$(b_s^1 \Leftrightarrow b_t^1) \wedge (b_s^0 \Leftrightarrow b_t^0) \Rightarrow Q$$

$$Q' := \underbrace{\text{dec}_n}_{\text{less stable}} \vee \left(\underbrace{\text{eq}_n}_{\text{equally stable}} \wedge Q^{\text{k-liveness}} \right)$$

HWMCC25



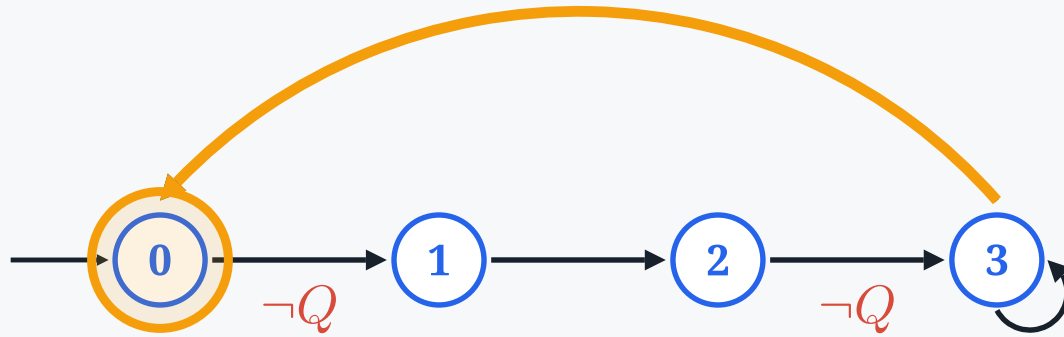
Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



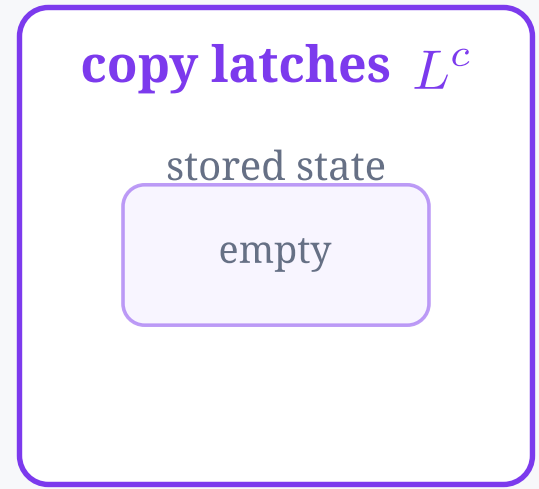
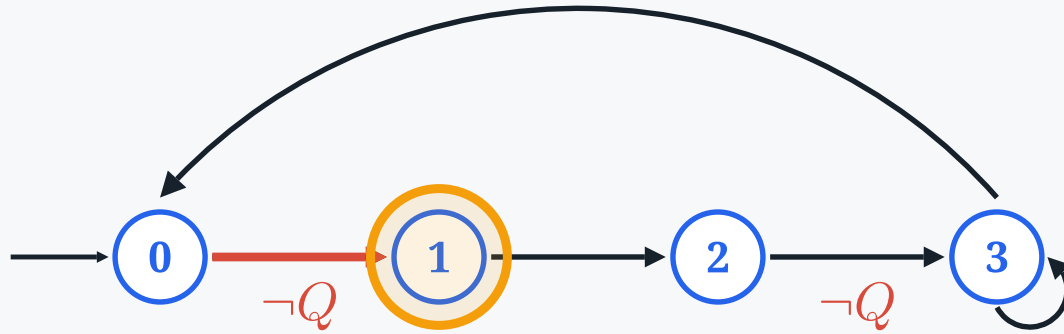
Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



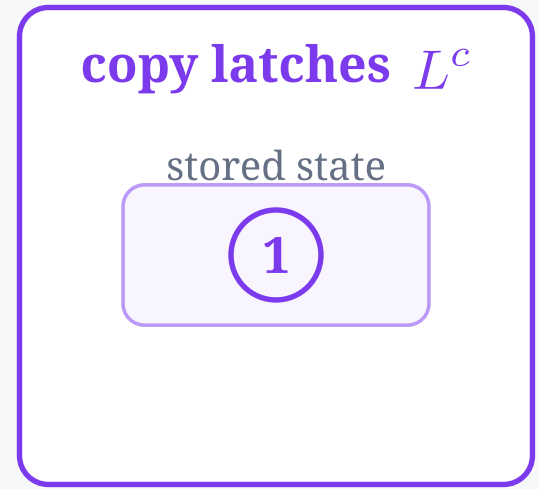
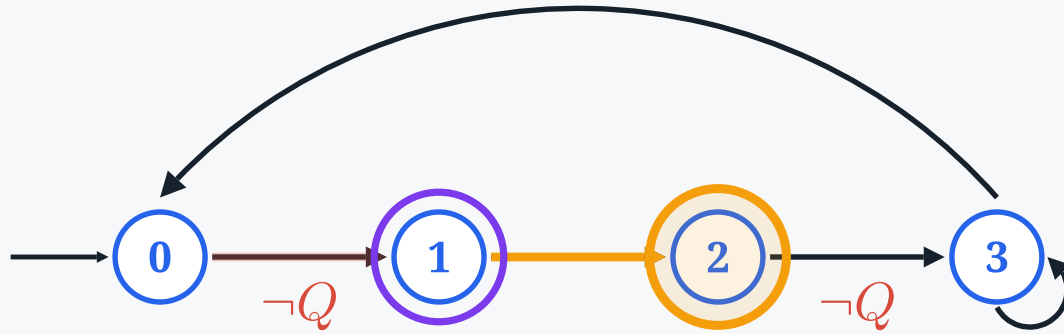
Liveness to Safety

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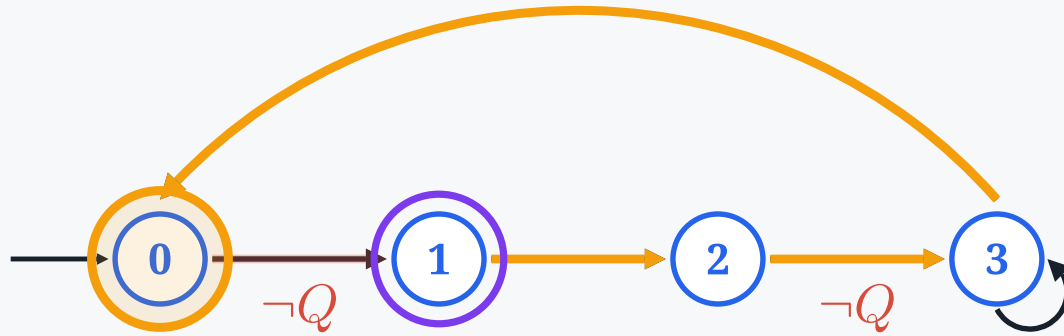
Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



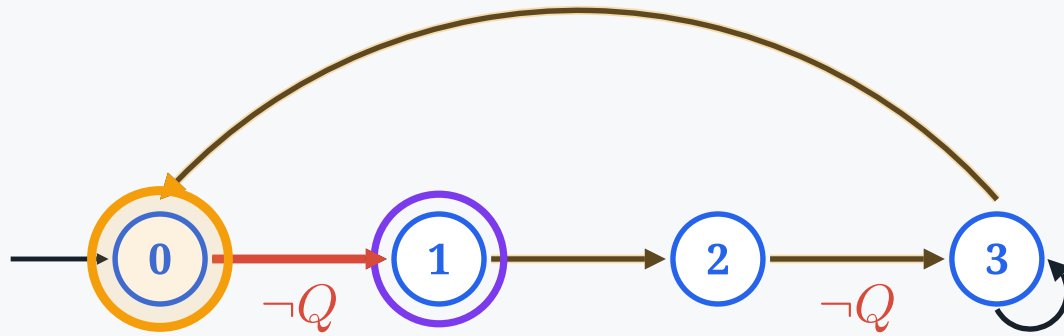
Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



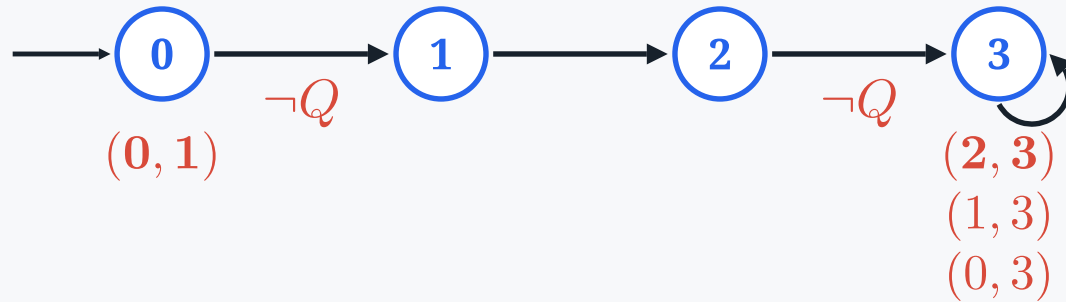
Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



Liveness to Safety

Biere, Artho & Schuppan [FMICS02]

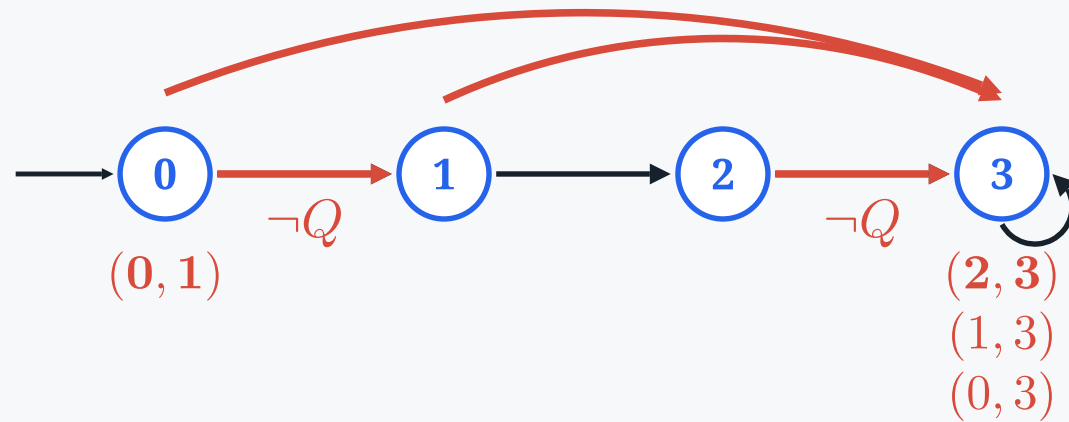


(L, L^c)
current state stored state

$$P_{st}^{L2S} : F_{st} \wedge (L_t = L^c) \Rightarrow Q_{st}$$

Liveness to Safety

Biere, Artho & Schuppan [FMICS02]

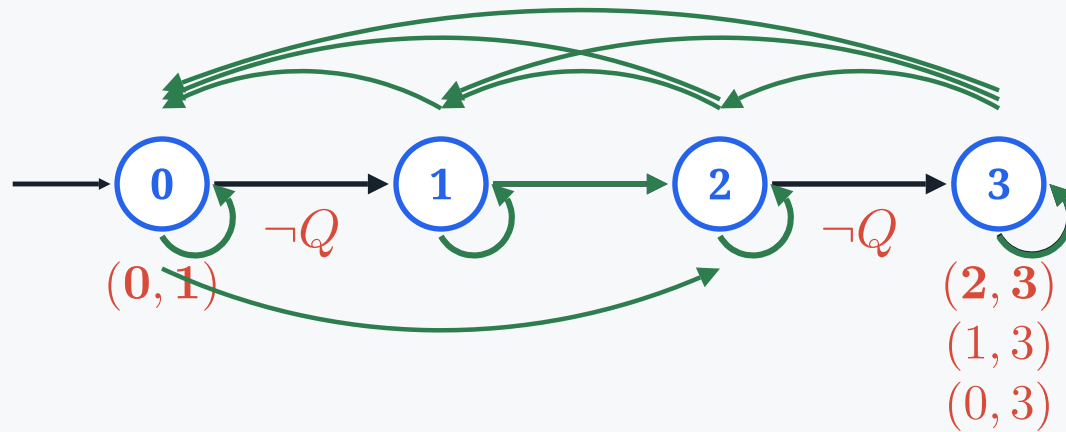


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Liveness to Safety

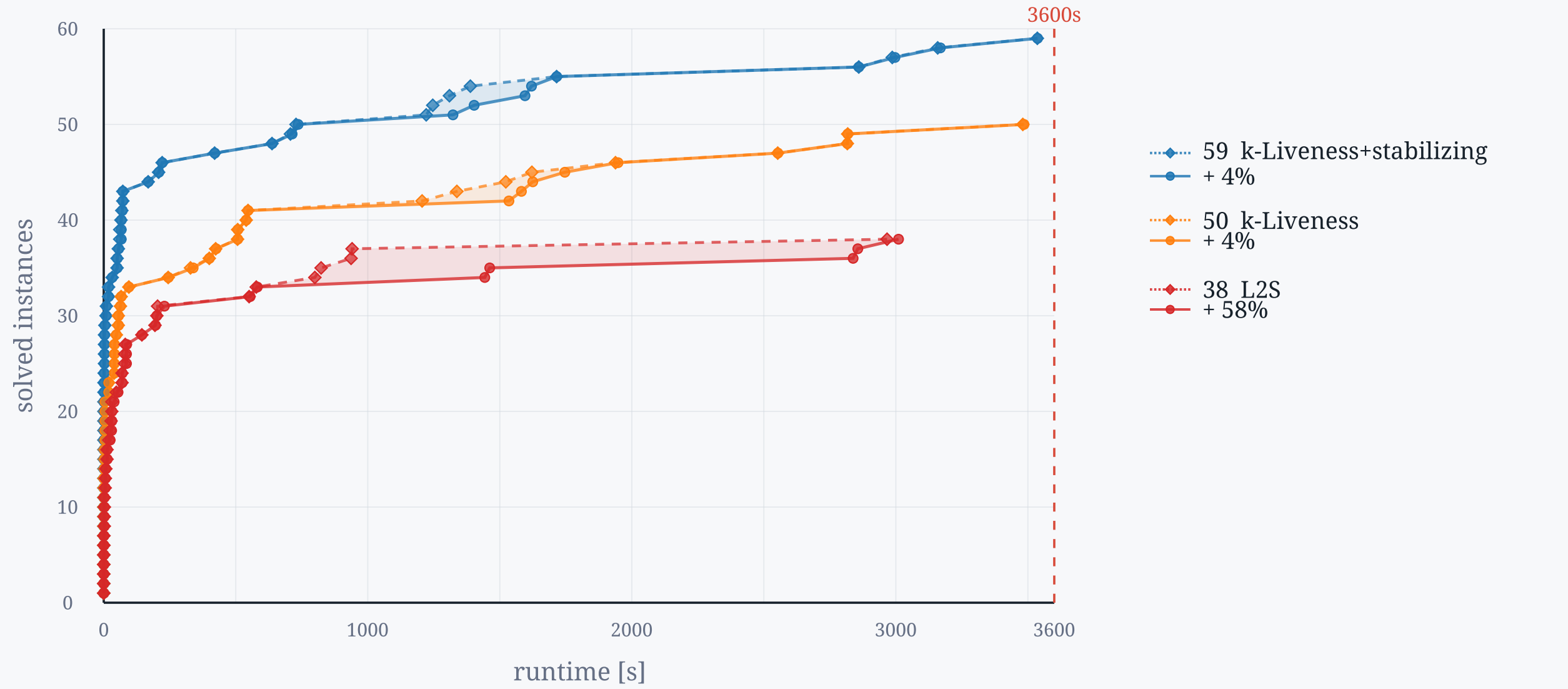
Biere, Artho & Schuppan [FMICS02]



(L, L^c)
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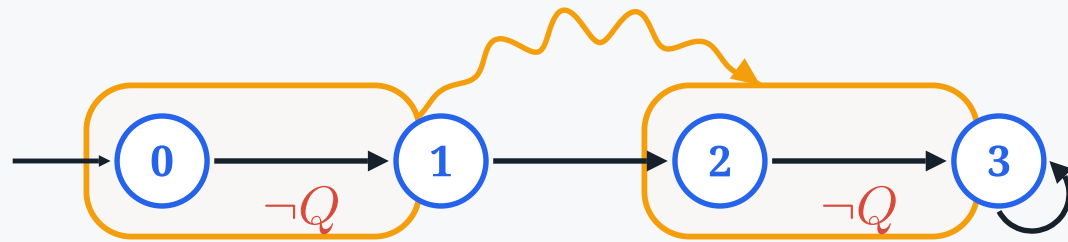
$$P_{st}^{L2S} : F_{st} \wedge (L_t = L^c) \Rightarrow Q_{st}$$

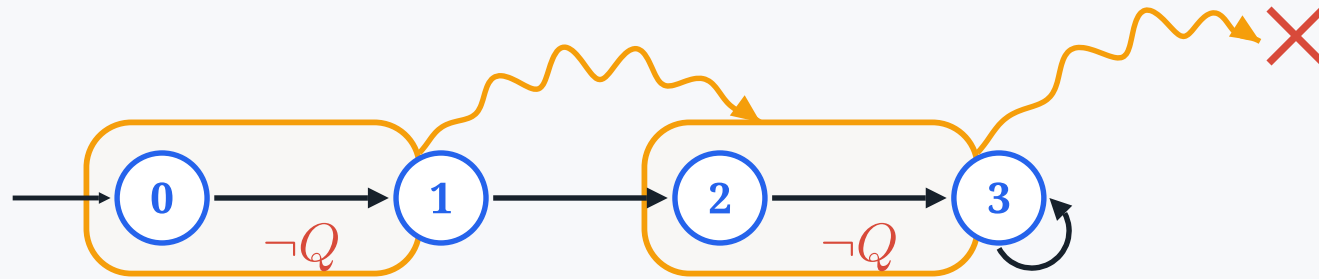
HWMCC25

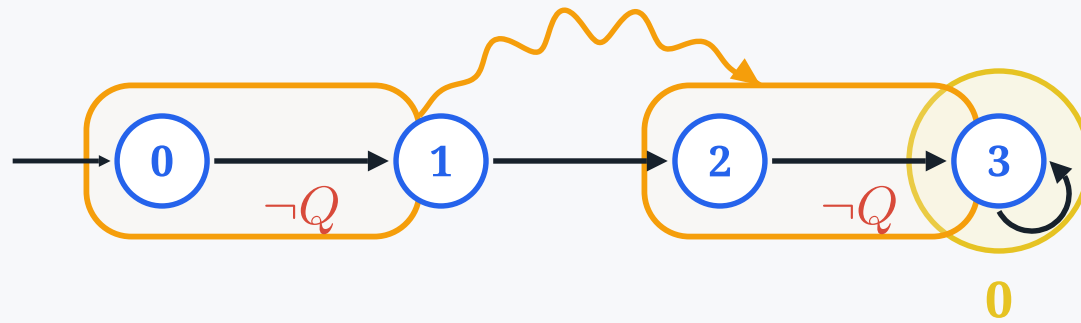


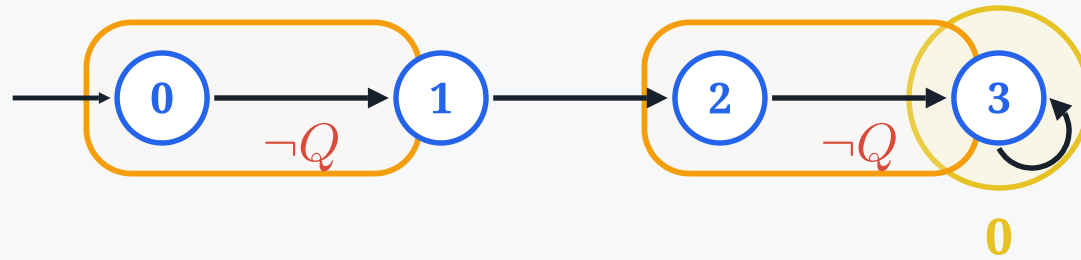


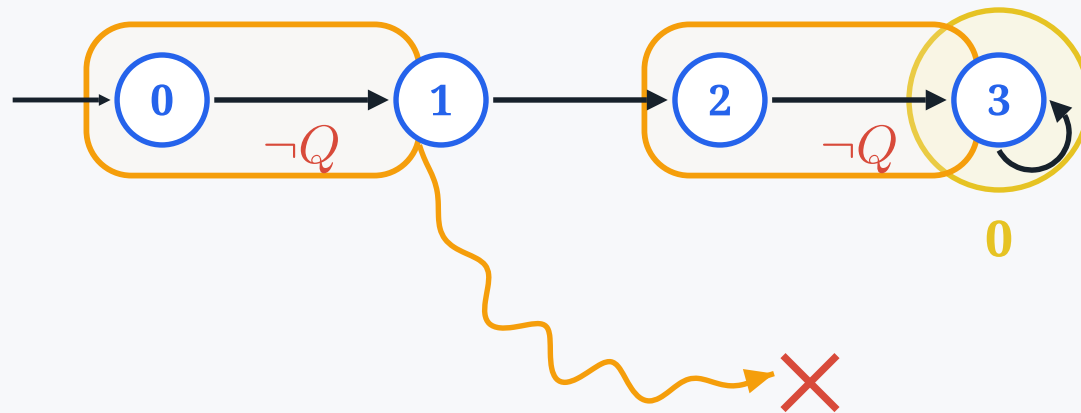


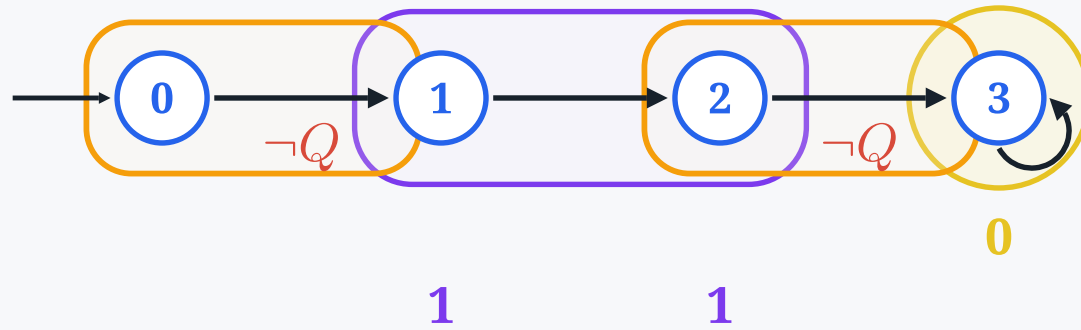


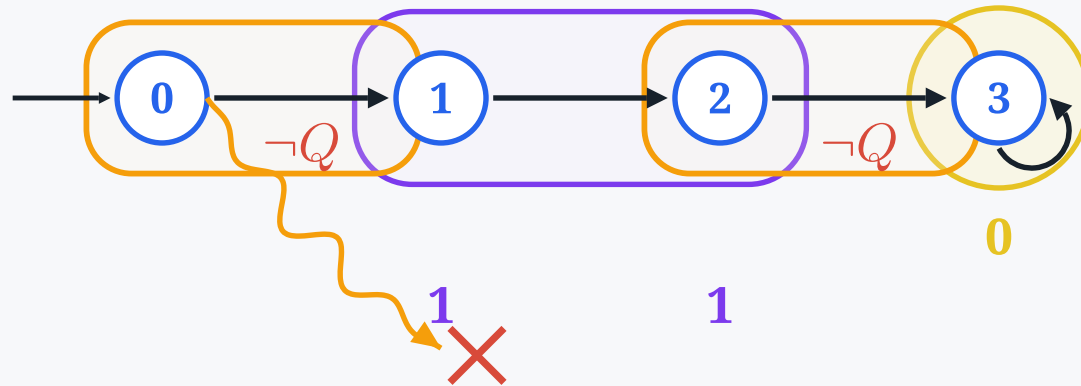


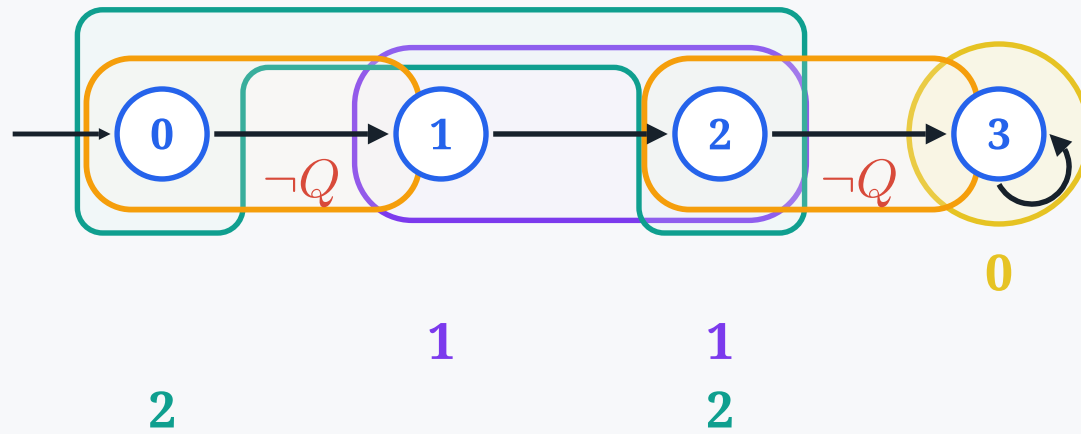


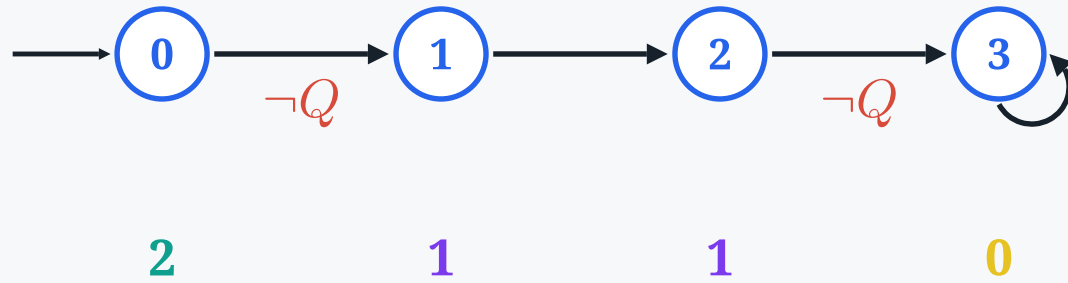


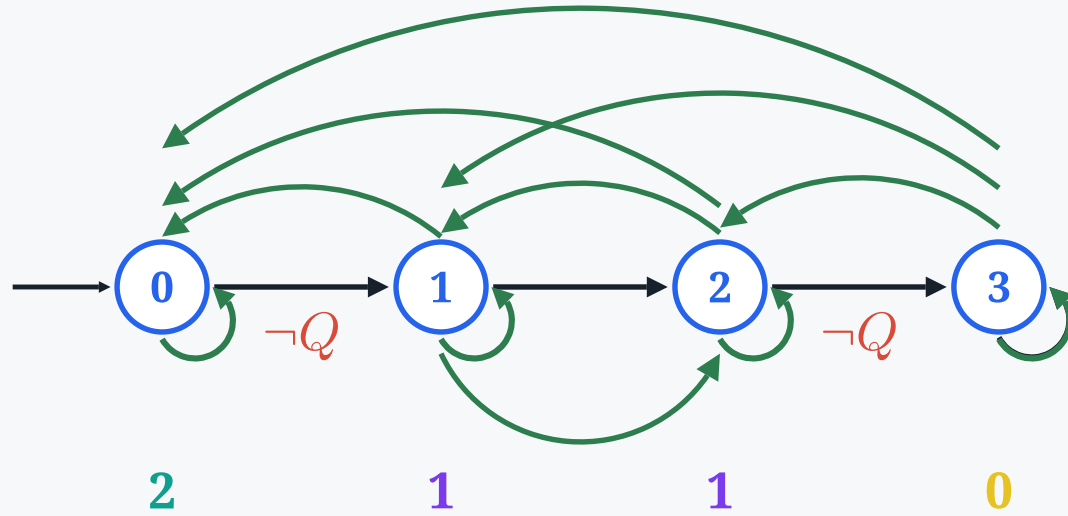




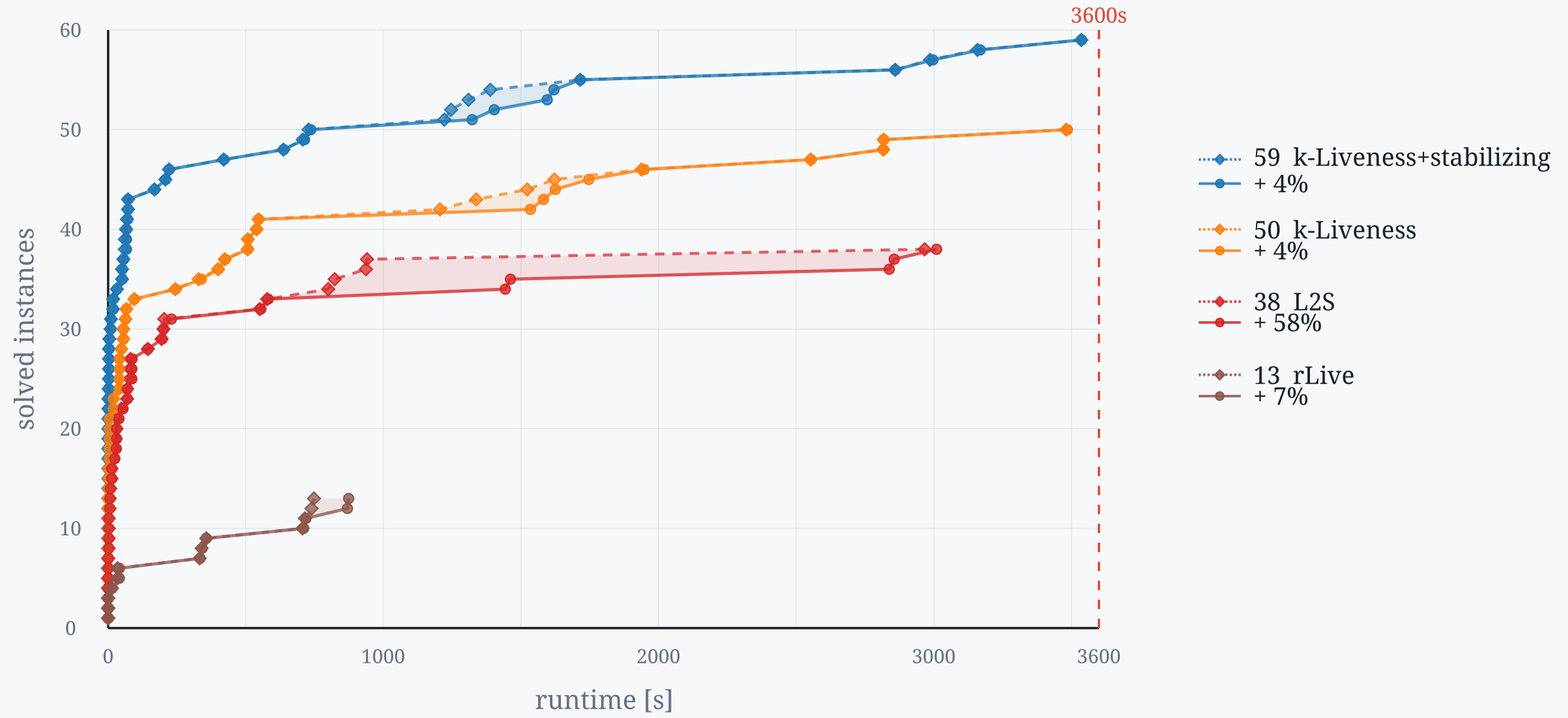




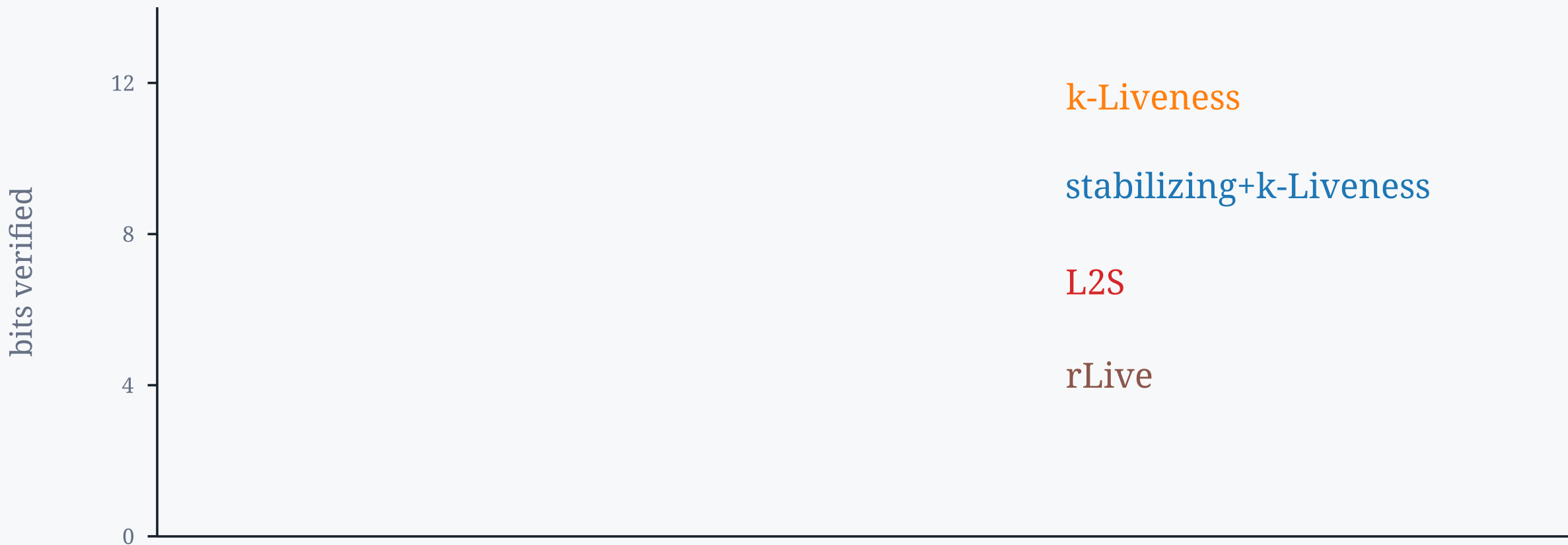




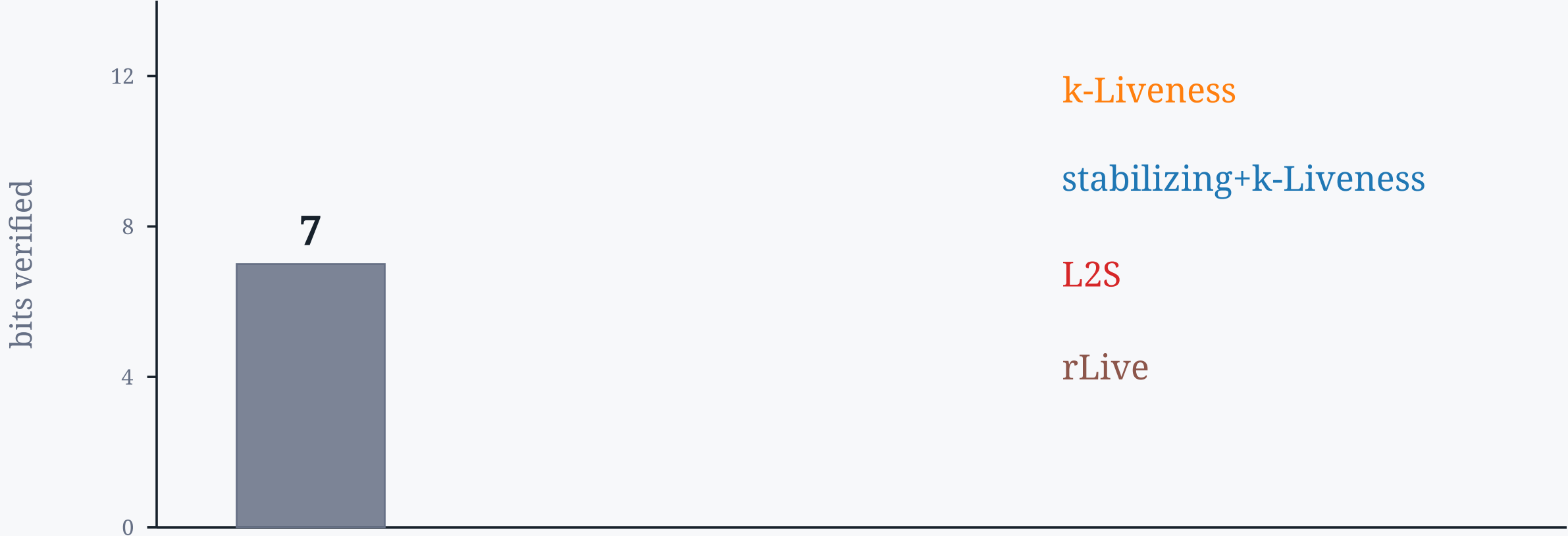
HWMCC25



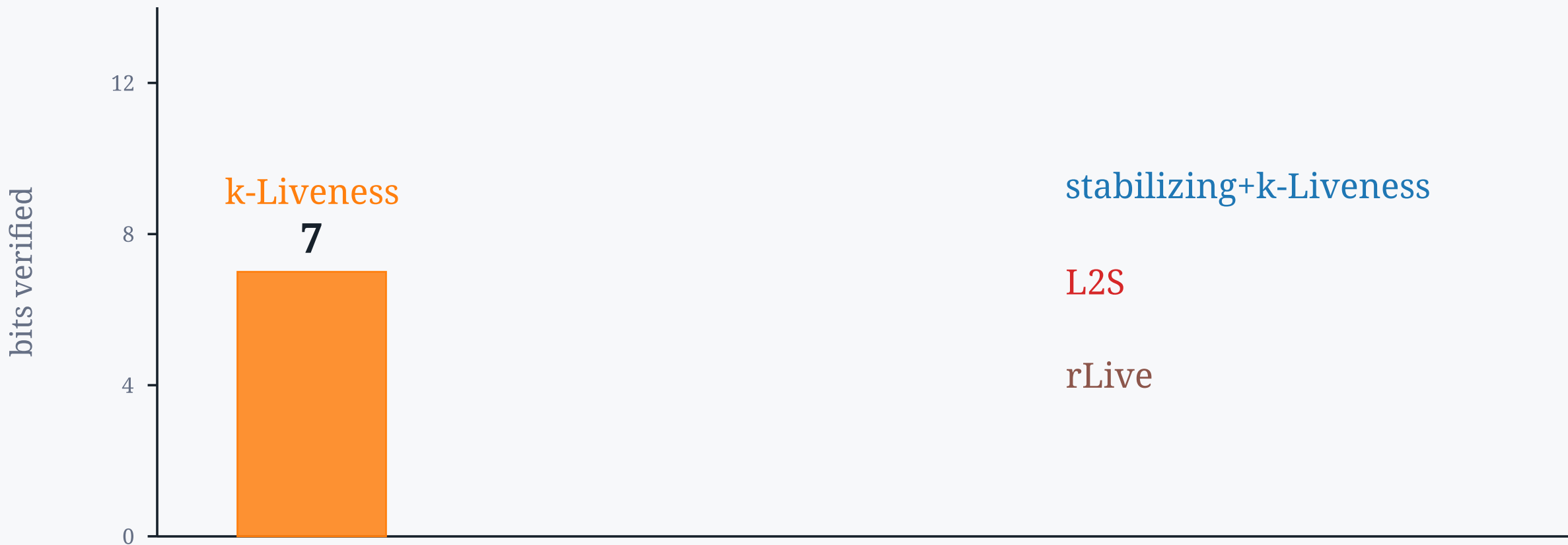
Performance Quiz: Saturating Counter



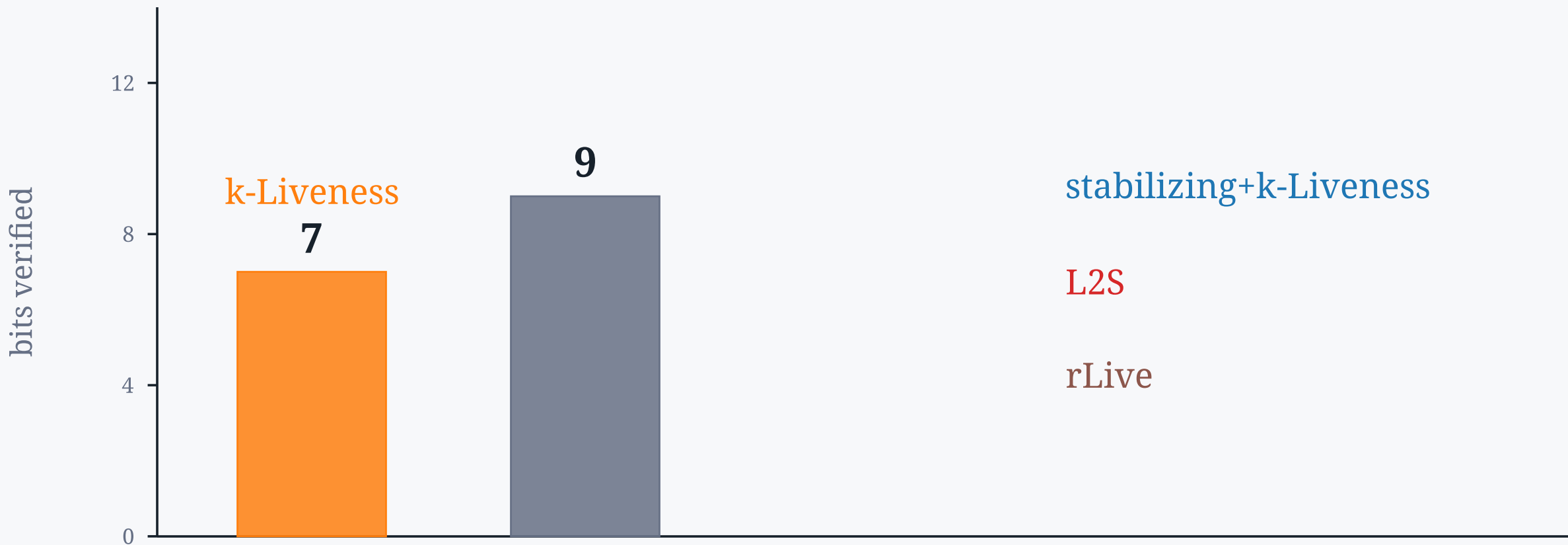
Performance Quiz: Saturating Counter



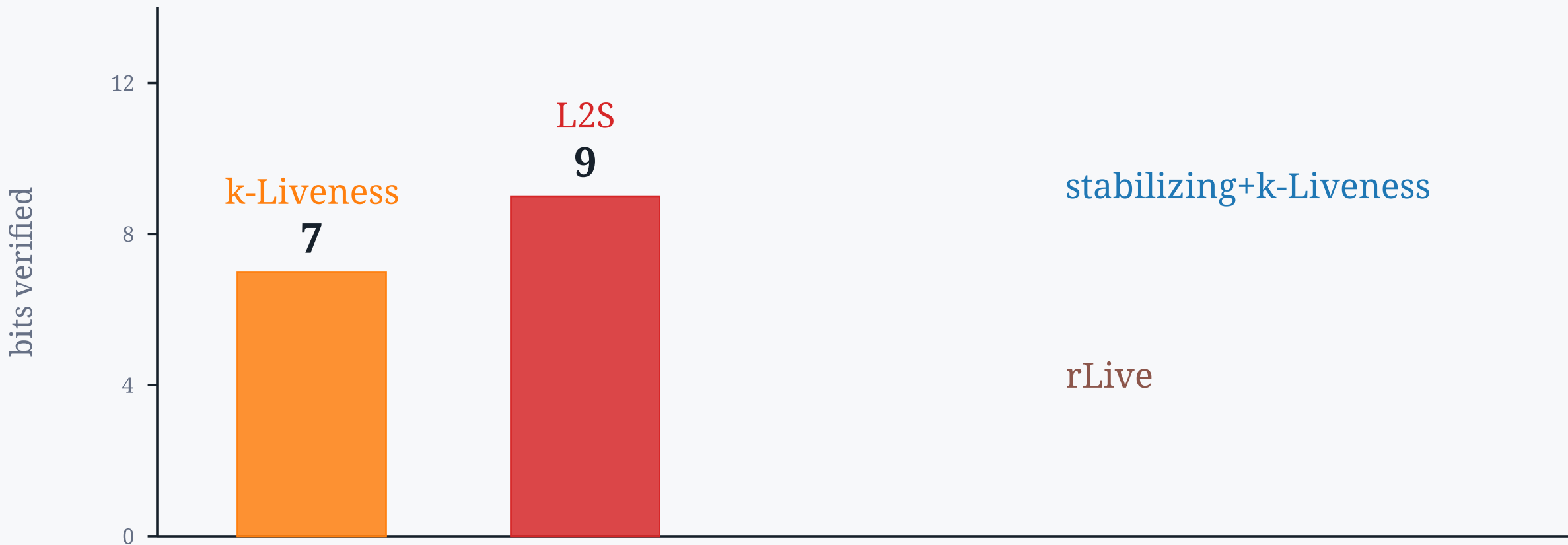
Performance Quiz: Saturating Counter



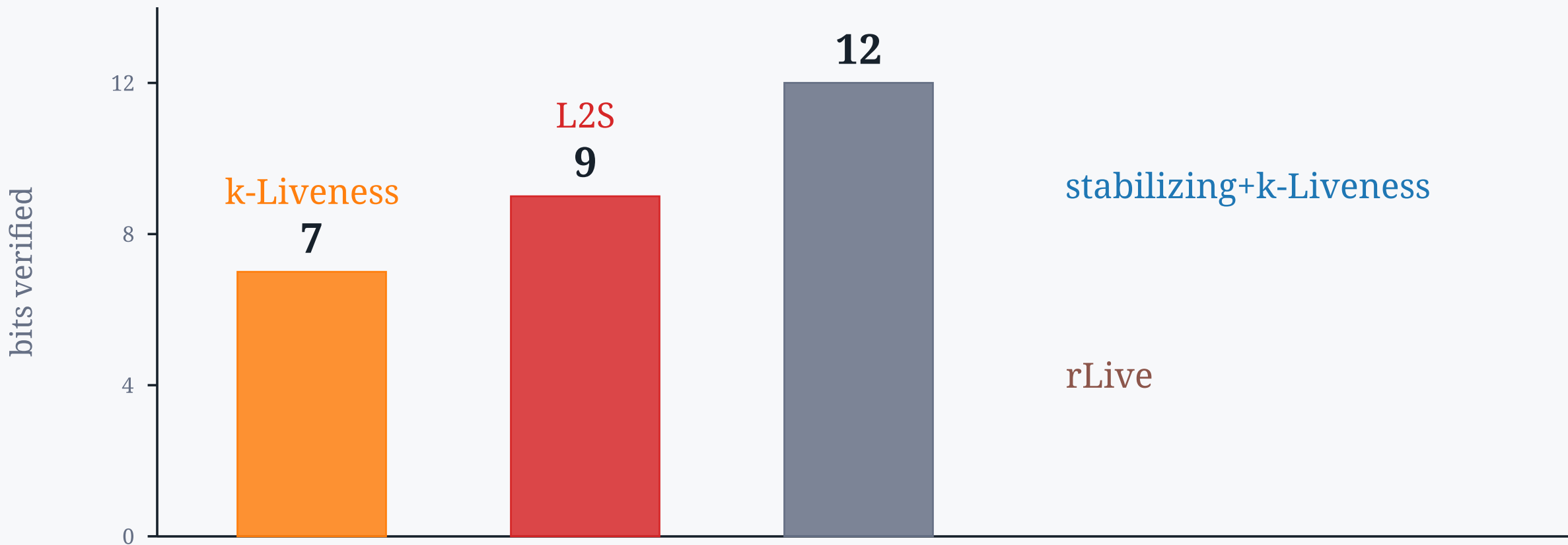
Performance Quiz: Saturating Counter



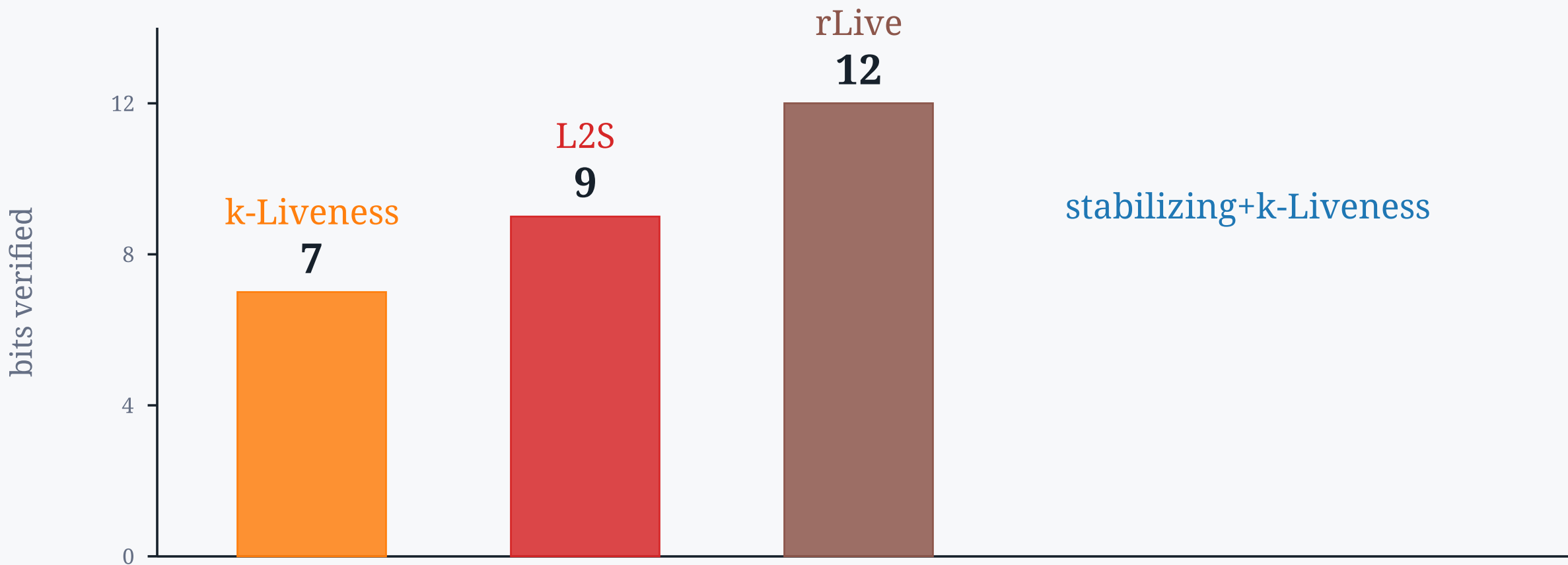
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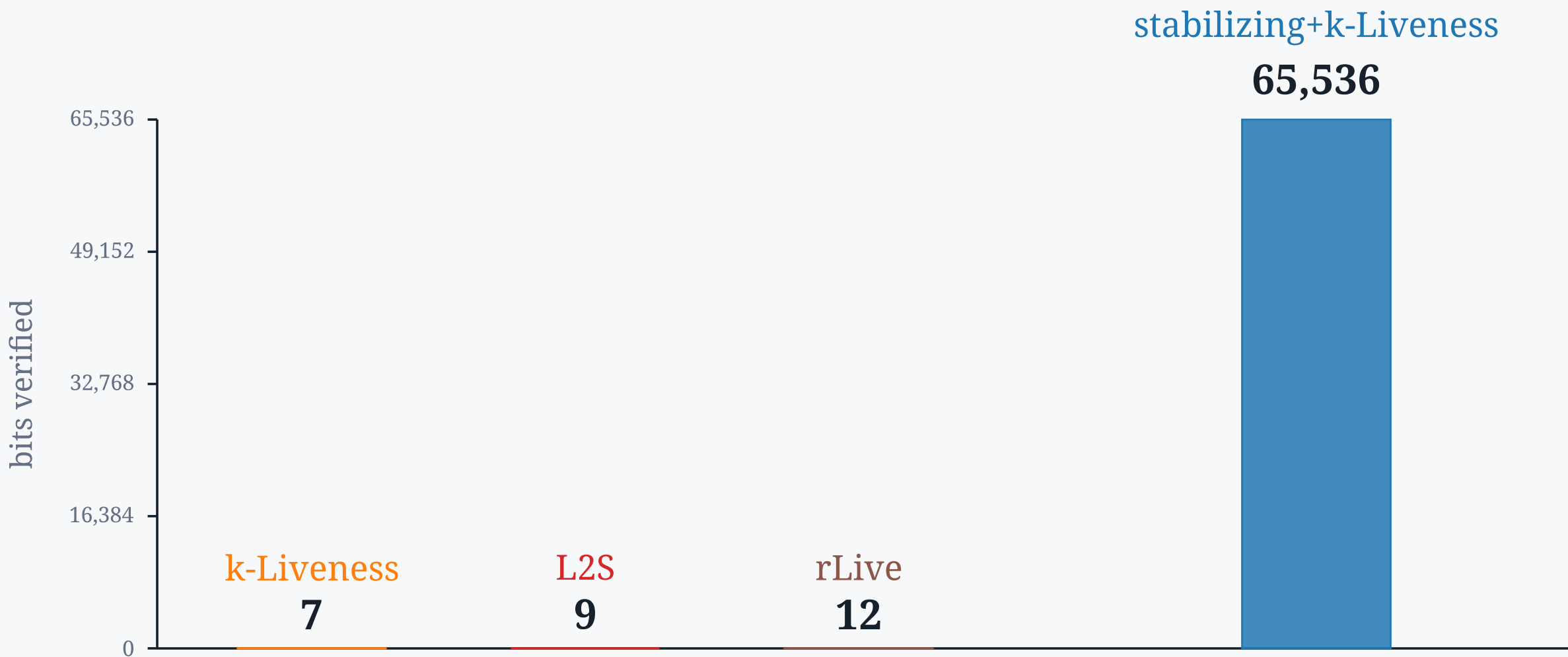
Performance Quiz: Saturating Counter



Performance Quiz: Saturating Counter



Performance Quiz: Saturating Counter



Takeaway

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

Base $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



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Takeaway

Reset $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

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Decrease $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

Closure $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



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References

- [\[CAV21\]](#) Yu, Biere & Heljanko, Progress in Certifying Hardware Model Checking Results, CAV 2021.
- [\[FMCAD22\]](#) Yu et al., Stratified Certification for k-Induction, FMCAD 2022.
- [\[FMCAD23\]](#) Yu et al., Towards Compositional Hardware Model Checking Certification, FMCAD 2023.
- [\[IJCAR24\]](#) Froleys et al., Certifying Phase Abstraction, IJCAR 2024.
- [\[CAV25\]](#) Froleys et al., Introducing Certificates to the Hardware Model Checking Competition, CAV 2025.
- [\[CAV26\]](#) Froleys et al., Liveness Proofs for Hardware Model Checking, CAV 2026.
- [\[FM26\]](#) Froleys et al., Certifying Constraints in Hardware Model Checking, FM 2026.
- [\[IJCAR26\]](#) Froleys & Yu, Hardware Model Checking Certification with Certifaiger and Cerbtora, IJCAR 2026.
- [\[BMC03\]](#) Eén & Sörensson, Temporal Induction by Incremental SAT Solving, BMC 2003.
- [\[VMCAI11\]](#) Bradley, SAT-Based Model Checking without Unrolling, VMCAI 2011.
- [\[FMCAD12\]](#) Claessen & Sörensson, A Liveness Checking Algorithm That Counts, FMCAD 2012.
- [\[FMICS02\]](#) Biere, Artho & Schuppan, Liveness Checking as Safety Checking, FMICS 2002.
- [\[CAV24\]](#) Xia et al., Avoiding the Shoals – A New Approach to Liveness Checking, CAV 2024.